GEK-36481

INSTRUCTIONS

ELECTROHYDRAULIC CONTROL

FOR

BOILER FEEDPUMP TURBINE

3S7513TC300A14

GENERAL ELECTRIC

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company

INTRODUCTION

The electronic portion of an electrohydraulic control system for a boiler feedpump turbine drive is described in these instructions.

Any adjustment or maintenance of this equipment should be performed either by qualified turbine personnel or supervised by such personnel.

This equipment is an electronic speed governor whose speed set point can be varied by an automatic boiler controller or by an operator from a remote control unit. Electronic redundancy and mechanical overspeed trip are used to provide a high degree of reliability and safety.

The elementary diagram and alignment diagram for this unit are included in these instructions. Since both the number and arrangement of auxiliary components vary with particular application, it is necessary to refer to the drawings furnished with each equipment to determine the correct connections and auxiliary components furnished in each instance.

RECEIVING, HANDLING AND STORAGE

RECEIVING AND HANDLING

Immediately upon receipt, the equipment should be carefully unpacked to avoid damaging the apparatus. Particular care should be exercised to prevent small parts being mislaid or thrown away in the packing material.

As soon as the equipment is unpacked it should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly.

STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If the equipment is to be placed in long term storage, the interior cabinet temperature should be 5°C above ambient.

DESCRIPTION

SPEED CONTROL

The speed control loop will be described first. Starting at zone B6 and B7 of the alignment drawing, the coil of the electrohydraulic servo valve receives a low power signal from the electronic unit and controls the flow of oil to a double acting hydraulic cylinder which positions the turbine steam control valve. Turbine speed is sensed by three magnetic sensors mounted near the periphery of a shaft-driven gear. Pulses whose frequency is directly proportional to turbine

DESCRIPTION (continued)

SPEED CONTROL (cont'd)

speed are generated as each gear tooth passes a sensor. Primary and secondary speed channels in the electronic control produce a DC voltage proportional to the frequency of the sensor outputs. Outputs of two speed channels are fed into high value gate number one, which rejects the lower of the two speed signals and passes on the higher valued speed signal. This redundancy permits one channel to fail without significantly affecting turbine speed. The secondary channel is adjusted for an output 1% less than the primary channel. It is therefore uncontrolling, except in case of failure of the primary channel or if a backup amplifier test switch in the cabinet is actuated which increases the output of the secondary channel to 1% greater than the primary channel. A test meter in the cabinet can be used to determine the controlling channel.

The speed signal from high value gate one is summed with the speed set point to obtain a speed error. (See zone B3 of the alignment diagram.) The speed error is passed through a stability compensation and error amplifier, which includes a regulation adjustment to obtain a valve lift demand signal. This is summed with a valve position signal to obtain a valve position error which is passed through the servo amplifier to the servo valve. Increasing the speed set point increases valve lift demand, valve position error, and valve opening. Speed will increase until the rise matches the set point increases.

The valve position signal is obtained from a Linear Variable Differential Transformer (LVDT) whose core is positioned by the control valve. Three kilohertz excitation for the LVDT is generated in the electronic cabinet. A demodulator in the cabinet produces a DC signal proportional to the AC output of the LVDT, which varies with core position.

A control valve closing bias is applied directly to the servo coil of the control valve in the tripped condition. Before the EHC can reset the feedpump turbine, a valve closing signal from the servo amplifier (2SA) must be present to operate the Close Signal Check (CSC) relay and remove the bias signal that had been previously applied directly to the servo valve coil. Relay circuitry also requires that the electrical connectors must be plugged into the servo valve, the LVDT, and the speed sensors before starting. The LVDT signal is biased to close the control valve if the LVDT connector is removed with the turbine running. The servo valve contains a 5% mechanical bias which will close the control valve in the absence of an electrical signal.

SPEED INDICATION

At zone B9 of the alignment diagram, a third magnetic sensor and circuitry identical to the previously described speed signal are used for speed indication. An analog signal is obtained for speed indication on the EHC remote control panel, local panel, and customer use.

DESCRIPTION (continued)

SPEED INDICATION (cont'd)

A clamp signal from the 10% speed voltage comparator holds the turbine to less than 10% speed if both the primary and secondary speed channels show less than 10% speed.

At 105% speed, a voltage comparator operates a relay that unlocks the lock out valve in the emergency trip exerciser scheme. At 110% speed another voltage comparator operates the 110SX relay to indicate that 110% speed has been reached. A contact of this relay is provided for the customer.

SPEED REFERENCE OR SET POINT

Identical motor driven LVDT's are used to generate the manual and automatic speed references. The core position is varied by a two-phase stepping motor which moves in discrete steps and locks in position when not running. Raise and Lower pushbuttons on the operator's manual control panel cause the manual reference to run from the LSS stop to the HSS stop in 80 to 90 seconds. Raise Fast and Lower Fast pushbuttons will run the manual reference from stop to stop in approximately 10 seconds. The automatic speed reference circuits contain interface circuitry for compatibility with a specific boiler controller. The auto speed reference is adjusted to run from the LSS to the HSS in 10 seconds. The automatic and manual speed references are fed into a low value gate which rejects the higher reference and passes on the lower reference to control turbine speed. A relay operated from this gate indicates which of the two references is controlling.

An overspeed test can be performed after first running both manual and automatic references to the high speed stop. The Overspeed Test pushbutton is then actuated, causing an additional speed reference to be generated which increases linearly with time at approximately 12% speed per minute. This is added to both manual and automatic references. If turbine speed rises to 120% of rated, a speed signal will override the ramp generator and prevent further speed increase. Upon release of the Test pushbutton, the reference ramps back to normal at the same rate as it increased.

HIGH PRESSURE/LOW PRESSURE STOP VALVE INDICATION

The EHC cabinet contains circuitry for driving HPSV and LPSV position indicators located on the control panel supplied with the EHC cabinet. The signals for these position indicators are obtained from LVDT's mounted on their respective stop valves. This function is for position indication only and is not part of the control circuitry for the feedpump. (Refer to zone C7, C8 of the alignment drawing.)

DESCRIPTION (continued)

LOW SPEED SWITCH

An electronic Low Speed Switch (LSS) is provided in the EHC cabinet to automatically, if selected, engage the turning gear on decreasing turbine speed less than 2 rpm.

If at turbine speeds greater than 200 rpm, pulses are lost from the LSPU shown at zone A9 of the alignment drawing, the LSS will prevent the LSSX relay from energizing and prevent automatic engagement of the turning gear by the EHC.

The LSS provides an alarm under the following conditions:

1. If the turbine is at standstill, 0 rpm, and then is increased in speed and pulses are not detected from the LSPU, the LSS 2K contacts will energize the LSS alarm X relay when 200 rpm is reached.

2. If the pick-up pulses are detected from the LSPU but lost before 200 rpm is reached, the LSS 2K contacts will energize the LSS alarm X relay when 200 rpm is reached.

3. If pick-up pulses are lost from the LSPU at turbine speeds greater than 200 rpm, the LSS 2K contacts immediately energize the LSS alarm X relay.

4. Once the LSS alarm X relay is energized, the automatic engagement of the turning gear is prevented until pulses from the LSPU are obtained.

POWER SUPPLIES

Two redundant power supplies, referred to as line A and line B, are provided, They are the full wave rectifier, capacitor filtered variety. Each receives its 120 VAC from the customer's power.

The 120 VAC is fed to a transformer that has three secondary windings. The output of each of the rectified, filtered supplies is connected to three identical series regulators that are adjusted for 30 VDC, 24 VDC and -22 VDC.

The outputs of the series regulators on the line A supply are adjusted slightly higher so they are carrying load and the line B supply is the backup power supply.

SERIES REGULATOR

Operation of the series regulators is described as follows. Output of the voltage sensing bridge consisting of 15R, 16R, 17R, 1P and 2ZD is fed to 7Q and 8Q. An increase in output voltage raises the voltage on 8Q, increasing its conduction and lowering conduction of 7Q. Decreased current in 7Q decreases current in 4Q, 5Q, and 6Q, dropping the output voltage until balance is restored. Excessive current output through 10R turns 3Q on, 2Q off, and 1Q on, grounding the base of 7Q. With 7Q off, series regulators 4Q, 5Q, and 6Q are off and the output goes to zero. When 1C discharges, 3Q is turned off and output is restored. The circuit will oscillate between off and on until the cause of the high current is removed.

MONITORING

Located in the front of the cabinet are two meter panels that contain several alarms, meters, indicating lights and test meter that will allow checking or monitoring of the power supplies and most amplifiers in the cabinet.

POWER SUPPLY ALARM

For example, an alarm will occur if the 30 volt line A power supply should exceed the 95% or 105% set point of the meter relay in the cabinet. There will be three lights that will be illuminated. First, the power lamp on the control panel supplied with the cabinet will change from a steady glow to a flashing indication, the alarm lamp in the cabinet will be illuminated and the illuminated pushbutton above the line A 30 volt meter will indicate either a high or low voltage.

The three alarm lamps can be cleared after the malfunction is corrected by depressing the lighted pushbutton above the line A 30 volt meter. Identical functions are provided for both the line A and line B 30 V, 24 V, and -22 V power supplies and 3 KHz 6 V RMS power supply in the EHC cabinet. If any of these power supplies are not at their proper voltage level when the 120 V 60 Hz is applied, startup of the feedpump will be prevented.

GROUND ALARM (24 VDC)

If a ground were to occur on the positive or negative side of either the line A or line B 24 volt power supply, the power lamp on the control panel would flash, the +24 V GND/-24 V GND lamp in the cabinet would be illuminated, and the alarm lamp in the cabinet would be illuminated. These lamps will be cleared when the ground is removed. Depressing the +24, -24 V lamp in the cabinet is not required to clear the lamp.

HSS/LSS ALARM

HSS and LSS indication is provided both in the cabinet and on the control panel for both the manual and auto speed reference travel limits. These lamps will be cleared when the respective speed reference is driven off the limit.

CABLE CHECK ALARM

The cable check will prevent startup of the feedpump if the cables are not properly connected. The power lamp on the control panel will flash, the cable check lamp in the cabinet will be illuminated, and the alarm lamp in the cabinet will be illuminated until the cable is properly reconnected.

LAMP TEST

A lamp test can be performed on the indicating lamps in the cabinet by depressing the Alarm pushbutton. There aren't any lamps in the Backup Speed Amplifier Test pushbutton.

MONITORING (continued)

LAMP TEST (cont'd)

Depressing the Power pushbutton on the control panel will test the lamps on the control panel. There aren't any lamps in the Lower Fast, Lower, Raise, Raise Fast pushbuttons.

TEST SWITCH

The amount of current being supplied by each series regulator can be observed by the test switch and meter in the cabinet. When measuring series regulator current, the meter is calibrated to read approximately 5 amps full scale.

The voltage output of amplifiers 1 thru 12 can be observed by using the test switch and meter. When measuring amplifier outputs, the meter is calibrated to read 10 volts full scale. The amplifier numbers and function are shown on the alignment drawing and elementary diagram.

INSTALLATION AND ADJUSTMENT

The electronic control cabinet should preferably be mounted in a clean, dry, air-conditioned environment. Ambient temperature should not exceed 104°F unless otherwise noted on the outline drawing for a specific unit.

Equipment adjustment and the required test equipment are covered in detail in an alignment diagram and alignment instructions furnished with each unit.

The location of subassemblies within the cabinet is shown on the elementary diagram furnished with each control. Printed circuit board locations are shown on sheets 1 and 2 of the elementary diagram.

OPERATION

Refer to the instruction book for the turbine.

MAINTENANCE

No periodic maintenance is required. Outage time due to component failure is best minimized by maintaining a complete set of spare parts. Elementary diagrams of the printed circuit boards are included in this instruction book. The assembly drawings showing component location on the printed circuit boards are furnished with each order except where component identifications are marked on the boards.

TROUBLESHOOTING

Most troubleshooting problems can be quickly isolated to a particular section of the control by use of the built-in test meter, the meter relays, and alarm lights. These, used in conjunction with the alignment diagram, indicate the operating condition of each section. An oscilloscope is recommended for pinpointing a defective component or components. Each printed circuit board can be removed from the rack for servicing without disconnection due to the use of terminal boards.

RENEWAL PARTS

When ordering renewal parts, address the nearest Apparatus Sales Office of General Electric Company and supply the following information.

a. Catalog number stamped on the part and a complete description of the part, including its use and location.

b. Complete nameplate data appearing on the assembly of which the part is a component.

c. If possible, data on original order on which equipment was first supplied, including all numerical references.

Since operation of a steam turbine-generator is dependent upon operation of the feedpump control, it is suggested that an adequate stock of spare parts be carried on hand to minimize outage time due to component failure.

METER RELAY LAMP REPLACEMENT

Type 195 Meter Relays have been made with two versions of lamps; one without leads attached to the lamp (GE Pt. No. 4868K32P001) and the other with leads attached to the lamp and lamp holder assembly (GE Pt. No. 4872K12G700). These replacement parts can be ordered directly from: the nearest General Electric Sales Office or Utility & Process Automation Products Department, General Electric Company, 40 Federal Street, West Lynn, Mass. 01910.

In the event of lamp failure, the applicable lamp or lamp assembly can be replaced when the control unit is detached from the rear of the indicator set point unit. Then, the lamp-access plate (see figure) must be removed (screws are beneath adhesive labels on some models) to expose the lamp holder which can be pulled out.

PT.NO. 4868K32P001 LAMP

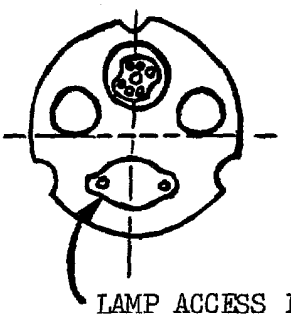
When installing the bayonet-type Pt. No. 4868K32P001 (without leads), twist the lamp in the holder assembly until the lamp filament will be in the horizontal position when the lamp holder is replaced in the indicator set point unit.

PT. NO. 4872K12G700 LAMP & HOLDER

The Pt. No. 4872K12G700 lamp has been fixed in the lamp holder at the factory so the filament position will be in the correct position when the keyed assembly is inserted in the indicator set point unit. Make sure the lamp-lead clips are engaged by the screws when the lamp access cover is replaced.

NOTE: If relay operation is not within $\pm$ 2% of set point index, the lamp holder position must be rotated 180°.

METER RELAY LAMP REPLACEMENT (continued)



REASSEMBLY

After the new lamp or lamp assembly has been installed, make sure the lamp access plate is replaced (and screws are tight) before control unit is reassembled on rear of indicator set point unit.

DRAWING LIST

DESCRIPTION	DRAWING NUMBER
Diode Board, 1DB	44A300243
Reference Low Value Gate, 2LVG	44A300281
Compensation and Regulation, 1CR	44A301400
Overspeed Integrator, 1OSI	44A302692
Signal Low Value Gate, 1LVG	44A304763
Alignment Instructions	44A307258
Series Regulator Board #1, 1REC	44A306392
Renewal Parts	44A335558
Cable Wiring Diagram, Signal & Low Voltage, 1CO	44B300423
Control Panel Outline	44B307044
Low Speed Switch Power Supply, LPS	44B306144
Cable Wiring Diagram Power, 2CO & 3CO	44B306148
Series Regulator Board #2, 1SR	44B306197
Detector Indicator, 1DI	44C300609
Operational Amplifier, 1OA	44C300610
Analog Output, 1AO2	44C301216
Analog Output, 3AO2	44C301622
Frequency to Voltage Converter, 2FVC	44C301682
Clock and Stepping Motor Drive, CL&SMD	44C303567
Servo Amplifier, 2SA	44C304444
3 kHz Oscillator, 6 OCS	44C304462
Signal Converter, 3SC	44C304898
Voltage Comparator, 1VC	44C305259
Elementary Drawing, Master	44C307556
Connection Drawing	44C307557
Low Speed Switch, LSS	44C306444
Boiler Control Interface, 5BCI	44C305210
Line A & Line B Power Supply	44C306526
Cabinet Outline	44D300043
Alignment Drawing	44D306685
Control Panel Connection Diagram	44D306687
Relay Panel Connection Diagram	44D308103
Summary	44X306880

GENERAL ELECTRIC

44A300243
CONT ON SHEET SH NO.

REV NO. 0

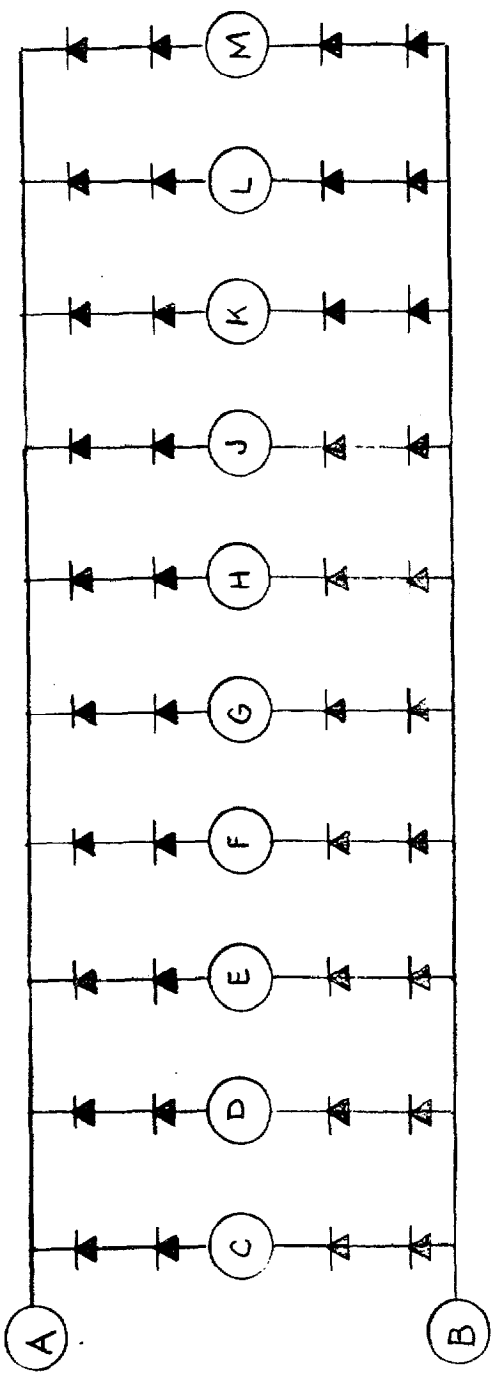
CONT ON SHEET SH NO.

TITLE

DIODE BOARD IDB

FIRST MADE FOR 44D236370-G01

REVISIONS



ALL DIODES 200 MA; 200 V.

11

DFB 12P

PRINTS TO

MADE BY	APPROVALS	DIV OR DEPT.	
R. Chambers 1/15/68	MBC	SPECIALTY CONTROL	44A300243
ISSUED MAR 12 1968		WAYNESBORO, VA. LOCATION	CONT ON SHEET SH NO.

CODE IDENT NO. 2

FF-803-WF (2-67)
PRINTED IN U.S.A.

GENERAL ELECTRIC

44A300281

CONT ON SHEET SH NO.

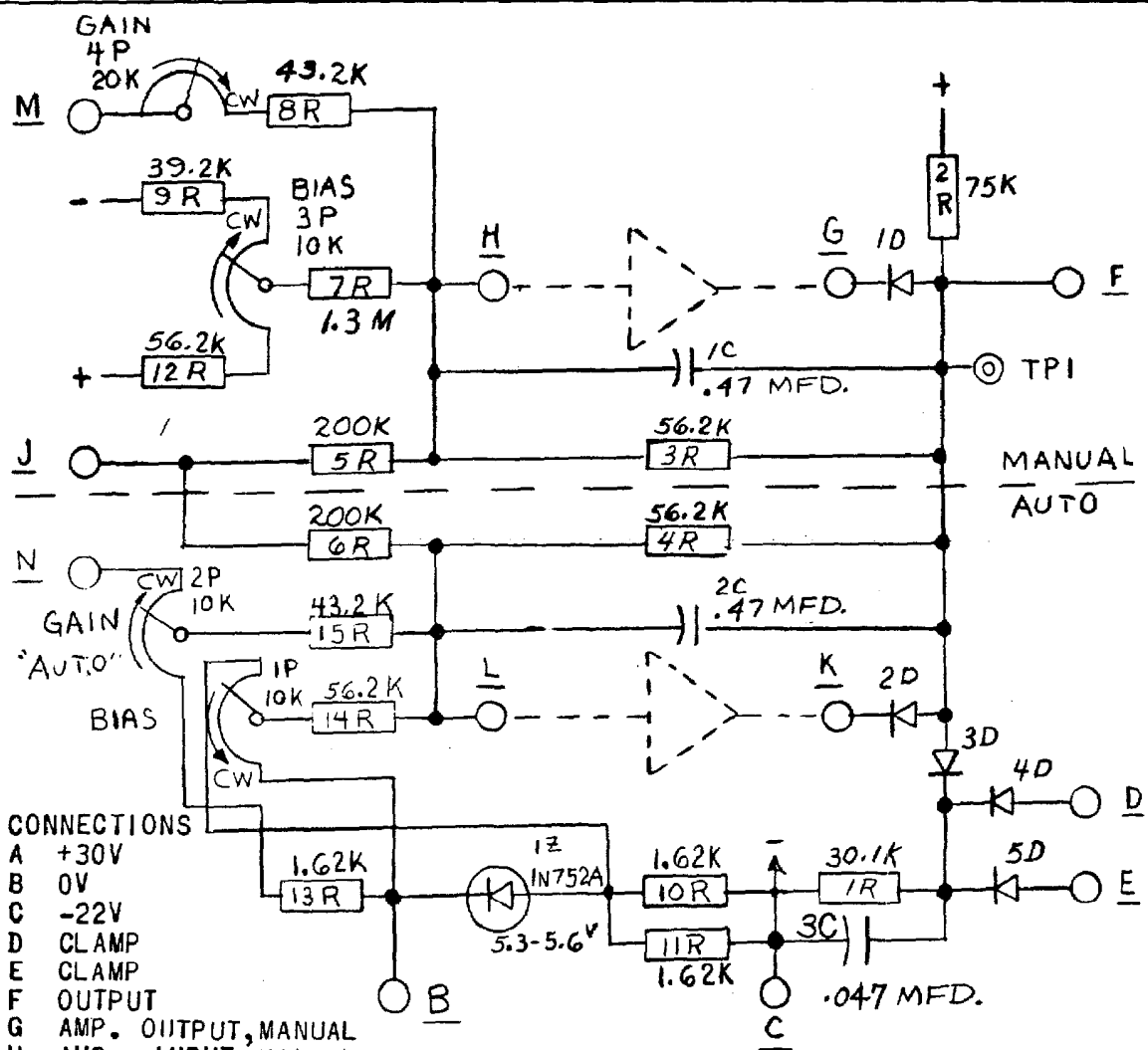
REV NO. (1) C. KENNEDY

CONT ON SHEET SH NO.

TITLE

REFERENCE LOW VALUE GATE (2LVG)

FIRST MADE FOR 44D236378-G01



CLAMP NOTE: ZERO VOLTS ON D OR E WILL HOLD OUTPUT (F) AT ZERO.

ADJUSTMENT:

1: WITH AUTO INPUT AT HSS AND MANUAL AT LSS, ADJUST BIAS FOR ZERO OUTPUT AT F.

2: WITH BOTH GAINS MAX. (CW) AND INPUTS AT HSS, ADJUST MANUAL GAIN FOR 5 VOLTS OUTPUT.

3: DECREASE AUTO GAIN UNTIL OUTPUT OF THE ASSOCIATED AMPLIFIER SWITCHES NEGATIVE TO POSITIVE.

4: AUTO BIAS IS SET CCW INITIALLY. FINAL SETTING PER APPLICATION REQUIREMENT.

REVISIONS

1	CHANGED 5R, 6R & 8P VALUES BUKOSKEY MAY 5 1969	
2	E.A. Lewis	OCT 19 1970
3	Updated Nov 17, 1971	

11

12P DFB

FG-W

PRINTS TO

MADE BY B Chambers Mod 11 1968

ISSUED APR 17 1968

APPROVALS R W Simmons

SPECIALTY CONTROL DIV OR DEPT.

WAYNESBORO, VA LOCATION

44A300281

CONT ON SHEET SH NO.

FF-803-WF (9-66)
PRINTED IN U.S.A.

CODE IDENT NO.

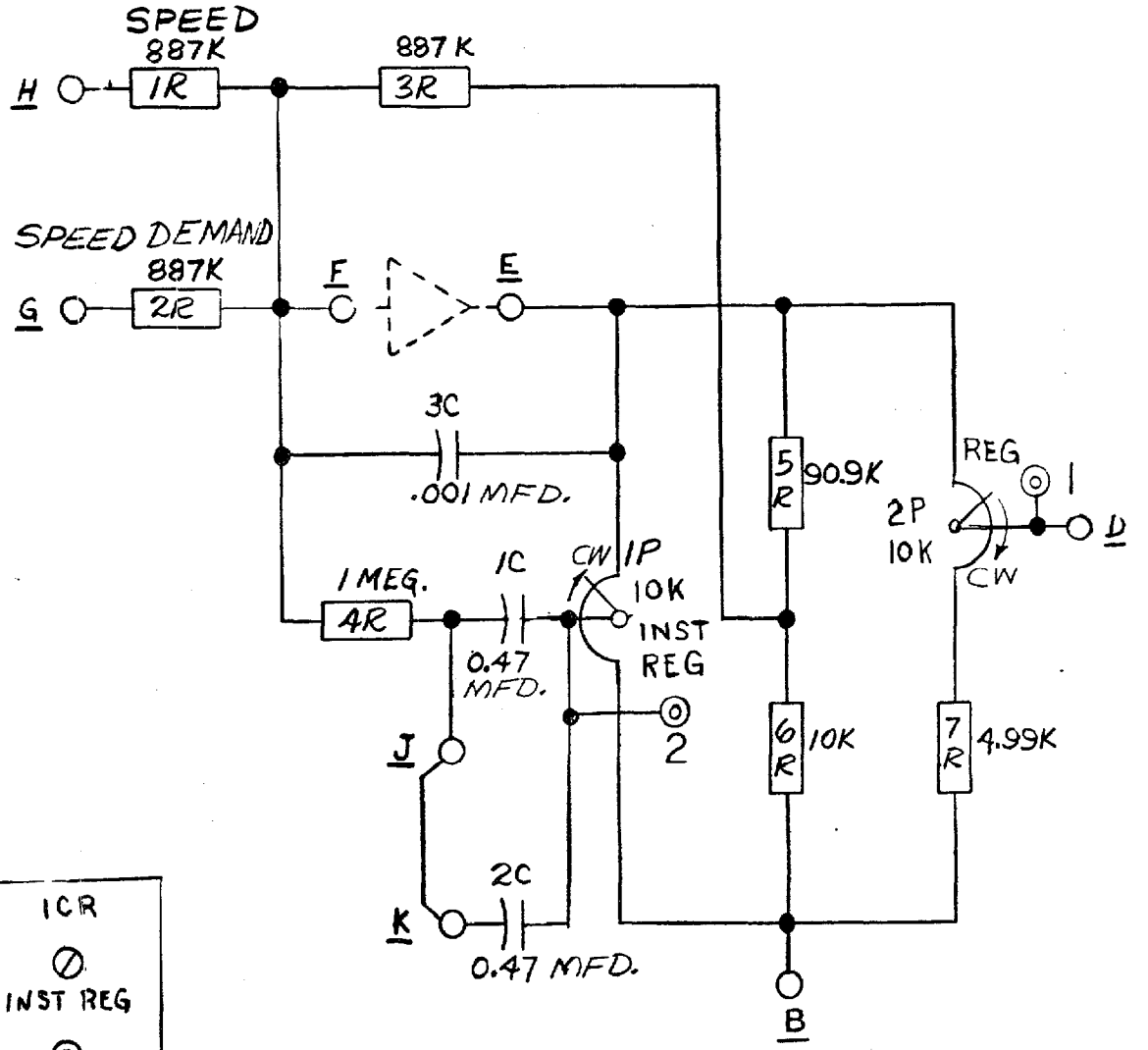
2

GENERAL ELECTRIC

REV NO.	44A301400 CONT ON SHEET	SH NO.

TITLE: COMPENSATION & REGULATION (ICR)

FIRST MADE FOR 44D236388-G01



ICR
INST REG
REG

REVISIONS			
1	R.M. BUKOSKEY APR 1 1969	2	E.A. Clemens Sept. 21, 1971
3	Nov. 17, 1971	4	H. Booth Aug. 28, 1972

11

PRINTS TO

MADE BY R.M. Bukoskey	APPROVALS R.W.	SPECIALTY CONTROL DIV OR DEPT.	44A301400 CONT ON SHEET SH NO.
ISSUED FEB 18 1969		WAYNESBORO, VA. LOCATION	CODE IDENT NO

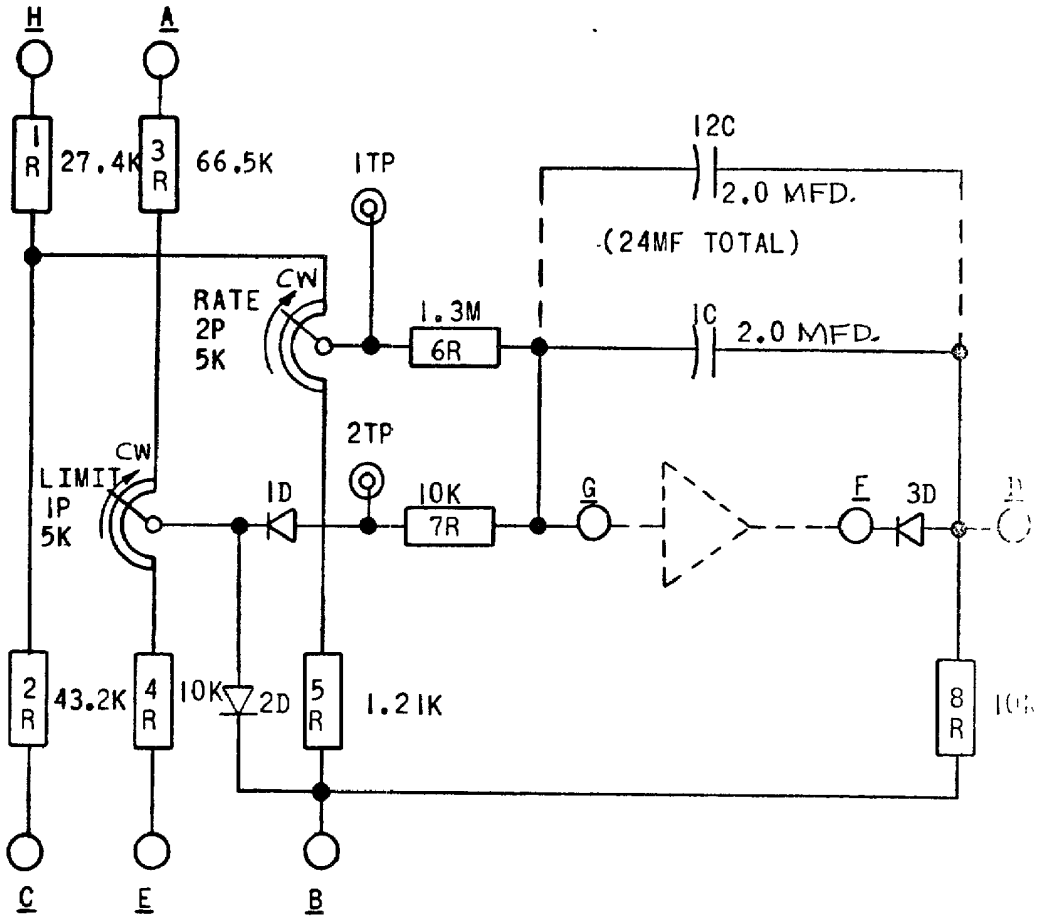
FF-803-WI* (9-67)
PRINTED IN U.S.A.

GENERAL ELECTRIC

44A302692

REV NO.	TITLE	CONT ON SHEET	SH NO.
① C KENNEDY	OVERSPEED INTEGRATOR		IØSI
44A302692 CONT ON SHEET SH NO	FIRST MADE FOR 44D236373-G02		

REVISIONS



CONNECTIONS

A + 30V
B 0V
C -22V
D OUTPUT
E SPEED SIGNAL
F AMPLIFIER OUTPUT
G AMPLIFIER INPUT
H POS. SUPPLY THROUGH RELAY
J N.C.

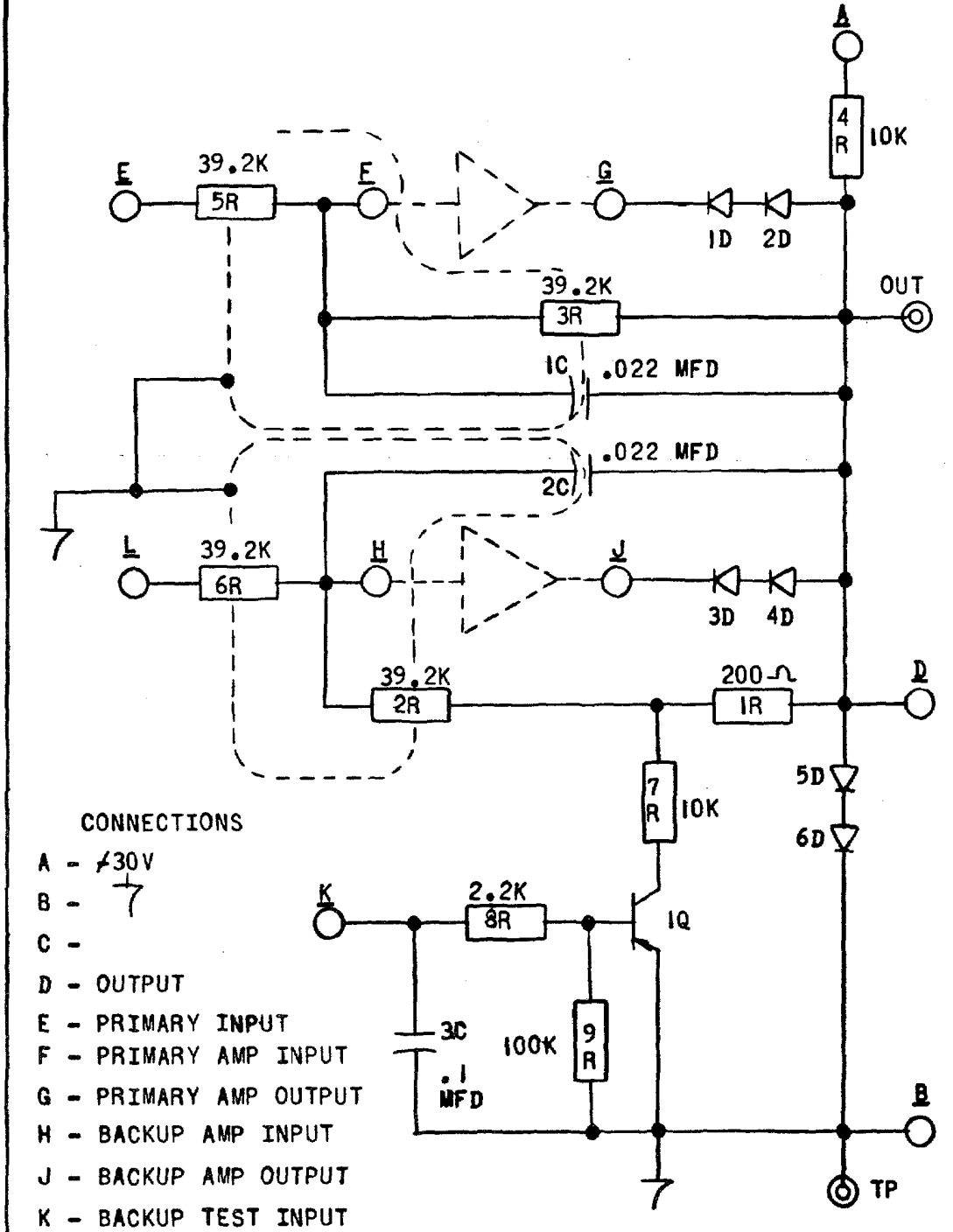
MADE BY	APPROVALS			
H. Templeton 3-16-70	R. Culver	Comm & Gov't Devices DIV OR DEPT	44A30?	
ISSUED APR 8 1970		WAYNESBORO, VA. LOCATION	CONT ON SHEET	

FF-503-WF (10-67)
PRINTED IN U.S.A.

2-1

GENERAL ELECTRIC

REV NO.	TITLE	44A304763 CONT ON SHEET SH NO.
44A304763 CONT ON SHEET SH NO.	SIGNAL LOW VALUE GATE FIRST MADE FOR 44D236372-G02	ILVG



CONNECTIONS

A - +30V
B - ⏚
C -
D - OUTPUT
E - PRIMARY INPUT
F - PRIMARY AMP INPUT
G - PRIMARY AMP OUTPUT
H - BACKUP AMP INPUT
J - BACKUP AMP OUTPUT
K - BACKUP TEST INPUT
L - BACKUP INPUT

REVISIONS

1	Nov. 18, 1971	

11
12P DFB
FG-W

PRINTS TO

MADE BY	APPROVALS		
H. Eggleston 8-13-71	R. Culver	COMM. & CONT. DEVICES DIV OR DEPT	44A304763 CONT ON SHEET SH NO.
ISSUED Aug. 16, 1971	D. L. Mays	WAYNESBORO, VA. LOCATION	

FF-503-WF (2-62)
PRINTED IN U.S.A.

TITLE GULF STATES, SABINE #5
ALIGNMENT INSTRUCTIONS
ELECTRONIC FEED PUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A14 TURBINE 122816

REVISIONS

REFERENCE ALIGNMENT DIAGRAM 44D306685

TEST EQUIPMENT

1. AUDIO OSCILLATOR: HEWLETT PACKARD 201C, 200CD; GENERAL RADIO 1210C, OR EQUIVALENT. OUTPUT MUST BE 10 VOLTS RMS MINIMUM INTO 600 OHMS.

2. AC/DC DIGITAL VOLTMETER: 0.1 MV TO 100 V MAX.

3. DIGITAL FREQUENCY COUNTER: 6 VOLT RMS SENSITIVITY. 30 TO 5000 HZ.

4. TEN TURN POTENTIOMETER; NOT MORE THAN 5000 OHMS RESISTANCE.

5. TWO INPUT OSCILLOSCOPE, CALIBRATED SWEEP: X1 AND X10 PROBES.

6. OPTIONAL: BRUSH RECORDER SPECIAL TESTS ARE DESIRED.

7. VOLT - OHM - MILLIAMMETER

8. TWO 100K RESISTORS, TWO 1 MFD CAPACITORS (TO ADJUST OSCILLATOR OFFSET)

I. PRELIMINARY

1. CHECK INTERCONNECTION OF CONTROL CABINET, OPERATOR'S CONTROL DEVICES, TURBINE DEVICES, ETC.

2. MECHANICALLY ZERO TEST METER IN CABINET AND OPERATOR'S TACHOMETER AND OTHER INSTRUMENTS.

II. LINE "A" AND LINE "B" POWER SUPPLIES LOC. 12 AND 13

1. SET UP CONNECT 120V, 60 HZ TO LINE "A" POWER SUPPLY TERMINALS T24 AND T25. REMOVE POWER FROM LINE "B" TERMINAL T22 & 23.

2. DC SERIES REGULATORS, ADJUST POWER SUPPLIES AND METER CALIBRATIONS PER CHART. IF THE VOLTAGES LISTED CANNOT BE OBTAINED IT MAY BE NECESSARY TO ADJUST THE TAPS ON TRANSFORMER 1T, LOC. 12 & 13.

LINE "A" SUPPLY	TEST POINTS	ADJUST
+30V	BUS BARS	0308
-22V	BUS BARS	0316
24V	+F LOC. 0322; -G, LOC. 0324	0324

11

12P-KFM 2

FG-W

PRINTS TO

REVISIONS

3. METER RELAY ALARMS VARY ALARM SET POINT OF LINE "A" POWER SUPPLY METER RELAYS TO CHECK ALARM OPERATION. SET ALARMS FOR 95% AND 105%.

4. CONNECT POWER TO LINE "B" POWER SUPPLY BY JUMPERING TERMINAL T24 TO T22 AND T25 TO T23, SO THAT POWER IS APPLIED TO BOTH SUPPLIES.

5. ADJUST THE INDIVIDUAL LINE "B" SERIES REGULATOR BOARDS SO THAT THEY ARE CARRYING ONLY 0.1 AMP AND THE LINE "A" SUPPLIES THE REMAINING LOAD. WHILE NOTING THIS CONDITION IS MAINTAINED MAKE FINAL ADJUSTMENTS IF NECESSARY PER CHART.

SUPPLY	LINE B	LINE A	TESTPOINTS
+30	0310	0308	BUS BAR
-22	0318	0316	BUS BAR
+24	0326	0324	LINE A, + F, 0322; -G, 0324 LINE B, + F, 0328; -G, 0326

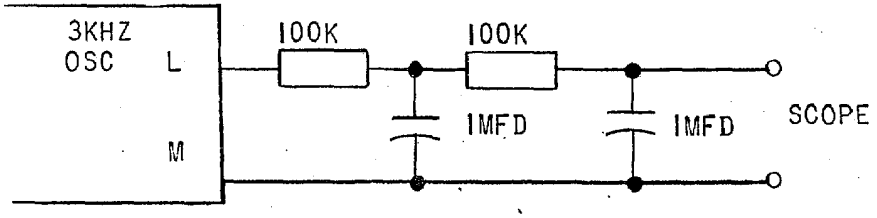
6. TEST AND SET LINE "B" METER RELAY ALARMS AT 95% AND 105%

III. 3KHZ OSCILLATOR, -0217

1. VERIFY OUTPUT FREQUENCY AT 0217 IS 3000HZ. ± 300 HZ. USING COUNTER AT TERMINALS L&M.

2. ADJUST VOLTAGE LEVEL FOR 6.000 ± .006V. SET METER FOR 100%. CHECK METER RELAY ALARMS AND SET AT 95 AND 105%.

3. OFFSET ADJUSTMENT, CONNECT A FILTER CIRCUIT AS SHOWN OR DIGITAL VOLTMETER TO L & M. VOLTMETER ON MILLIVOLT SCALE



3.1 REMOVE 3KHZ OSC BD FROM RACK AND ADJUST OFFSET FOR ZERO D.C. VOLTS ± 1 MV.

IV. SSPU SPEED INDICATION CIRCUITS

1. SET UP, REMOVE LEAD FROM T28 AND T29 AND CONNECT AUDIO OSCILLATOR. SET OSCILLATOR TO SSPU RATED HZ.

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12P·KFM²

FG-W

PRINTS TO

2. FREQUENCY TO VOLTAGE CONVERTER

2FVC LOC. 0224. ADJUST COARSE AND FINE POTS TO OBTAIN ⁄5.00V AT TERMINAL J.

3. VOLTAGE COMPARATORS IVC LOC., 0218, 0219, 0221. SET EACH VOLTAGE COMPARATOR (IVC) PER CHART ON ALIGNMENT DRAWING. RED LIGHT ON IVC SHOULD BE ON WHEN RELAY ENERGIZED. CHECK EACH IVC SETTING BY SLOWLY VARYING FREQUENCY ABOUT THE OPERATING POINT.

4. TACHOMETER 3AØ2 LOC. 0108 WITH OSCILLATOR AT ZERO FREQUENCY ADJUST 2A BIAS FOR ZERO INDICATION ON REMOTE CONTROL PANEL. SET OSCILLATOR TO RATED SSPU FREQUENCY AND ADJUST GAIN AND I OUT AS REQUIRED FOR CORRECT INDICATION OF TURBINE RPM.

4.1 REPEAT ZERO FREQUENCY AND RATED SPEED SETTINGS AND MAKE FINAL ADJUSTMENTS IF REQUIRED.

5. SECOND TACHOMETER 3AØ2 LOC. 0110 WHERE REQUIRED ADJUST 1A & 2A USING PROCEDURE IN STEPS 4 AND 4.1 ABOVE.

6. REMOVE OSCILLATOR FROM T28 AND T29 AND RECONNECT INCOMING LEADS.

V. SSPU SPEED MEASURING CIRCUITS

1. SET-UP JUMPER T4 TO T7 AND T5 TO T8. CONNECT OSCILLATOR AND COUNTER TO T4 AND T5.

2. FREQUENCY TO VOLTAGE CONVERTERS 2FVC, LOC. 0203 AND 0206 HIGH SPEED GATE 1LVG, LOC. 0209.

2.1 PRIMARY AT RATED SHAFT SPEED PICKUP FREQUENCY, ADJUST 2FVC LOC. 0203 FOR -5.000 VOLTS AT OUTPUT TESTPOINT OF 1LVG, 0209. (OBSERVE WITH TEST METER IN POSITIONS 6 AND 7 THAT AMPLIFIER 7 IS MORE POSITIVE THAN 6; IF NECESSARY ADJUST 2FVC, 0206, TO OBTAIN THIS CONDITION).

2.2 BACKUP DEPRESS BACKUP SPEED AMPLIFIER TEST SWITCH AND ADJUST 2FVC, 0206, FOR -5.05 VOLTS AT OUTPUT TP, 0209.

3. REMOVE TERMINAL POINT JUMPERS AND OSCILLATOR AND COUNTER LEADS.

VI. CONTROL VALVE RAM SERVO LOOP

1. CHECK LVDT CORE PER LEVER DIAGRAM.

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2. DETECTOR INDICATOR IDI, LOC. 0216. WITH CONTROL VALVE FULL OPEN ADJUST NULL OF IDI FOR 0.00 ± 0.01 VOLTS AT OUT TP. NOTE VOLTAGE READING ON ALIGNMENT DIAGRAM. WITH VALVE IN CLOSED OVERTRAVEL POSITION RECORD OUT TP VOLTAGE ON ALIGN. DIAG. ADJUST ZERO FOR 0 INDICATION OF CONTROL VALVE POSITION ON OPERATOR'S CONTROL CONSOLE. RETURN VALVE TO FULL OPEN POSITION AND ADJUST CALIBRATE FOR 100% POSITION INDICATION.

3. DETERMINE GAIN OF DETECTOR CIRCUIT IN VOLTS PER INCH BY DIVIDING THE DIFFERENCE IN OUT TP VOLTS AT EXTREME VALVE POSITIONS BY THE TOTAL VALVE TRAVEL IN INCHES, AND RECORD THIS FIGURE ON THE ALIGNMENT DIAGRAM.

3.1 ANALOG CONTROL VALVE POSITION SIGNAL

1AØ2 LOC. 0106. WITH VALVE CLOSED ADJUST 1A BIAS FOR ZERO POSITION ON OPERATORS METER.

3.2 WITH VALVE OPEN ADJUST 1A GAIN AND I OUT FOR FULL OPEN INDICATION.

3.3 REPEAT 3.1 AND 3.3 AS NECESSARY.

4. MAKE ELECTRICAL CONNECTIONS TO SERVO VALVE.

5. SERVO AMPLIFIER, 2SA, LOC. 0214.

5.1 PS10 TRIP CHECK FOR APPROX. -0.7 VOLTS AT "I" TP UNDER TRIP CONDITION.

5.2 CLOSE SIGNAL CHECK WITH TURBINE TRIPPED, DISCONNECT DEMAND TERMINAL E AND LIFT TERMINAL D OF 2SA. CONNECT TEST POT WITH NEGATIVE OUTPUT TO TERMINAL D AND SET POT FOR ABOUT -3 VOLTS AT LIFT TP. THE OUTPUT OF AMPLIFIER 9 ON TEST METER POSITION 9 SHOULD BE ABOUT -10 VOLTS. TURN BIAS ADJUSTMENT CCW UNTIL AMP. 9 GOES TO ≠10 VOLTS. THERE SHOULD BE LITTLE OR NO CONTROL OF 9 AT INTERMEDIATE VOLTAGES. LEAVE 9 AT ≠10 VOLTS AND RESET TURBINE. SLOWLY REDUCE TOWARDS 0 THE TEST POT SETTING AND NOTE THAT 9 SWINGS FROM ≠10 TO A MINIMUM VALUE OF APPROX. -6.2 VOLTS. NOW APPLY MORE NEGATIVE VOLTAGE FROM THE TEST POT AND NOTE THAT 9 HAS A MAXIMUM VALUE OF APPROX. ≠6.2 VOLTS.

5.3 GAIN REFER TO ALIGNMENT DRAWING TO OBTAIN LIFT TEST POINT VOLTAGE REQUIRED TO PRODUCE 1 VOLT AT I TEST POINT, WHEN GAIN IS PROPERLY ADJUSTED.

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REVISIONS

5.31 VARY THE TEST POT SETTING UNTIL I TEST POINT IS 0.000 VOLTS AND NOTE LIFT TEST POINT VOLTAGE. CHANGE THIS VOLTAGE WITH THE TEST POT BY PLUS LIFT TEST POINT VOLTAGE FROM STEP 5.3, ADJUST GAIN UNTIL I TEST POINT READS MINUS 1.000 VOLTS.

5.32 RECHECK ZERO POINT BY RETURNING TEST POT SO THAT ZERO VOLTS IS ONCE AGAIN AT I TEST POINT AND TEST POT VOLTAGE IS THE SAME AS NOTED IN 5.31.

5.33 ADJUST TEST POT FOR MINUS LIFT TEST POINT VOLTAGE AND NOTE THAT I TEST POINT IS PLUS 1.000 VOLT.

5.4 BIAS DISCONNECT TEST POT AND RECONNECT THE LEAD REMOVED FROM D. WITH THE VALVE RAM AT ITS INTERNAL STOP AND ZERO VOLTS A DEMAND TEST POINT SET BIAS FOR -0.35 ± .01V AT I TEST POINT. RECONNECT TERMINAL E.

6. AFTER REMOVING THE ELECTRICAL CONNECTOR, MOUNT THE SERVO VALVE AND VERIFY THAT THE MECHANICAL BIAS IS IN THE VALVE CLOSING DIRECTION.

VII. REGULATION AND COMPENSATION ICR, LOC. 0212

1. REGULATION CONNECT THE OUTPUT OF THE TEST POT TO SUPPLY A NEGATIVE VOLTAGE TO TERMINAL E. ADJUST REGULATION TO OBTAIN VALVE TRAVEL FROM NO LOAD LIFT TO FULL LOAD LIFT FOR 3.00 VOLTS CHANGE IN THE OUTPUT OF THE TEST POT.

2. COMPENSATION VERIFY THAT JUMPER J-K IS NOT CONNECTED.

3. INSTANTANEOUS REGULATION SET TEST POT FOR -1 VOLT OUTPUT AND ADJUST INST. REG. FOR VOLTAGE EQUAL TO 0.0 VOLTS AT INST REG. T.P.

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PRINTS TO

TITLE GULF STATES, SABINE #5
ALIGNMENT INSTRUCTIONS
ELECTRONIC FEED PUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A 14 TURBINE 1228J16

REVISIONS

VIII. MANUAL SPEED REFERENCE (STAND-STILL CHECKS)

1. THIS REFERENCE IS OBTAINED FROM THE UPPER OF THE TWO MOTOR DRIVEN LVDT'S. WHEN POWER IS APPLIED IT SHOULD RUN TO LSS AND LIGHT THE LSS INDICATING LIGHT.

2. SLOW REFERENCE SPEED, CL & SMD, LOC. 0118. SHUT OFF HIGH PRESSURE OIL SUPPLY OR DISCONNECT SERVO VALVE BEFORE RAISING MANUAL REFERENCE. ACTUATE RAISE PUSHBUTTON AND TIME TRAVEL TO HSS LIGHT INDICATION. TIME SHOULD BE 80-90 SECONDS.

3. FAST REFERENCE SPEED, CL & SMD, LOC. 0118. ACTUATE LOWER FAST PUSHBUTTON AND TIME MOTOR TRAVEL FROM HSS LIGHT TO LSS LIGHT. IT SHOULD BE 8 TO 10 SECONDS.

4. BIAS AND GAIN, MANUAL REFERENCE, 2LVG, LOC. 0116. DISCONNECT K OF 2LVG. RAISE MANUAL REFERENCE OFF LSS AND BACK INTO IT AT SLOW RATE WITH LOWER PUSHBUTTON. ADJUST MANUAL BIAS FOR 0.000 VOLTS AT OUTPUT TP, 0116 RAISE TO HSS AT SLOW RATE AND ADJUST MANUAL GAIN FOR OUTPUT VOLTAGE AT TP 0116 AS SPECIFIED BY ALIGNMENT DIAGRAM.

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REVISIONS

IX. AUTO SPEED REFERENCE, (STAND-STILL CHECK)

1. BOILER CONTROLLER INTERFACE, 5BCI, LOC. 0123.
CONFIRM THAT THE INTERCONNECTING CABLE BETWEEN THE FEEDWATER CONTROLLER AND EHC CABINET IS A SHIELDED CABLE, AND THE SHIELD OF THE CABLE IS CONNECTED TO COMMON ON THE FEEDWATER CONTROLLER AND TO T34 ON THE EHC CABINET.

   THE ADJUSTMENT OF IP (CMR) IS TO ELIMINATE ANY NOISE THAT MAY BE PICKED UP BETWEEN THE FEEDWATER CONTROLLER AND THE EHC CABINET.

2. USING A SCOPE, DETERMINE IF NOISE IS PRESENT BETWEEN TERMINALS B AND D OF 5BCI, LOC. 0123 IF A NOISE SIGNAL (PROBABLY 60 HZ OR 120 HZ) EXISTS, SET FEEDWATER CONTROLLER SIGNAL FOR LSS AND ADJUST IP FOR MINIMUM NOISE AT 1A OUT.

3. IF THERE ISN'T ANY NOISE SIGNAL BETWEEN TERMINALS D AND B OF 5BCI LOC. 0123, THEN DISCONNECT CUSTOMER'S INCOMING SHIELDED CABLE AT EHC TERMINALS T32, T33, AND T34 AND TEST EHC AS FOLLOWS. PLACE A JUMPER BETWEEN T32 AND T33 AND CONNECT OSCILLATOR BETWEEN T32 AND T34. SET OSCILLATOR AT 6 V RMS, 60 HZ AND ADJUST IP FOR MINIMUM AT 1A OUT. REMOVE OSCILLATOR AND JUMPER AND RECONNECT CUSTOMER LEADS TO T32, T33 AND T34.

4. WITH FEEDWATER CONTROLLER AT LSS, ADJUST 2P BIAS FOR OVDC AT 1A OUT.

5. SIGNAL CONVERTER 3SC, LOC. 0125

   WITH FEEDWATER CONTROLLER AT LSS. CHECK THAT 2LS LIMIT SWITCH CONTACTS BETWEEN 1TB(D)& 1TB(E) ON AUTO SPEED REF. LOC. 0005 ARE CLOSED. IF NECESSARY, REDUCE INPUT FROM FEEDWATER CONTROLLER UNTIL 2LS LIMIT SWITCH CLOSES. JUMPER SIGNAL FAILURE HOLD CONTACTS T77 AND T78 WITH 2LS LIMIT SWITCH CLOSED AND FEEDWATER CONTROLLER SIGNAL AT LSS. CONNECT VOLTMETER TO 1TP OF 3SC LOC. 0125 AND CHECK FOR ZERO VOLTS. IF NECESSARY, OBTAIN ZERO VOLTS BY WITHDRAWING 3SC FROM RACK AND ADJUSTING 5P.

6. REMOVE T77-T78 JUMPER AND SET FEEDWATER CONTROLLER SIGNAL TO HSS. CHECK THAT 1LS LIMIT SWITCH CONTACTS BETWEEN 1TB(D) AND 1TB(C) ON AUTO SPEED REF. LOC. 0005 ARE CLOSED. IF NECESSARY, ADJUST IP CW ON 3SC LOC. 0125 AND/OR 2P GAIN UNTIL 1LS LIMIT SWITCH CLOSES. WITH FEEDWATER CONTROLLER AT HSS AND 1LS LIMIT SWITCH CLOSED, JUMPER SIGNAL FAILURE HOLD CONTACTS T77-T78 AND ADJUST 2P GAIN FOR MINUS 5.00 V ±.01 VDC AT 1TP.

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7. WITH T77-T78 STILL JUMPERED, ADJUST IP B.C. REF. FOR ZERO VOLTS AT 2TP. REMOVE JUMPER.

8. WITH FEEDWATER CONTROLLER AT LSS CHECK FOR ZERO VOLTS AT ITP AND WITH FEEDWATER CONTROLLER AT HSS CHECK FOR ZERO VOLTS AT 2TP.

9. ADJUST 3P DEADBAND FOR MINUS 2VDC (0.1% DEADBAND) AT 3TP. DISCONNECT WIRE AT OUTPUT TERMINAL F OF 3SC LOC. 0125.

10. ADJUST "BOILER CONTROLLER" INPUT TO LSS (-9±1VDC AT 2TP). CONNECT AN OSCILLOSCOPE TO TERMINAL F AND CHECK FOR A PULSE WIDTH OF 10 MSEC. IF NECESSARY, WITHDRAW BOARD AND ADJUST FREQUENCY CONTROL POTENTIOMETER 4P. RECONNECT WIRE TO TERMINAL F. AUTO SPEED REFERENCE SHOULD DRIVE FROM LSS TO HSS IN 10 SECONDS.

11. AUTO SPEED REFERENCE STOPS, LSS, 2LVG, LOC. 0116 DISCONNECT G, 0116 AND WITH AUTO REFERENCE AT LSS ADJUST SC BIAS, 0116 FOR OUTPUT TEST POINT VOLTS PER ALIGNMENT DIAGRAM.

12. HSS, 2LVG, LOC. 0116. WITH AUTO REFERENCE AT HSS ADJUST SC GAIN, 0116 FOR OUTPUT TEST POINT VOLTS PER ALIGNMENT DIAGRAM. RECONNECT G.

13. LOCAL/REMOTE CONTROL LIGHT, IVC, LOC. 0117. ADJUST REFERENCE, 0117 FOR 6.5 VOLTS AT L TP. CHECK THAT LOCAL CONTROL INDICATION IS OBTAINED WHEN AUTO REFERENCE IS AT LSS AND MANUAL REFERENCE IS LOWERED SO THAT 2LVG OUTPUT IS LESS THAN APPROXIMATELY 1 VOLT.

14. AUTO SPEED REFERENCE INDICATION, IDI LOC. 0128. ADJUST ZERO AT LSS AND CAL AT HSS TO OBTAIN 0 AND 100 INDICATIONS ON OPERATORS POSITION INDICATOR.

X. OVERSPEED TEST INTEGRATOR, 10SI, LOC. 0112

1. RATE CONNECT VOLTMETER TO OUTPUT TEST POINT 2LVG, 0116. WITH MANUAL AND AUTO REFERENCE AT HSS ACTUATE OVERSPEED TRIP TEST PUSHBUTTON VOLTMETER READING SHOULD INCREASE 0.6 ± 0.1 VOLTS PER MINUTE FOR A (12% ± 2%) SPEED INCREASE PER MINUTE. ADJUST RATE, 0112 AS REQUIRED.

2. LIMIT CONNECT OSCILLATOR AND COUNTER TO T4 AND T5. SET FREQUENCY FOR 120% SSPU FREQUENCY. OBSERVE OUTPUT OF AMPLIFIER 5 WITH TEST METER. ACTUATE OVERSPEED TRIP TEST PUSHBUTTON AND ADJUST LIMIT, 0112 FOR POINT AT WHICH AMPLIFIER SWITCHES FROM POSITIVE SATURATION TO ZERO VOLTS. CHECK BY REDUCING OSCILLATOR FREQUENCY AND SLOWLY INCREASING IT TO THE SWITCHING POINT FREQUENCY.

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PRINTS TO

3. 120% SPEED INDICATION, IVC, LOC. 0210.:
WITH OSCILLATOR AT 120% SSPU FREQUENCY ASJUST REFERENCE, FOR DROPOUT. CHECK BY VARYING FREQUENCY.
4. HIGH SPEED REFERENCE LIGHT, IVC, LOC. 0121.:
ADJUST REFERENCE OF IVC FOR +5 VOLTS. CHECK FOR DROPOUT AT OUTPUT TEST POINT WHEN AMPLIFIER 5 SWITCHES UPON ACTUATING OTT.

XI. LP STOP VALVE POSITION INDICATOR IDI, LOC. 0102. WITH STOP VALVE FULLY CLOSED ADJUST DIFF. TRANS. NULL FOR -3 VOLTS AT OUT TP AND METER ZERO FOR 0 POSITION. ADJUST CAL. FOR 100% WITH VALVE FULL OPEN.

XII. HP STOP VALVE POSITION INDICATOR IDI, LOC. 0104. SAME AS XI.

XIII. LOW SPEED SWITCH POWER SUPPLY, LPS, LOC 0227.
1. CHECK FOR 5.6 VDC AT TERMINAL D.
2. CHECK FOR 12VDC AT TERMINAL E.

XIV. LOW SPEED SWITCH LSS LOC. 0225
1. STANDSTILL CHECKS - CHECK THAT TERMINALS F&G ARE +5.6VDC WITH RESPECT TO B. THIS WILL CONFIRM ITDX & 200RPM RELAYS ARE ENERGIZED.
2. RELAY CONTACTS AT TERMINAL D AND E SHOULD BE CLOSED.
3. ITP SHOULD BE 5.6 VDC.
4. REMOVE WIRE FROM F AND JUMPER F TO B. RELAY CONTACTS AT H & K CLOSE. REMOVE JUMPER AND CONNECT WIRE TO F.
5. OPERATE TURNING GEAR TO OPEN CONTACTS AT H & K.

OPERATIONAL CHECKS
1. CHECK FOR 0-5 VOLT PULSE AT TERMINAL N WHEN TURNING GEAR OPERATED.
2. POSITIVE PULSES SHOULD BE PRESENT AT 2TP WHEN TURNING GEAR OPERATED.
3. ITP SHOULD BE ZERO VOLT WHEN TURNING GEAR OPERATED.
4. RELAY CONTACTS AT D & E SHOULD BE OPEN WHEN TURNING GEAR OPERATED.
5. WITH TURNING GEAR OFF AND TURBINE AT STANDSTILL RELAY CONTACTS AT D & E SHOULD BE CLOSED.

XV. SEE LEVER DIAGRAM FOR RUNNING CHECKS OF SPEED REGULATION.

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PRINTS TO

GENERAL ELECTRIC

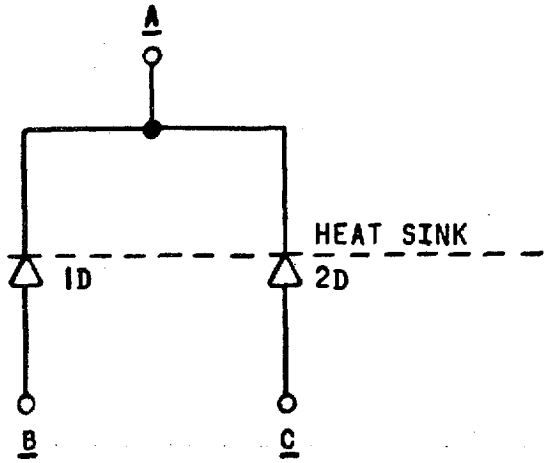
44A306392
CONT ON SHEET SH NO.

REV NO.
44A306392
CONT ON SHEET SH NO.

TITLE
RECTIFIER 1 REC

FIRST MADE FOR 44B317827-G01

REVISIONS



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12A-DFR2

PRINTS TO

MADE BY Eleanor Ellison 4/24/73
ISSUED D. Moyer April 26, 1973
APPROVALS R.G. Sloan RJ Culver
DRIVE SYSTEMS PROD. DEPT OR DEPT.
WAYNESBORO, VA. LOCATION
44A306392
CONT ON SHEET SH NO.

FF-803-WF (8-71)
PRINTED IN U.S.A.

CODE IDENT NO.
3

3S75$_{13}$TC3$_{00}$A$_{14}$ FHC BOILER FDPUMP

QTY	PART NUMBER	DESCRIPTION
2	3S75$_{13}$KF3$_{00}$A5	LINE POWER PANEL
1	3S75$_{13}$KF3$_{0}$6A9	METER PANEL
1	3S75$_{13}$KF3$_{0}$6A$_{11}$	METER PANEL ASM
1	3S75$_{13}$KF3$_{0}$7A$_{15}$	RELAY PANEL ASM
6	44B33$_{1}$76$_{1}$-G$_{01}$	PWB VOLT COMPARATR
1	44D236389-G$_{02}$	PWB SERVO AMPLI
5	44D38$_{1}$585-G$_{01}$	PWB DETECTOR IND
9	44D38$_{1}$589-G$_{01}$	PWB OPER AMPLIFIER
1	44B33$_{1}$7$_{1}$7-G$_{01}$	PWB 3KHZ OSC
1	44D236379-G$_{01}$	PWB ANALOG OUTPUT
2	44D236379-G$_{03}$	PWB ANALOG OUTPUT
2	44D33$_{1}$9$_{1}$8-G$_{01}$	PWB CL MOTOR DRIVE
3	44D236365-G$_{02}$	PWB 2FREQ VOLT CNV
1	44B33$_{1}$743-G$_{01}$	PWB SIG CONV 3SC
1	44B33$_{1}$746-G$_{01}$	PWB BOILR CON INTF
3	44D23637$_{0}$-G$_{01}$	PWB DIODE BOARD
6	44B3$_{1}$78$_{2}$7-G$_{01}$	PWB SERIES REG 1
6	44D23637$_{1}$-G$_{01}$	PWB SERIES REG 2
1	44D236373-G$_{02}$	PWB OVERSPD INTEG
1	44D23637$_{2}$-G$_{02}$	PWB SIG LO VAL
1	44D236388-G$_{01}$	PWB COMP+REG ICR
2	44D3$_{15}$$_{1}$78-G$_{01}$	PWB DIFF TRANS DR
1	44D236378-G$_{01}$	PWB REF LOW VALUE
1	44B337328-G$_{01}$	PWB LPF PWR SUPPLY
1	44B337327-G$_{01}$	PWB LOW SPD SW LSS
3	44C3$_{19}$112-G$_{01}$	RACK ASM PWB
2	K97747$_{19}$L5$_{0}$	CAP 5$_{00}$UF 5$_{0}$VDC

3S75$_{13}$KF3$_{00}$A5 LINE POWER PANEL

QTY	PART NUMBER	DESCRIPTION
2	44B334726-G$_{01}$	SUPPLY TRANSF
18	44A3$_{16}$724-M$_{19}$	CAP 1$_{900}$UF 75VDC
6	44A333453-D36	RES 36OHM 100W
6	44B2145$_{18}$-G$_{01}$	REACTOR

3S75$_{13}$KF3$_{0}$6A9 METER PANEL

QTY	PART NUMBER	DESCRIPTION
5	44A3$_{1}$7759-$_{01}$8	SWITCH
20	GE387	LAMP

3S75$_{13}$KF3$_{0}$6A$_{11}$ METER PANEL ASM

QTY	PART NUMBER	DESCRIPTION
1	128A4$_{0}$57H2$_{0}$	RES 200K OHMS ,5 W
2	44A3$_{1}$7759-$_{01}$8	SWITCH
2	44A3$_{1}$7759-$_{01}$9	SWITCH
1	44A3$_{1}$7759-$_{01}$2	SWITCH
3	44A3$_{1}$7759-$_{02}$0	SWITCH
3	128A4$_{0}$57G3$_{01}$	RES 30.1K OHMS ,5W
20	GE387	LAMP

REVISIONS

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3S7513KF307A15 RELAY PANEL ASM

QTY	PART NUMBER	DESCRIPTION
33	3S2791G138D4	MINIATURE RELAY
2	3SAA1343A2	RELAY
1	12HMA11B3	RELAY
1	44B312431-005	TIME DELAY RELAY
2	44A230536-001	SIGNAL FLASHER
47	44B232019-004	RECT 1A 400V
2	44A317994-G01	COMP BD ASM
4	3S2791G138D3	MINIATURE RELAY
2	K9774723K40	CAP 40UF 450VDC
2	44B314000-G01	D C REACTOR
2	K590830G1	T B ASM
39	9338545P2	RELAY SOCKET
14	44B232011-001	RECT 200MA 200VDC

3S7513KL300A12 CONTROL PANEL ASM

QTY	PART NUMBER	DESCRIPTION
1	44A317759-007	PUSHBUTTON SWITCH
2	44A317759-006	PUSHBUTTON SWITCH
2	44A317759-005	PUSHBUTTON SWITCH
1	44A317759-001	PUSHBUTTON SWITCH
2	44A317759-003	PUSHBUTTON SWITCH
1	44A317759-004	PUSHBUTTON SWITCH
1	44A317759-008	PUSHBUTTON SWITCH
1	44A317759-052	ILLUM PB SWITCH
1	44B337344-G01	PWB RES BD DIM BRT
26	GE387	LAMP
1	44A317759-026	PB SWITCH
2	CR2940U200AC	RING
1	CR2940U200AD	RING
1	CR2940U200AF	RING
4	CR2940UA200A	OPERATOR
1	CR2940U207	CONTACT BLOCK
3	CR2940U206	BLOCK
1	6105700G43G	IND.LIGHT

REVISIONS	PRINTS
	11
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GENERAL ELECTRIC

44B300423 CONT ON SHEET SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

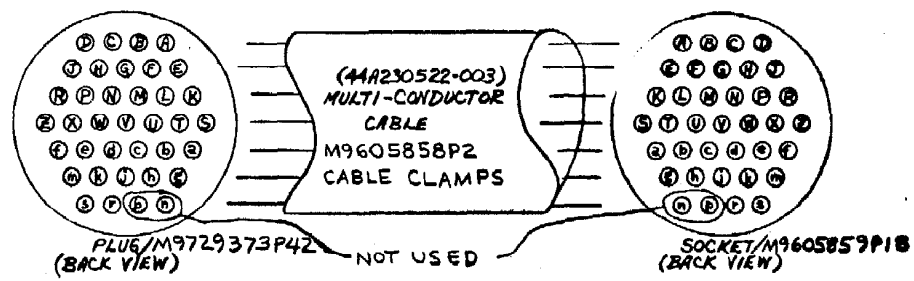
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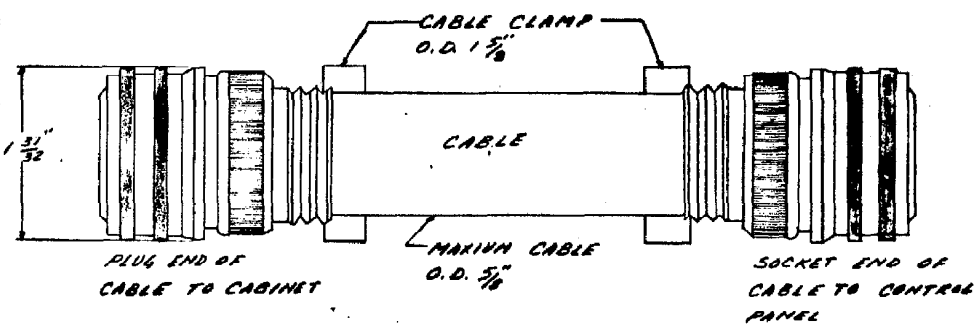
TITLE: CABLE WIRING

FIRST MADE FOR 3S7513TC300A2

SIGNAL & LOW VOLTAGE

	PLUG TERMINAL	WIRE	SOCKET TERMINAL
SHIELDED CABLE	V	SHIELD	V
	M	WHITE/BLACK/VIOLET	M
	N	BLACK	N
	W	BROWN	W
SHIELDED CABLE	d	WHITE/BLACK/GRAY	d
	c	SHIELD	c
	U	RED	U
SHIELDED CABLE	T	WHITE/BROWN/RED	T
	L	SHIELD	L
	F	ORANGE	F
SHIELDED CABLE	G	WHITE/BROWN/ORANGE	G
	H	SHIELD	H
	P	WHITE	P
SHIELDED CABLE	X	WHITE/BLACK	X
	e	SHIELD	e
	k	WHITE/BROWN/YELLOW	k
SHIELDED CABLE	j	WHITE/BROWN/GREEN	j
	h	SHIELD	h
	b	WHITE/BROWN	b
SHIELDED CABLE	g	WHITE/RED	g
	a	SHIELD	a
	S	WHITE/ORANGE	S
	K	WHITE/YELLOW	K
	E	WHITE/GREEN	E
	A	WHITE/BLUE	A
	B	WHITE/VIOLET	B
	C	WHITE/GRAY	C
	D	WHITE/BLACK/BROWN	D
	J	WHITE/BLACK/RED	J
	R	WHITE/BLACK/ORANGE	R
	Z	WHITE/BLACK/YELLOW	Z
	f	WHITE/BLACK/GREEN	f
	m	WHITE/BLACK/BLUE	m
	s	WHITE/BROWN/BLUE	s
	r	WHITE/BROWN/VIOLET	r





NOTES:

1. USE PLASTIC SLEEVING OVER WIRE AND TERMINALS TO INSULATE CONNECTIONS FROM EACH OTHER.

REVISIONS

1.	Work Order Aug 19 1969	
2	[illegible]	
3	ADDED NOTE: 3-12-13 J.K. MOYER 1973	FG-N

PRINTS TO: 11

MADE BY R.M. Bukosky ISSUED MAY 22 1968

APPROVALS R.M.B.

SPECIALTY CONTROL DIV OR DEPT. WAYNESBORO, VA. LOCATION

44B300423 CONT ON SHEET SH NO.

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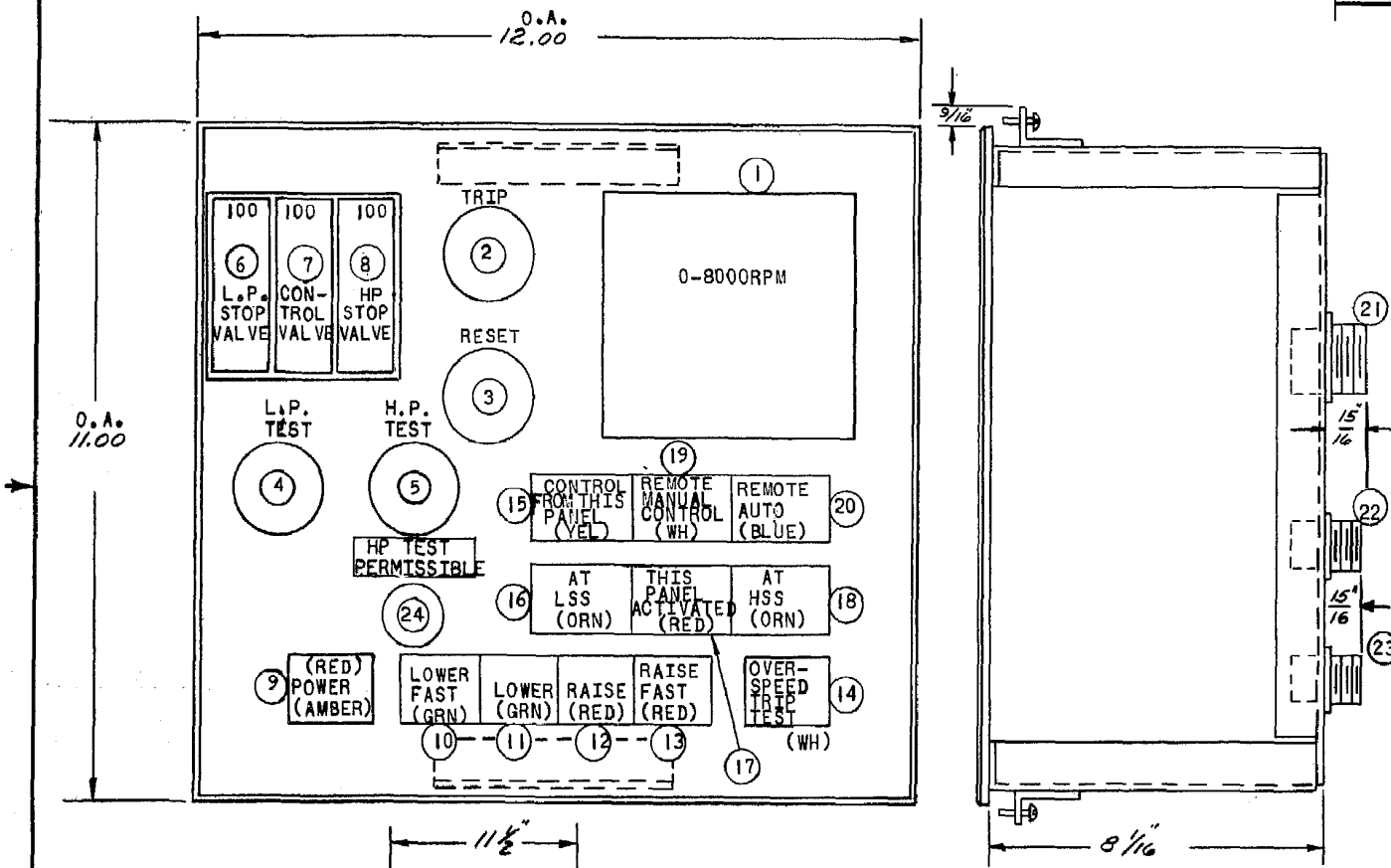
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GENERAL ELECTRIC

44B307044 CONT ON SHEET SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—					REV NO.
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS			
		FRACTIONS	DECIMALS	ANGLES	
	✓	±	±	±	CONT ON SHEET SH NO.

TITLE
OUTLINE
FEEDPUMP TURBINE CONTROL PANEL
FIRST MADE FOR 3S7513KL300A12 122816
GULF STATES SABINE #5 091-41289



PARTS LIST

A. 1 INSTRUMENTS, G.E. TYPE DB40.
B. 4,5 G.E. CR2940UA202C PUSH BUTTON WITH GUARD, WITH CR2940-U207 BLOCK.
C. 6,7,8 INSTRUMENTS, G.E. TYPE 185.
D. 9 ILLUMINATED PUSH BUTTON RED, POWER ON, CONDITION NORMAL; YELLOW, POWER ON, CONDITION ABNORMAL.
E. 10-13, & 20 PUSH BUTTONS W/DIM ILLUMINATION
F. 16-19, INDICATING LIGHTS.
G. 9-20, MASTER SPECIALTIES SERIES 12, VERTICAL BARRIER.
H. 21,22,23 CONNECTORS
J. 2,3 G.E. CR2940UA203C PUSH BUTTON WITH GUARD.
K. 14,15 ILLUMINATED PUSH BUTTON.
L. 9 & 14-20 HAVE DIM-BRIGHT FEATURE.
M. 2 HAS 5/8" DEEP RED COLOR RING GUARD.
3 HAS 5/8" DEEP GREEN COLOR RING GUARD.
4,5 HAS 5/8" DEEP BLACK COLOR RING GUARD .
N. APPROX. WIEGHT 50 POUNDS.
P. 24, ET6 GREEN LAMP, TEST PERMISSIBLE

REVISIONS	PRINTS TO
1 Lmm Sept 9, 1974	11
	12P-KFM
	FG-W

MADE BY	APPROVALS	DIV OR DEPT. LOCATION	
Elmer Ellison 4/17/74 ISSUED F. Marie Smith 4/17/1974	R.G. Sloan	DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.	44B307044 CONT ON SHEET SH NO.

FOLD

FF-806-F (12-67) PRINTED IN U.S.A.

3

GENERAL ELECTRIC

44B306144

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING.—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

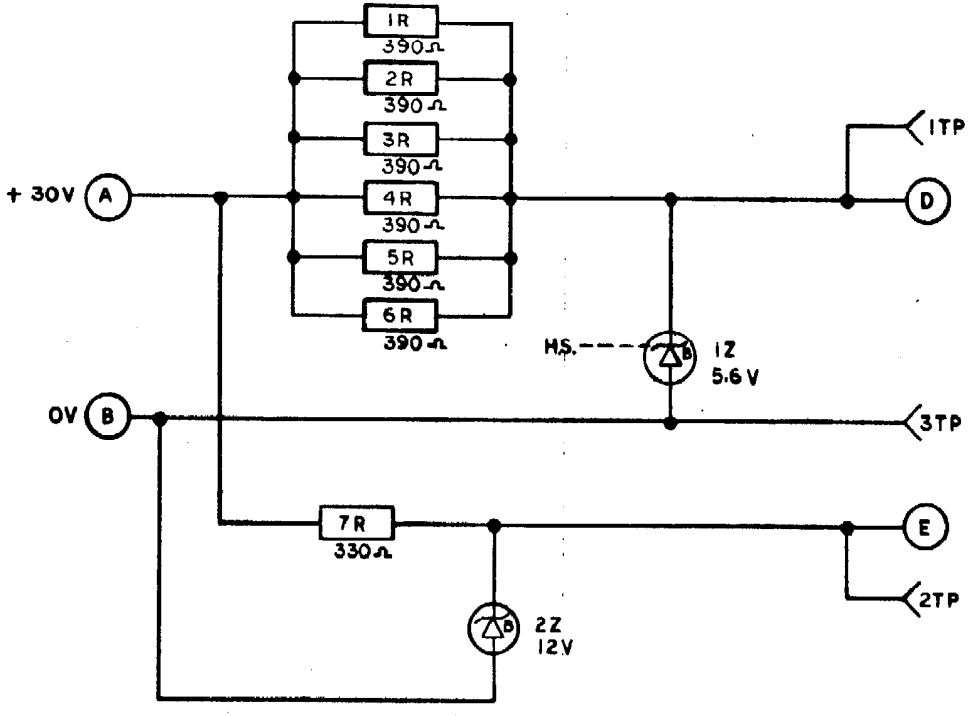
REV NO. | CONT ON SHEET | SH NO.

TITLE

LSS PWR SUPPLY

FIRST MADE FOR

ELEM DIAGRAM



REVISIONS

PRINTS TO

//

12P-3PB

FG-W

MADE BY Guy Rouzie 9-11-72

ISSUED D.K. Meyer Sept. 12, 1972

APPROVALS R Cofer 9/12/72

DRIVE SYSTEMS PROD. DEPT. DEPT

WAYNESBORO, VA. LOCATION

44B306144 CONT ON SHEET SH NO.

3

FP-909-F (10-71) PRINTED IN U.S.A.

GENERAL ELECTRIC

44B306148

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV NO

TITLE
POWER CABLE WIRING, ELECTRONIC BOILER FEED PUMP TRUBINE CONTROL

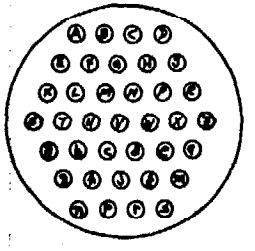
FIRST MADE FOR 3S7513TC300A10

2CO, 3CO POWER CABLE

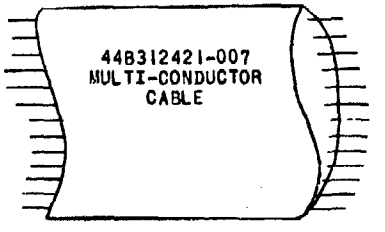
ASSEMBLY NOTES

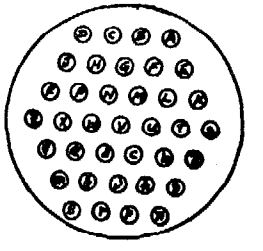
THE SPECIFIC COLOR OF WIRE CONNECTED TO A PIN IS NOT IMPORTANT, BUT THE SAME WIRE SHOULD BE CONNECTED TO THE SAME PIN AT THE OTHER CABLE END AND CONNECTOR.
EXAMPLE: IF A RED WIRE IS CONNECTED TO PIN "A" ON THE CONNECTOR PLUG, THE OTHER END OF THAT RED WIRE MUST BE CONNECTED TO PIN "A" OF THE SOCKET CONNECTOR AT THE OTHER END.

CONNECT SAME COLOR WIRE TO THESE PINS AT BOTH CABLE ENDS		DO NOT CONNECT ANY WIRES TO THESE PINS	
A		B	4
C		D	
T		G	
E	4	L	
F		M	
		N	
H	X	P	
J	Z	R	
K		S	
		U	
		V	
		W	

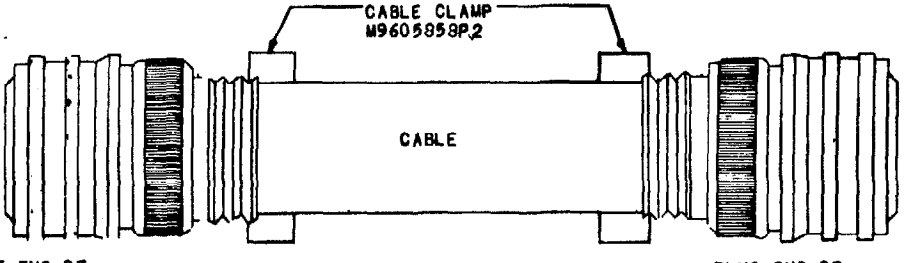


SOCKET
M9605859P18





PLUG
M9605859P42



SOCKET END OF
CABLE TO EHC REMOTE CONTROL PANEL.

PLUG END OF
CABLE TO EHC CABINET

REVISIONS	PRINTS TO
	11
	RPVE
	FG-W

MADE BY: [illegible] 10-11-72
BY: S. Morgan Oct. 20, 1972

APPROVED: R Culver

COMM. & CONT. DEVICES DIV OR DEPT
WAYNESBORO, VA. LOCATION

44B306148 CONT ON SHEET SH NO

FF 506 P 112-601
PRINTED IN U.S.A.

DRAWING NO. 44B306197 SH NO. CONT ON SHEET

GENERAL ELECTRIC

44B306197 CONT ON SHEET SH NO.

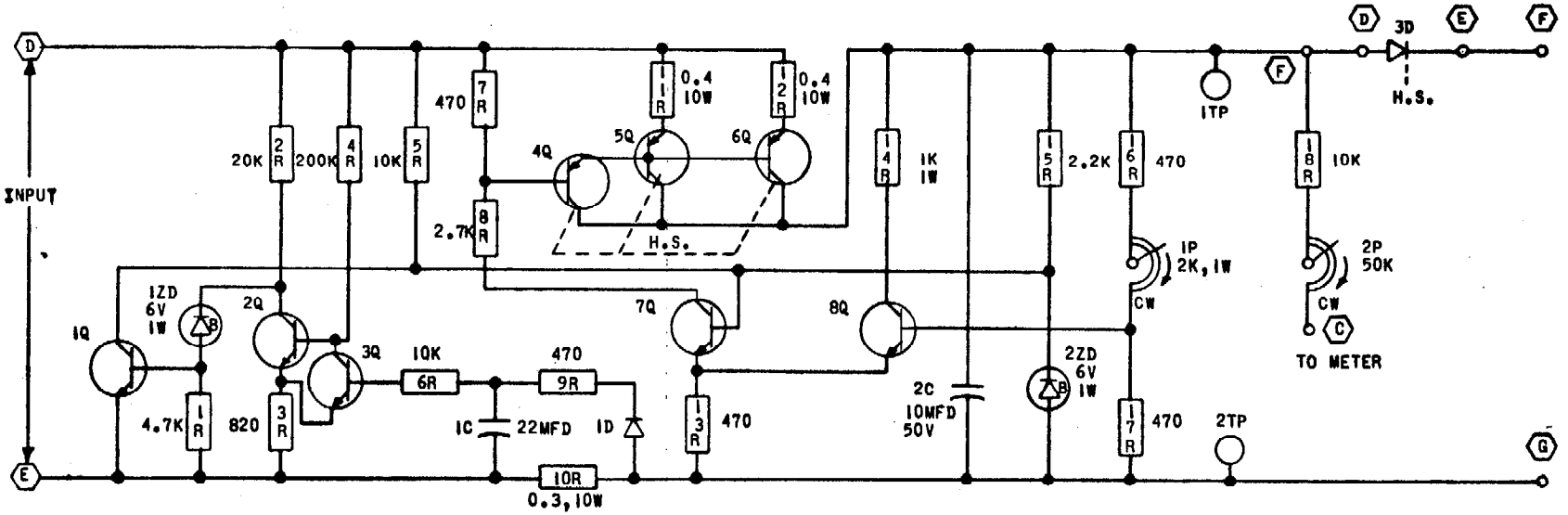
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV NO. 44B306197 CONT ON SHEET SH NO.

TITLE: SERIES REGULATOR ISR

FIRST MADE FOR 44B236371-G01



REVISIONS

PRINTS TO

11
FG-W
12P-DATA

MADE BY Clinton Ellison 4/24/73

ISSUER B. Moyer April 26, 1973

APPROVALS R C Fillyaw, R J Gilmer

DRIVE SYSTEMS PROD. DEPT. DIV. OR DEPT. WAYNESBORO, VA. LOCATION

44B306197 CONT ON SHEET SH NO.

FF-806-F (8-72) PRINTED IN U.S.A.

3

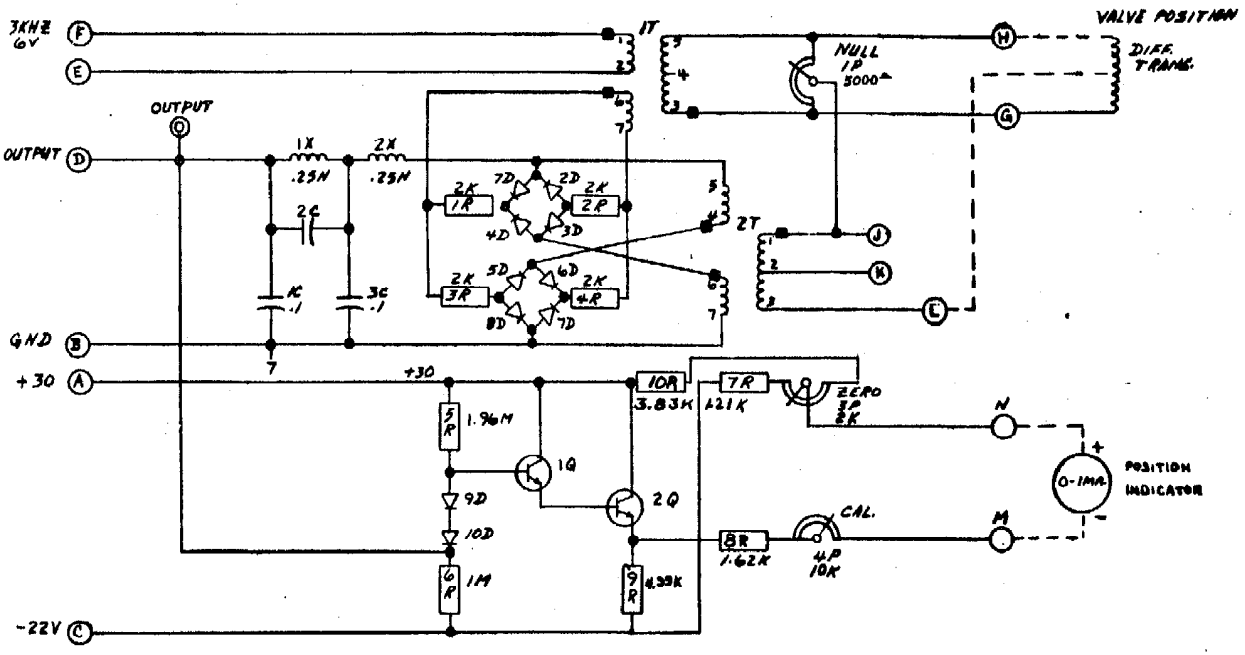
GENERAL ELECTRIC

44C300609

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: DETECTOR INDICATOR 1DI

FIRST MADE FOR 44D381585-G01



NOTES:
1. CONNECTIONS

A	+30V	G, H, J, K, L	DIFFERENTIAL TRANSFORMER CONNECTIONS
B	GROUND		
C	-22V		
D	OUTPUT		
E, F	6V 3 KHZ		
M, N			OUTPUT METER 0-1MA

44C300609 REV NO. 0

REVISIONS			PRINTS TO
	1	R.M.G. MAY 13 1969	N
	2	C.M.G. MAY 15 1969	FG-N
	3	R. Martin OCT 19 1969	12P-DEB
	4	J.P. DEC 18 1969	
8	5	D.K.L. MAY 18 1970	
7	6	C.A.C. OCT 19 1970	

C. M. Templeton JAN 26 1968 ISSUED
APR 17 1968
APPROVED
SPECIALTY CONTROL DIV. OR DEPT. WAYNESBORO, VA. LOCATION
44C300609 CONT ON SHEET

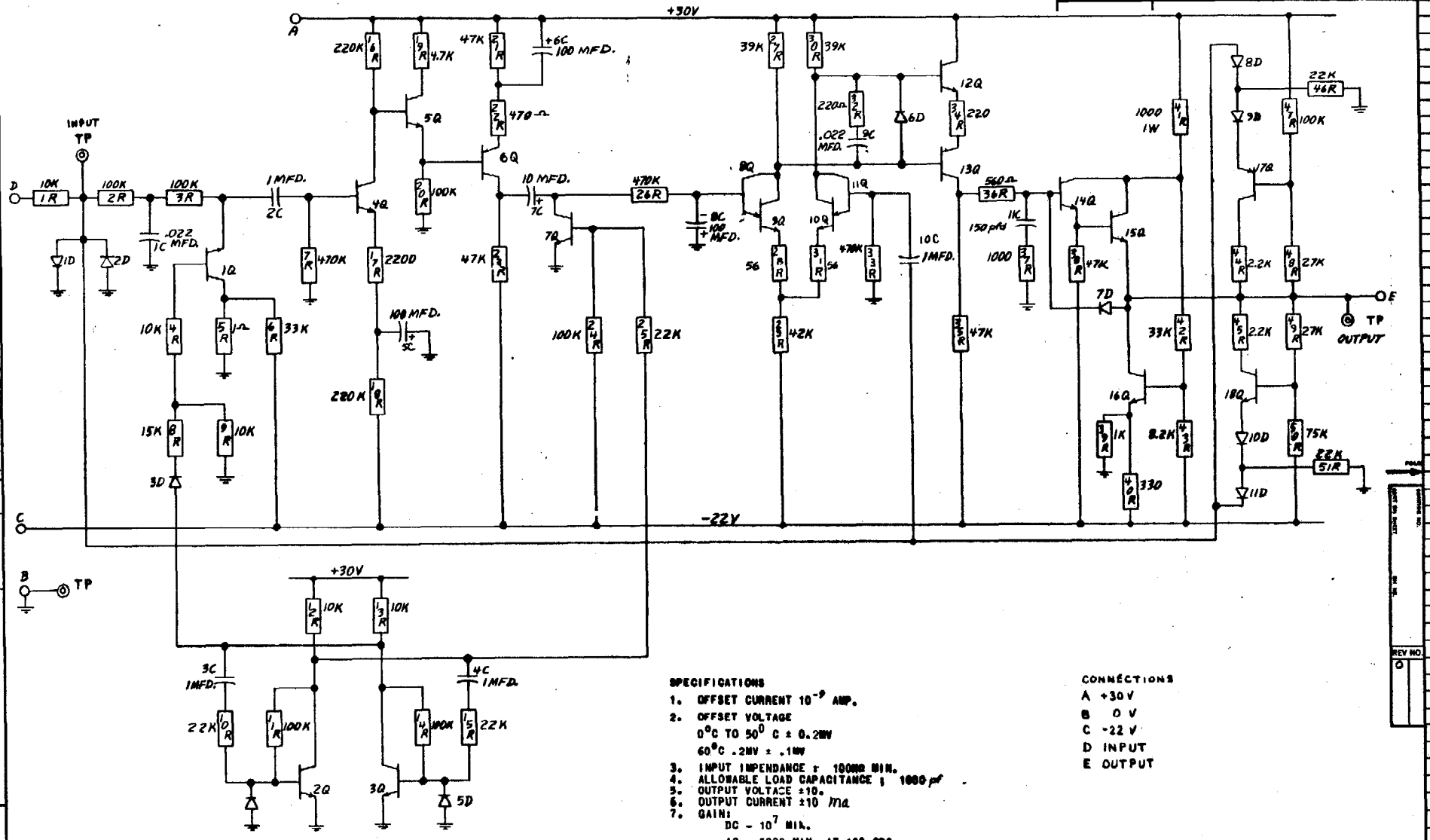
GENERAL ELECTRIC

44C300610

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
CHOPPER STABILIZED
OPERATIONAL AMPLIFIER
FIRST MADE FOR 44D381589-G01

10A



SPECIFICATIONS

1. OFFSET CURRENT 10^{-9} AMP.
2. OFFSET VOLTAGE
   0°C TO 50° C ± 0.2MV
   60°C .2MV ± .1MV
3. INPUT IMPENDANCE ; 100MΩ MIN.
4. ALLOWABLE LOAD CAPACITANCE ; 1000 pf
5. OUTPUT VOLTAGE ±10.
6. OUTPUT CURRENT ±10 ma
7. GAIN:
   DC - 10^7 MIN.
   AC - 5000 MIN. AT 100 CPS
   CHARACTERISTIC: BREAK FREQUENCY - 250 CPS
   20 db PER DECADE TO CROSSOVER.
8. CHOPPER STABILIZER RECOVERY TIME: 1 MINUTE
9. POWER REQUIREMENTS: 30 VOLTS - 25 ma
   -22 VOLTS - 20 ma
10. AMPLIFIER PROTECTION:
    A. OUTPUT CURRENT LIMITED TO APPROX 10 ma
    B. OUTPUT LIMITED TO 10 VOLTS
    C. ALLOWABLE INPUT VOLTAGE BEFORE DAMAGE - 70 VOLTS.

CONNECTIONS

A +30 V
B 0 V
C -22 V
D INPUT
E OUTPUT

REVISIONS		PRINTS TO
1.	Nov 17, 1958	11
2.	Aug 7, 1959	12P-DFD

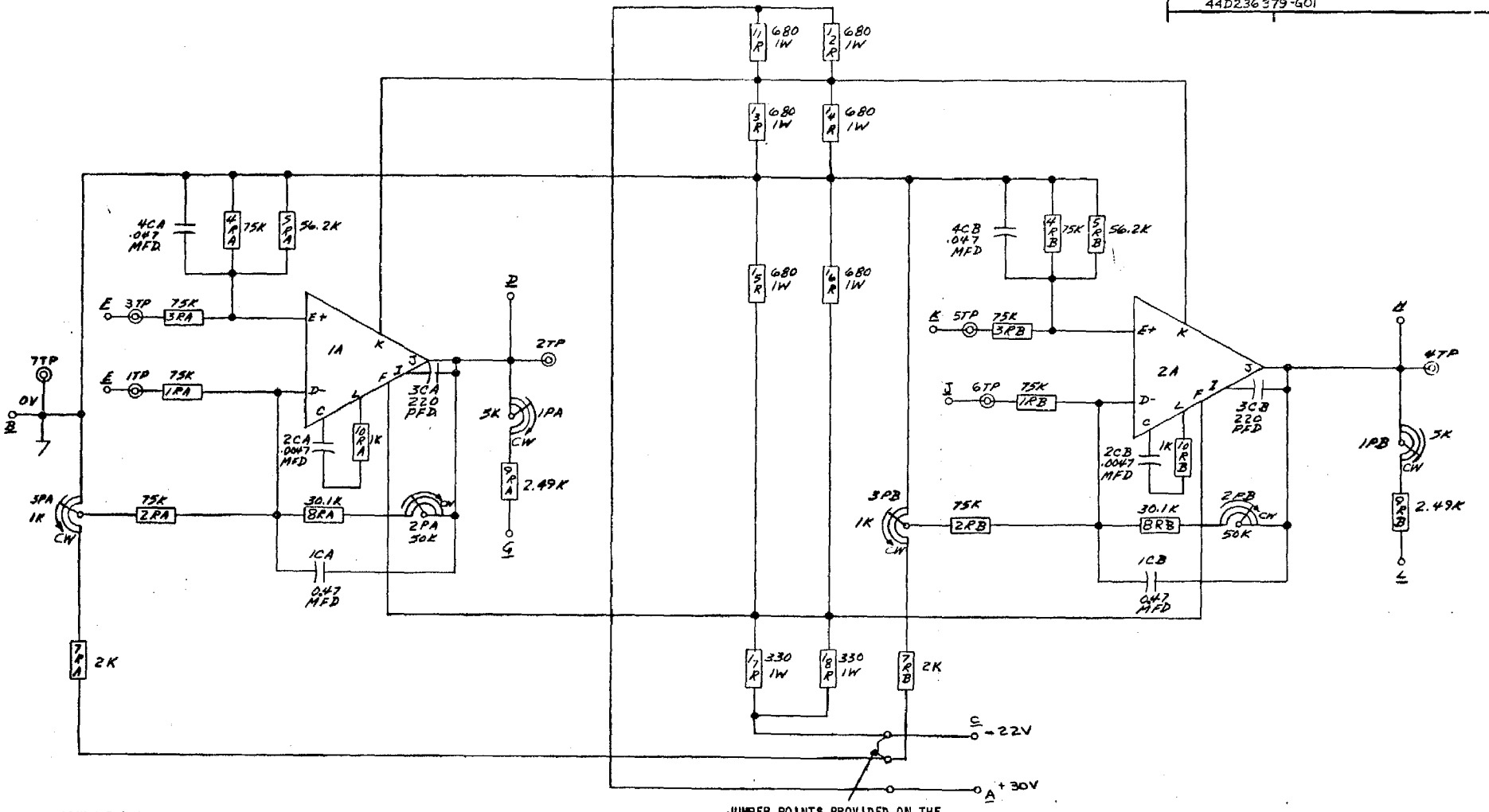
MADE BY J. C. Glenn JAN 22 1958
ISSUED MAR 12 1958
APPROVED MBC
SPECIALTY CONTROL DIV OR DEPT
WAYNESBORO, VA LOCATION
44C300610

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ANALOG OUTPUT 1AØ2

FIRST MADE FOR
44D236379-G01



CONNECTIONS

A = +30V
B = 0V
C = -22V
D = 1A VOLTAGE OUTPUT
E = 1A INPUT
G = 1A CURRENT OUTPUT
H = 2A VOLTAGE OUTPUT
J = 2A INPUT
L = 2A CURRENT OUTPUT

REVISIONS			PRINTS TO
1	RTD	SEP 30 1968	11
2	[illegible]	OCT 9 1969	2F DA B
3	[illegible]	FEB 3 1970	FIG-W
4	[illegible]	NOV 17 1971	1VN

R. L. Templeton SEP 16 1968 ISSUED SEP 24 1968 APPROVALS 18e
SPECIALTY CONTROL DIV. OR DEPT. WAYNESBORO, VA LOCATION
44C301216 CONT ON SHEET

DRAWING NO 44C301216 CONT ON SHEET SH NO REV NO.

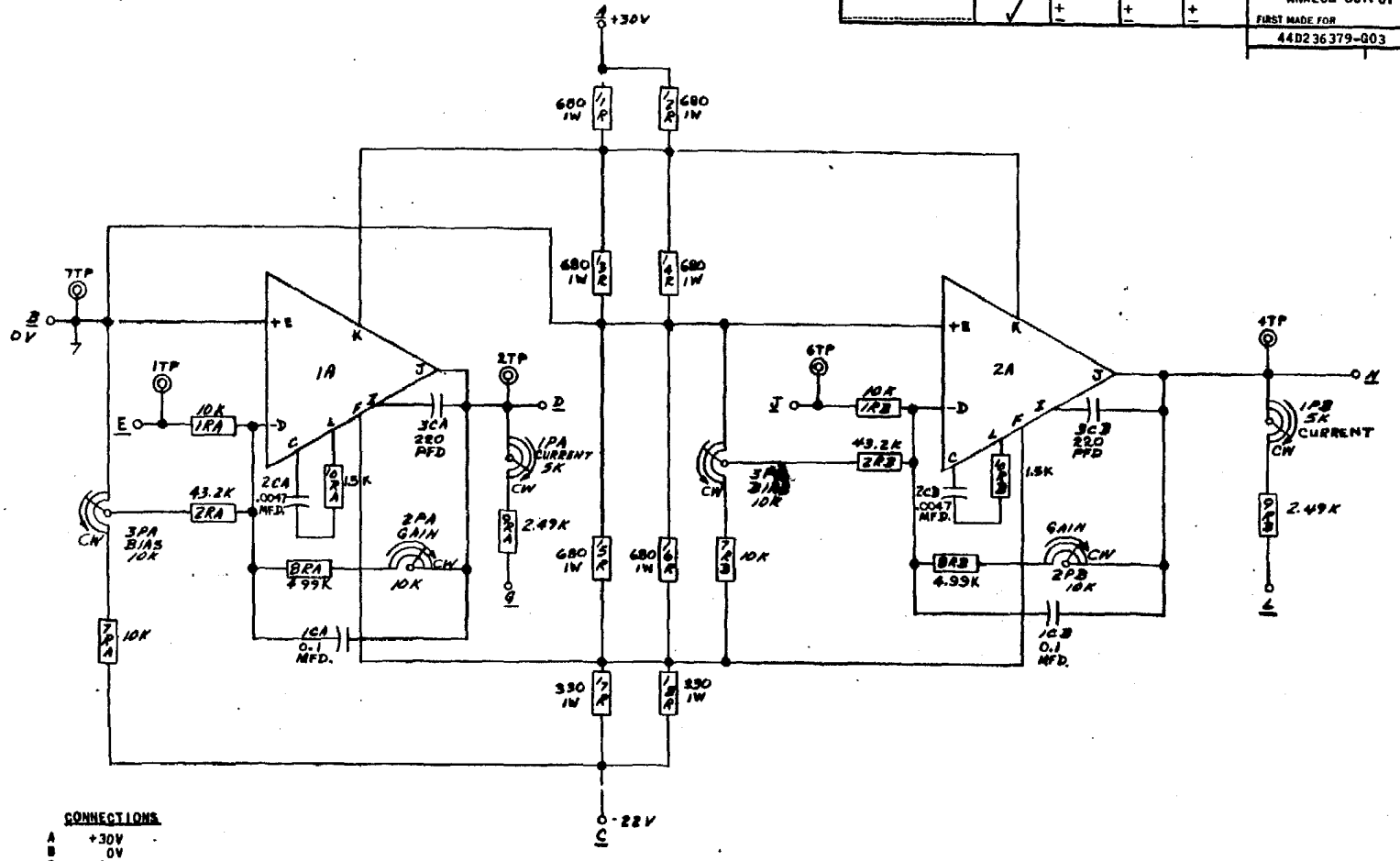
GENERAL ELECTRIC

44C301622

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ANALOG OUTPUT 3A#2

FIRST MADE FOR
44D236379-G03



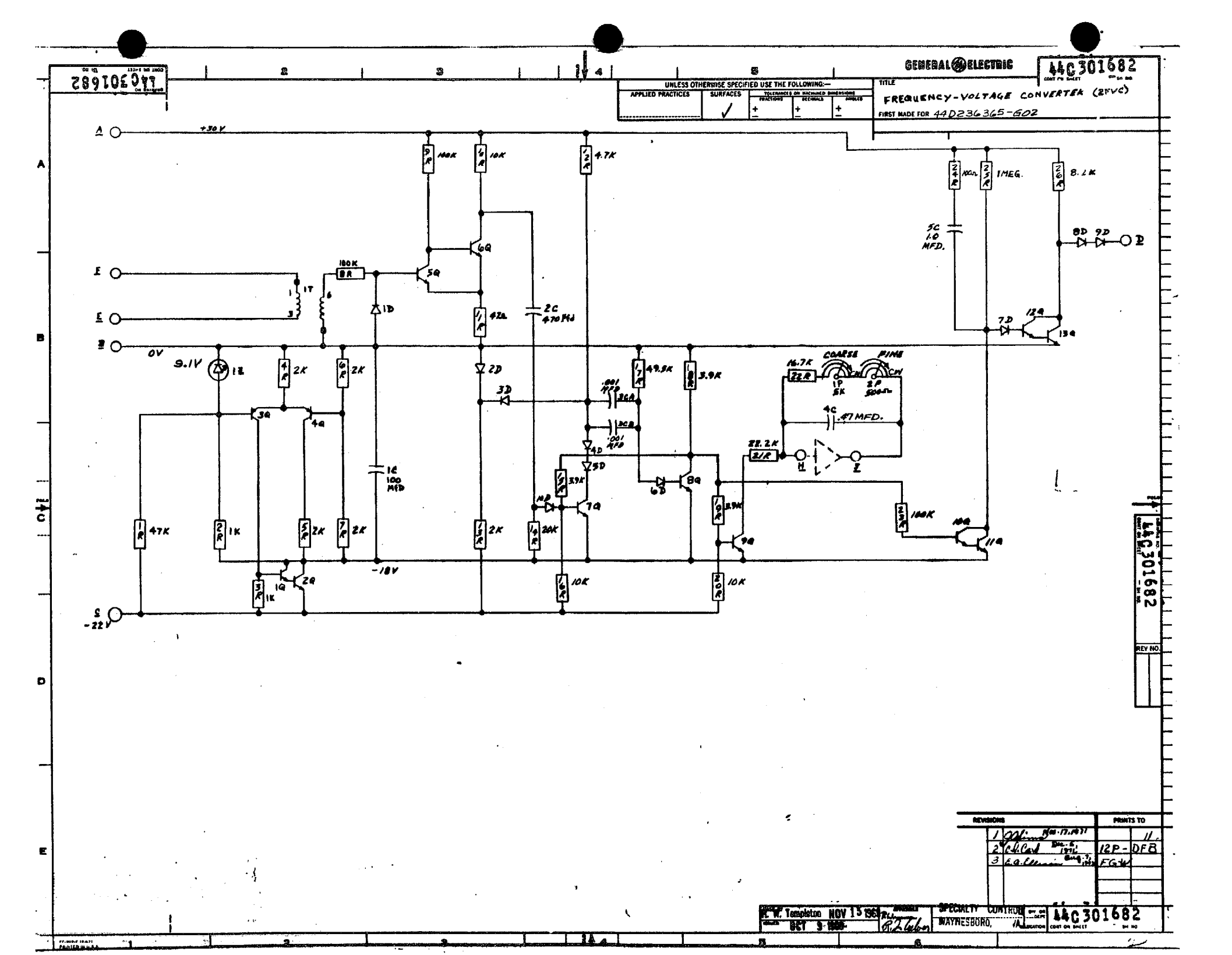
CONNECTIONS

A	+30V
B	0V
C	-22V
D	1A VOLTAGE OUTPUT
E	1A INPUT
G	1A CURRENT OUTPUT
H	2A VOLTAGE OUTPUT
J	2A INPUT
L	2A CURRENT OUTPUT

44C301622
REV NO.

REVISIONS		PRINTS TO-
1	A.M. [illegible] 1968	11
2	[illegible] Nov. 17, 1971	
3	C.A. Lewis Aug. 7, 1974	

Completion SEP 17 1968
SEP 19 1968
APPROVED MBC
SPECIALTY CONTROL DEPT. WAYNESBORO
44C301622


GENERAL ELECTRIC
44C301682
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
FREQUENCY-VOLTAGE CONVERTER (2FVC)
FIRST MADE FOR 44D236365-G02
+30V
0V
-18V
-22V
9.1V
180K
10K
4.7K
100K
1MEG.
8.2K
5C 1.0 MFD.
49.9K
3.9K
16.7K
22.2K
COARSE
FINE
4C .47 MFD.
47K
1K
2K
39K
20K
180K
1C 100 MFD
2C 470 MFd
.001 MFD
1Q
2Q
3Q
4Q
5Q
6Q
7Q
8Q
9Q
10Q
11Q
12Q
13Q
1D
2D
3D
4D
5D
6D
7D
8D
9D
REVISIONS
PRINTS TO
12P - DFB
FG-W
R. W. Templeton NOV 15 1968
OCT 9 1968
R. L. Tabor
SPECIALTY CONTROL
WAYNESBORO, VA
44C301682

GENERAL ELECTRIC

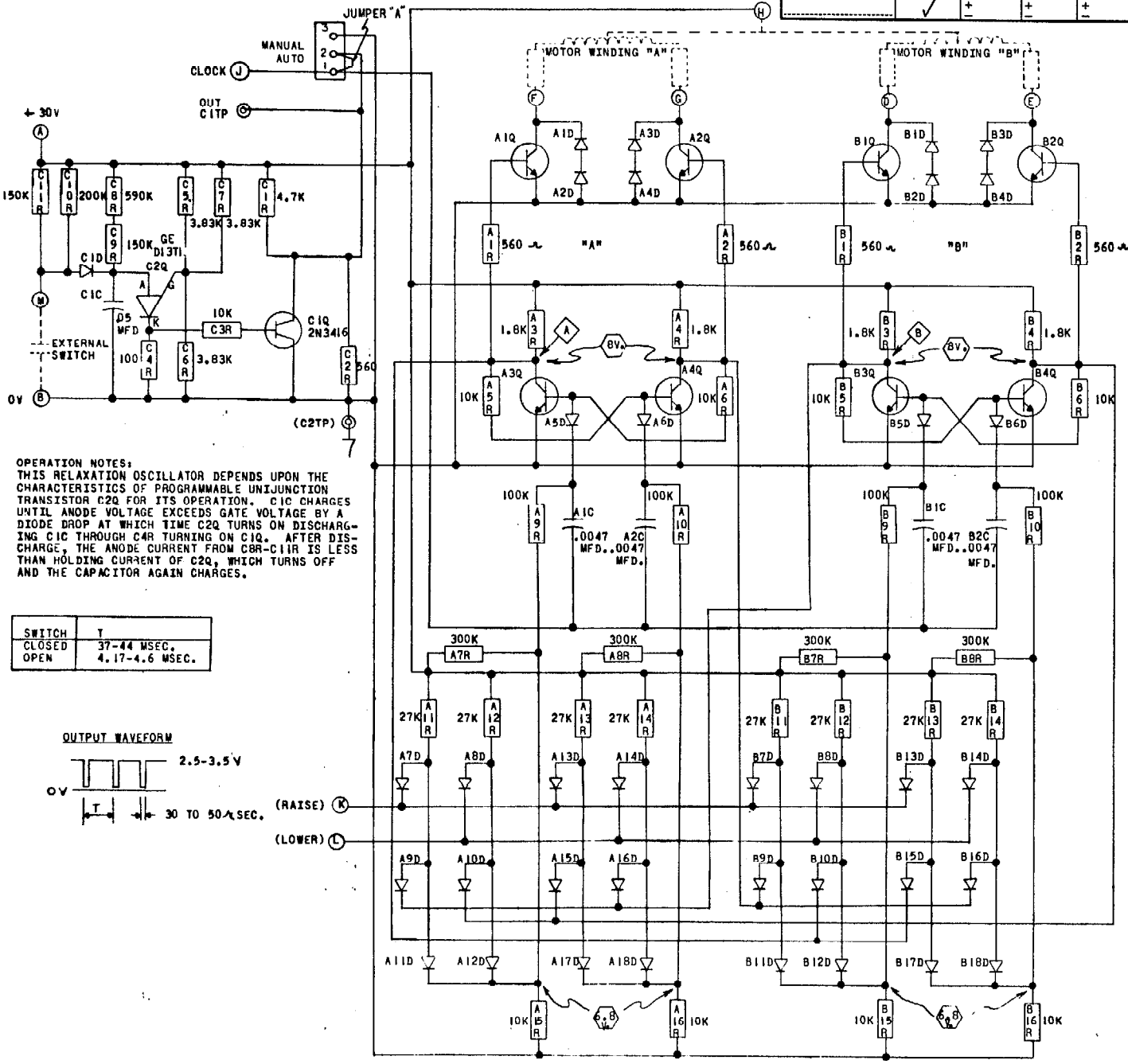
44C303567

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS, DECIMALS, ANGLES

TITLE

CLOCK & STEPPING MOTOR DRIVE

FIRST MADE FOR 44D331918-G01

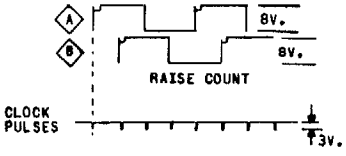


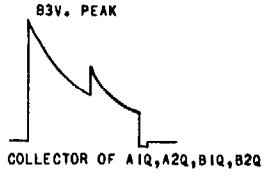
1. THIS BOARD CONSISTS OF A RELAXATION OSCILLATOR CLOCK IDENTIFIED BY "C" COMPONENTS AND ALSO TWO FLIP-FLOPS "A" AND "B" WITH DIODE LOGIC TO FORM A REVERSIBLE COUNTER. OUTPUT OF THE FLIP-FLOPS FEEDS POWER AMPLIFIERS A1Q, A2Q, B1Q & B2Q. THE STEPPING MOTOR WINDINGS ARE CONNECTED DIRECTLY TO THE COLLECTORS OF THESE FOUR TRANSISTORS.
2. THE CLOCK PULSES ARE THREE VOLTS AMPLITUDE WITH FAST FALL TIME. NORMAL CLOCK FREQUENCY IS 240 PULSES PER SECOND.
3. RAISE COUNTING RESULTS FROM 0 TO 1 VOLT ON TERMINAL (K) & +6 VOLTS OR MORE ON TERMINAL (L). REVERSING THE VOLTAGE ON TERMINALS (K) & (L) RESULTS IN LOWER COUNTING. A POSITIVE VOLTAGE ON TERMINAL (K) & (L) WILL INHIBIT COUNTING. 0 TO 1 VOLT ON BOTH (K) & (L) IS NOT PERMISSIBLE.
4. CONNECT JUMPER "A" TO 1 & 2 FOR MANUAL SPEED REFERENCE; CONNECT JUMPER "A" TO 2 & 3 FOR AUTO SPEED REFERENCE.

OPERATION NOTES:
THIS RELAXATION OSCILLATOR DEPENDS UPON THE CHARACTERISTICS OF PROGRAMMABLE UNIJUNCTION TRANSISTOR C2Q FOR ITS OPERATION. C1C CHARGES UNTIL ANODE VOLTAGE EXCEEDS GATE VOLTAGE BY A DIODE DROP AT WHICH TIME C2Q TURNS ON DISCHARGING C1C THROUGH C4R TURNING ON C1Q. AFTER DISCHARGE, THE ANODE CURRENT FROM C8R-C1IR IS LESS THAN HOLDING CURRENT OF C2Q, WHICH TURNS OFF AND THE CAPACITOR AGAIN CHARGES.

SWITCH	T
CLOSED	37-44 MSEC.
OPEN	4.17-4.6 MSEC.

OUTPUT WAVEFORM





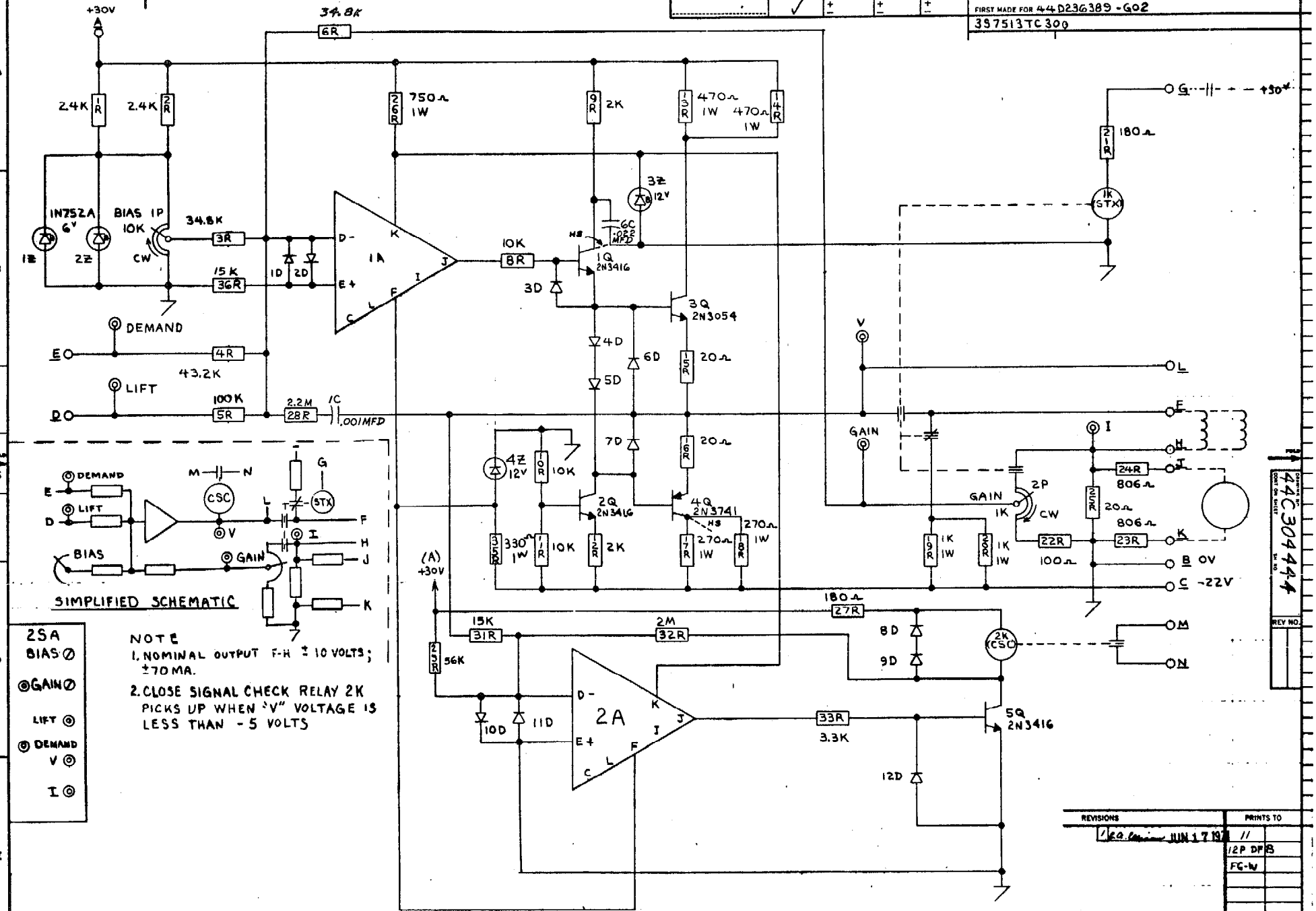
REVISIONS			PRINTS TO
1	S. MAR. 9 1970		11
2	[illegible] July 24, 1971		12P-DFB²
3	[illegible] Nov. 17, 1971		F3-W

MADE BY [illegible] ISSUED MAR 3 1970 | APPROVALS MBC | COMM. & CONT. DEVICES DIV. OR DEPT. WAYNESBORO, VA. LOCATION | 44C303567

GENERAL ELECTRIC 44C304444

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
SERVO AMPLIFIER (2SA)
FIRST MADE FOR 44D236389-G02
397513TC300



2SA
BIAS ⊘
GAIN ⊘
LIFT ◎
DEMAND ◎
V ◎
I ◎

NOTE
1. NOMINAL OUTPUT F-H ± 10 VOLTS; ±70 MA.
2. CLOSE SIGNAL CHECK RELAY 2K PICKS UP WHEN "V" VOLTAGE IS LESS THAN -5 VOLTS

REVISIONS
1 EO [illegible] JUN 17 1971

PRINTS TO
11
12P DP B
FC-W

MADE BY [illegible] 2/1/71 ISSUED MAR 26 1971
APPROVALS [illegible]
COMM. & CONT. DEVICES WAYNESBORO, VA.
44C304444
CONT ON SHEET
2

GENERAL ELECTRIC

44C304462

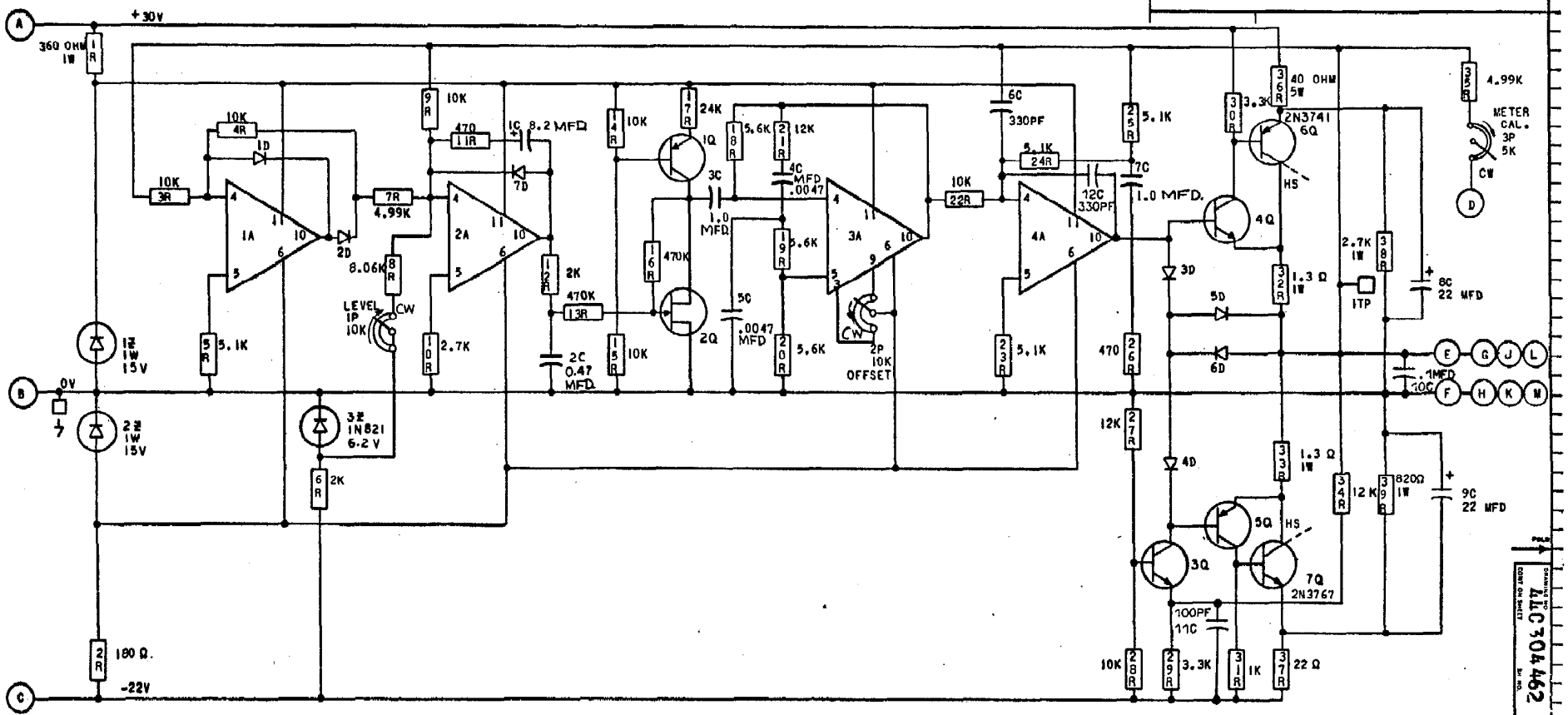
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

3KHZ OSCILLATOR

FIRST MADE FOR EHC 6-OSC 44B331717-G01



FREQUENCY: 3000HZ± 300HZ
AMPLITUDE: 6.00 VRMS (17V PEAK TO PEAK)
DC OFFSET: ADJUSTABLE TO ZERO
MINIMUM LOAD IMPEDANCE: 25Ω
DRIFT: 0.1% FIFTEEN HOURS
AMBIENT TEMPERATURE

		V	%
DISTORTION	2ND HARMONIC	.056	0.93
	3RD HARMONIC	.0083	0.14
	4TH HARMONIC	.0006	0.01
	5TH HARMONIC	.0022	0.04

DRAWING NO. 44C304462 REV NO.

REVISIONS			PRINTS TO
1		Nov 17, 1971	11
2	E.A. Ellison	Oct. 8, 1971	12-DFB
3	A.D. Card	Nov. 9, 1972	FG-W
4		May 24, 1973	

MADE BY [signature] 5-18-71 APPROVED [signature] MAY 20 1976

COMM. & CONT. DEVICES DIV. 38 DEPT. WAYNESBORO, VA. LOCATION

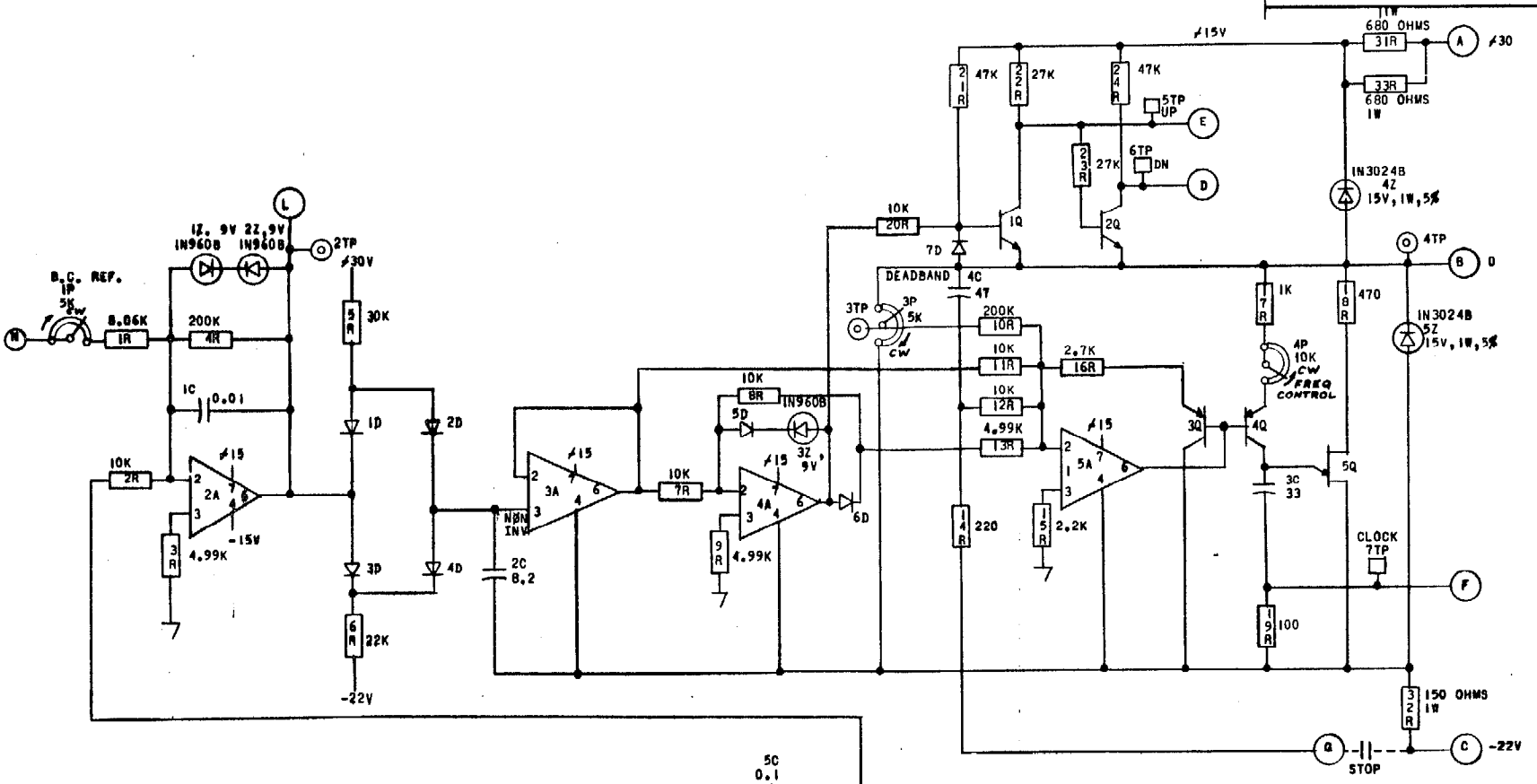
44C304462

GENERAL ELECTRIC

44C304898 FNL 1

TITLE: EHC SIGNAL CONVERTER

FIRST MADE FOR



REVISIONS		PRINT
1	per Temp. Ltr. May 26 1971	11
2	per G.W. Work Spec 1071	FG-W
3	E.C. Release (Jan. 5) 1972	12-3-FB

MADE BY: [illegible] July 27, 1971

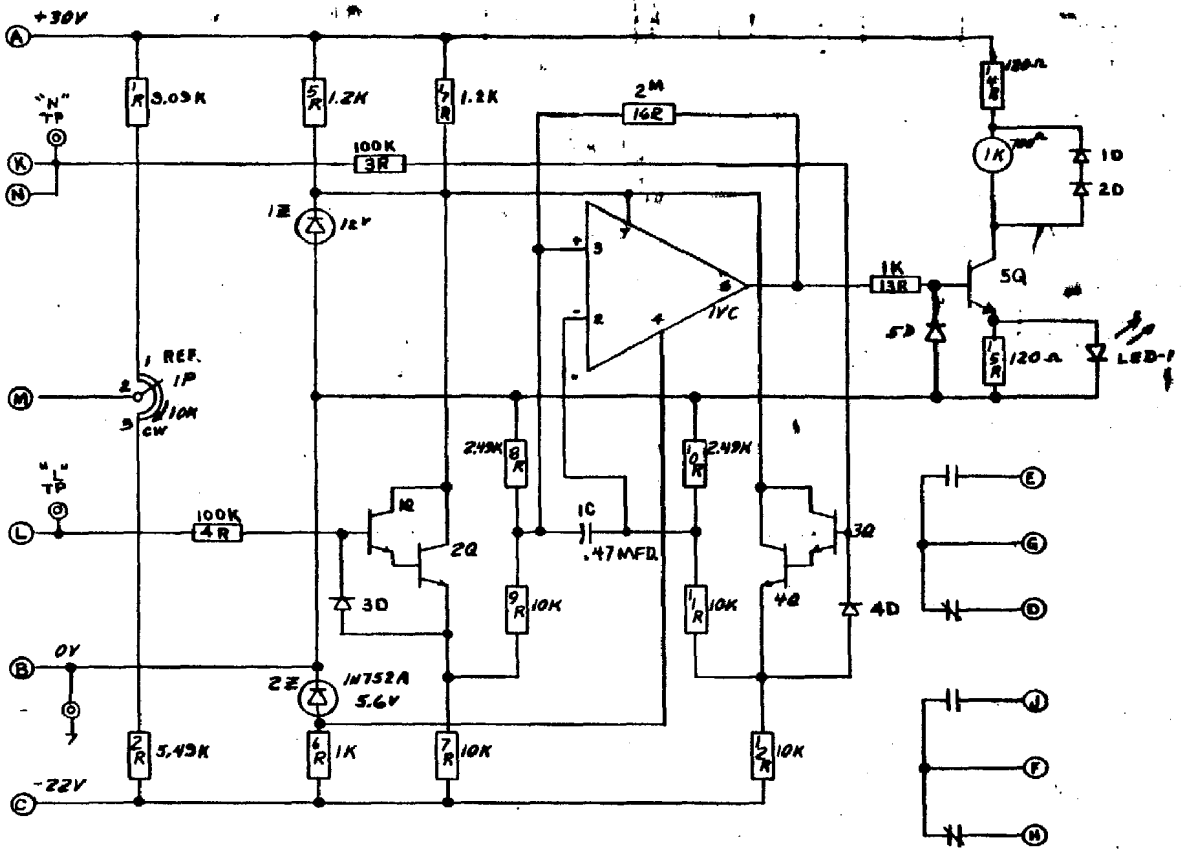
COMM. & CONT. DEVICES WAYNESBORO, VA.

GENERAL ELECTRIC

44C305259

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
VOLTAGE COMPARATOR
1VC
FIRST MADE FOR 44B33176L-G01

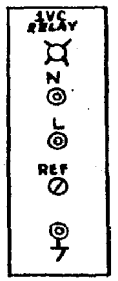


CONNECTIONS

A	+30V	N, K	INPUT
B	0 V	L	SEE NOTE 2
C	-22V	M	
D			
E			
F			
G			
H			
J			

NOTES:

1. IN NORMAL OPERATION THE RELAY IS ENERGIZED. IT CAN BE ADJUSTED TO DROPOUT ON EITHER INCREASING OR DECREASING SIGNALS BETWEEN PLUS AND MINUS 10 VOLTS.
2. DROPOUT ON — INCREASING SIGNAL: INPUT N, JUMPER L–M, REF. TP M; DECREASING SIGNAL: INPUT L, JUMPER N–M, REF. TP N. DROPOUT N>L
3. RELAY CONTACTS RATED 2A, 28VDC.
4. RELAY ENERGIZED WHEN LIGHT IS ON.



FACEPLATE

44C305259 REV NO.

REVISIONS	PRINTS TO
1 E.O. C... Aug. 7, 1973	11
	12P DR
	FG-IN

ISSUED BY	APPROVED	SPECIALTY CONTROL DEPT.	44C305259
[illegible] Dec 8, 197[illegible]	[illegible]	WAYNESBORO, VA.	CONT DW SHEET

GENERAL ELECTRIC

44C307556 CONT ON SHEET 1A SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A14 REQN. 091-4128

ELEMENTARY DIAGRAM GULF STATES, SABINE #5

PRINTED CIRCUIT BOARD LOCATIONS

RACK 1

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
LP STOP VALVE		NP STOP VALVE		CONTROL VALVE		SPEED SIGNAL		SPEED SIGNAL		OVERSPEED TEST		REF. LVG				LOCAL	MANUAL REFERENCE			HI REF	AUTO REFERENCE						
1DI		1DI		1AØ2		3AØ2		3AØ2		1ØSI	1ØA	1ØA	1ØA	2LVG	IVC	CL & SMD	1DI		IVO	5BCI		3SC		CL & SMD	1DI	

RACK 2

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
PRI SPEED			SEC. SPEED			SIGNAL LVG			0'SPD 120%	REGULATION & COMPENSATION		SERVO VALVE LOOP				3KHZ OSC	SPEED SENSING 10% 105%			200 RPM	SPEED INDICATION AND SWITCHING			LOW SPEED SW.		LSS PWP SUP	
1ØA	2FVC		1ØA	2FVC		1ØA	1ØA	1LVG	IVC	1ØA	1ØR	2SA	1DI			6OSC	1VC	1VC		1VC	1ØA	2FVC		LSS		LPS	

RACK 3

1	2	3		5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
				30 VOLT SUPPLY LINE A			LINE B					22 VOLT SUPPLY LINE A			LINE B					24 VOLT SUPPLY LINE A			LINE B				
1DB	1DB	1DB		1REC		1SR		1SR		1REC		1REC		1SR		1SR		1REC		1REC		1SR		1SR		1REC	

FRONT VIEW

REVISIONS
1 Rev. 44C307556 E.O. Clinton Jr. 9/19/74

PRINTS TO
11
12P-KFM²
FG-W

MADE BY ISSUED E. Clinton 9/15/74
APPROVALS E. Clinton 9/19/74 E. G. Wilson 9/19/74
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C307556 CONT ON SHEET 1A SH NO. 1

FORM 44C307556 P 12-69 PRINTED IN U.S.A.

3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A14 REQN. 091-41289
ELEMENTARY DIAGRAM GULF STATES, SABINE #5

PRINTED CIRCUIT BOARD AND SUB. ASM. LIST

PRINTED CIRCUIT BOARD

SYMBOL	TITLE	ELEM. DIAG.	ORDER NO.	QTY.
1AØ2	ANALOG OUTPUT BOARD	44C301216	44D236379-G01	1
CL& SMD	CLOCK & STEPPING-MOTOR DRIVE	44C303567	44D331918-G01	2
1CR	COMPENSATION & REGULATION	44A301400	44D236388-G01	1
1DB	DIODE BOARD	44A300243	44D236370-G01	3
1DI	DETECTOR INDICATOR	44C300609	44D381585-G01	5
3AØ2	ANALOG OUTPUT	44C301622	44D236379-G03	2
1LVG	SIGNAL LOW VALUE GATE	44A304763	44D236372-G02	1
2FVC	FREQUENCY TO VOLTAGE CONVERTER	44C301682	44D236365-G02	3
1REC	SERIES REG. BD. #1	44A306392	44B317827-G01	6
1SR	SERIES REG. BD. #2	44B306197	44D236371-G01	6
1ØA	OPERATIONAL AMPLIFIER	44C300610	44D381589-G01	9
1ØSI	OVERSPEED INTEGRATOR	44A302692	44D236373-G02	1
2SA	SERVO AMPLIFIER	44C304444	44D236389-G02	1
3SC	SIGNAL CONVERTER	44C304898	44B331743-G01	1
LPS	LOW SPEED SWITCH POWER SUPPLY	44B306144	44B337328-G01	1
LSS	LOW SPEED SWITCH	44C306444	44B337327-G01	1
1VC	VOLTAGE COMPARATOR	44C305259	44B331761-G01	6
5BCI	BOILER CONTROL INTERFACE	44C305210	44B331746-G01	1
2LVG	REF. LOW VALUE GATE.	44A300281	44D236378-G01	1
60SC	3KHZ. OSC. BOARD	44C304462	44B331717-G01	1

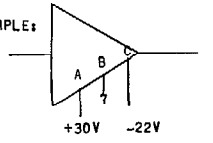
SUB. ASSEMBLIES - PANELS -

TITLE	ORDER NO.	CONNECTION DIAGRAM
CONTROL PANEL	3S7513KL300A12	44D306687
POWER PANEL, LINE A&B	3S7513KF300A5	44C306526
METER PANEL	3S7513KF306A9	44C305834
METER PANEL	3S7513KF306A11	44C306262
RELAY PANEL	3S7513K307A15	44D308103
DIFF. XFMR. DRIVE	44D315178-G01	44C300882

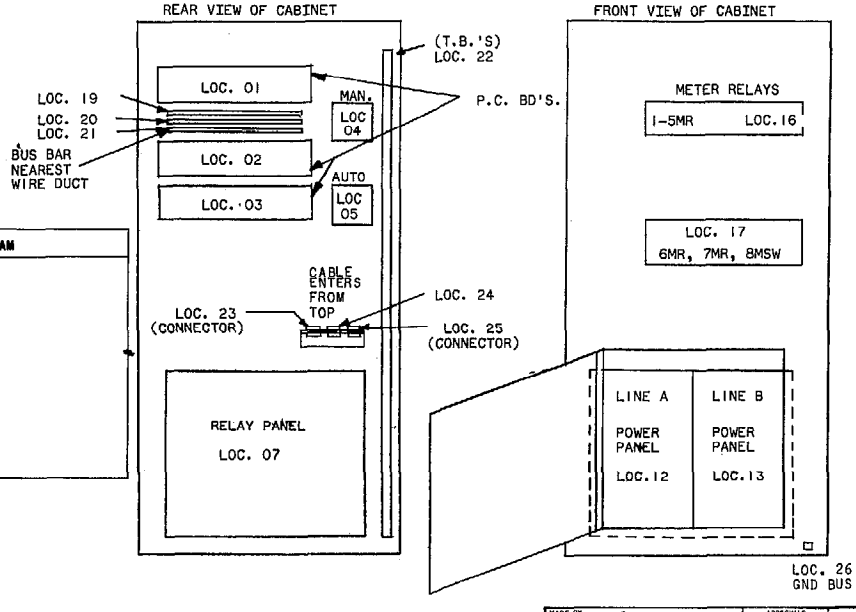
CUSTOMER NOTES:

1. ⏚ INDICATES ZERO REFERENCE POTENTIAL, NOT CASE GROUND.
2. ⏚ INDICATES STATION GROUND.
3. ALL OPERATIONAL AMPLIFIERS HAVE ADDITIONAL CONNECTIONS NOT SHOWN ELSEWHERE

EXAMPLE:



4. THE LOCATION OF THE PRINTED CIRCUIT BOARDS IS SHOWN BY THE NUMBERS APPEARING IN OR NEAR THE SQUARE WHICH REPRESENTS THE P.C. BD. EXAMPLE 0215 WOULD BE RACK #2 AND SLOT #15.
5. 1MR TO 7MR (SHEET 10) ARE METER RELAYS.
6. * INDICATES ITEMS NOT FURNISHED BY WAYNESBORO.
7. INDICATES SHIELDED CABLE.
8. INDICATES TWISTED WIRES.
9. INDICATES TERMINALS AT EHC CASE FOR CUSTOMER CONNECTION
10. INDICATES TERMINALS AT TURBINE
11. INDICATES SUB-PANEL & PC BOARD TERMINALS IN EHC CASE
12. CONTACTS FROM VOLTAGE COMPARATOR RELAY (1VC)



REVISIONS
1 E.O. [illegible] 44C307556 [illegible] 9/1974
2 [illegible] Oct 6/1974

PRINTS TO
11
12P-XMM²
FG-W

MADE BY [illegible] 4/15/74 APPROVING O.P. Calvert
ISSUED [illegible] 9/1974 [illegible] 6/6/1974
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C307556 CONT ON SHEET 3 SH NO 1A

GENERAL ELECTRIC

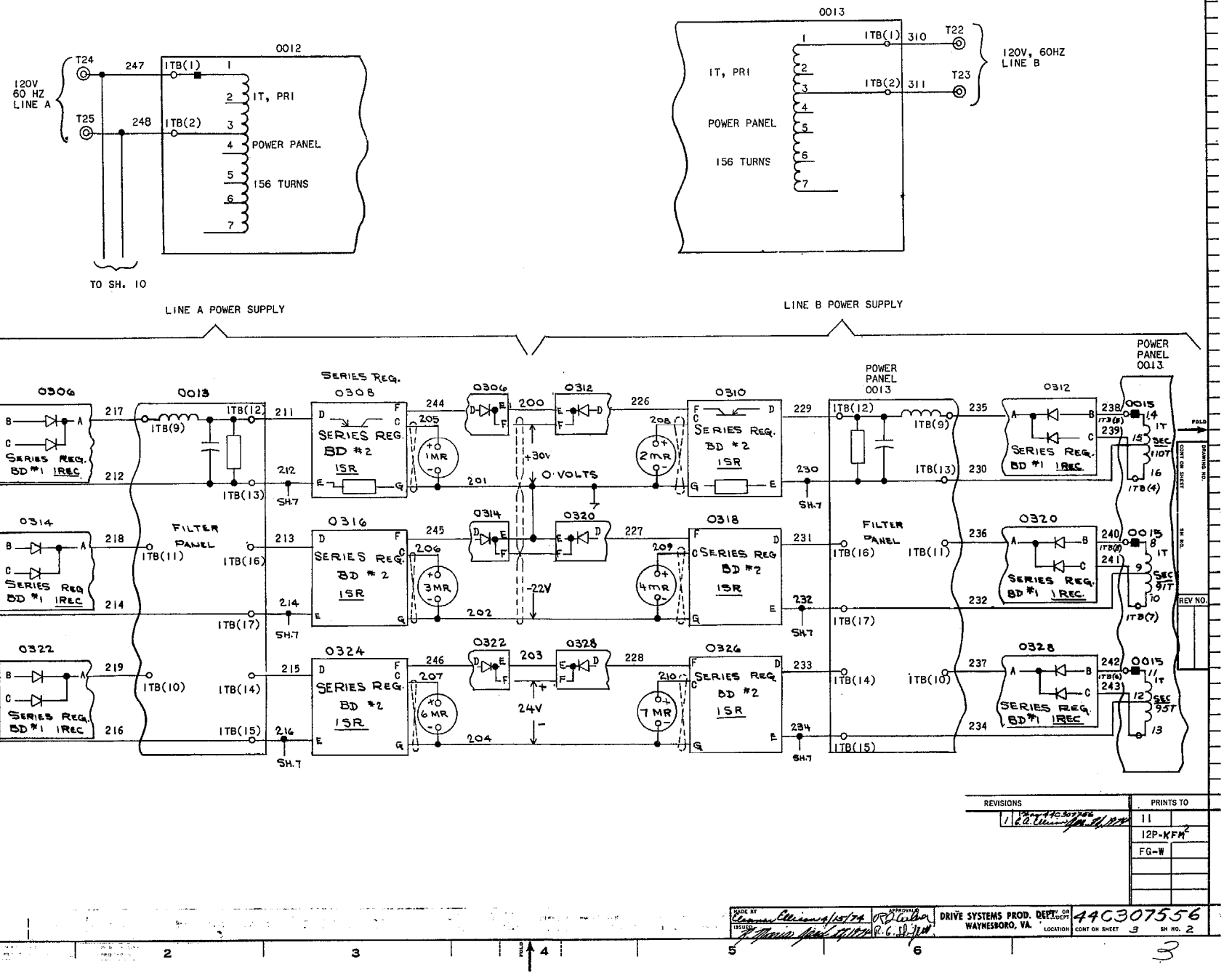
44C307556 CONT ON SHEET 3 SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TG300A14 REQN. 091-41299

ELEMENTARY DIAGRAM GULF STATES, SABINE #5



REVISIONS

1	Rev 44C307556 G.A. Ellis 1/24/77

PRINTS TO

11	
12P-KFM²	
FG-W	

MADE BY Eleanor Ellis 9/15/76 | APPROVED R. Ellis | ISSUED P. Morris 9/16/76 P.C. 9/16/76

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C307556 CONT ON SHEET 3 SH NO. 2

FF-809-F (2-68) PRINTED IN U.S.A.

3

GENERAL ELECTRIC

44C307556 CONT ON SHEET 4 SH NO. 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	√	±	±	±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A14 REQN. 091-41289

ELEMENTARY DIAGRAM GULF STATES, SABINE #5

0108 — 3AØ2 — IN, GAIN, OUT, I OUT, BIAS — T105 370, T93 201 — 0-1MA FOR CUST. USE — ICO-M 337, ICO-N 201, ICO-V — 0-8000 RPM — A B C 200 +30V, 201 0V, 202 -22V

0110 — 3AØ2 — IN, GAIN, OUT, I OUT, BIAS — T91 335 NIO, T92 N9, NII — TI-1 0-1MA — T94 339 — 0-1MA — T93 201 — A B C 200 +30V, 201 0V, 202 -22V

0224 COARSE A B C FINE — 30V 200, 0V 201, -22V 202 — 0222 — 1ØA — 317 — 12 — D E H J — 329 F — SPEED SIGNAL FROM SH.6 — 330 E — SH. 7 — 2FVC — 316

0218 — +30 200, 0V 201, -22 202 — L M N A B C — 10% SPEED — IVC — D G E H F J — 505 SH. 5 — 321 SH. 5

SPEED SIGNAL FROM SH. 6 610

0210 — N M L — SPEED 120% — IVC — A B C 200 +30V, 201 0V, 202 -22V — 1298 — 1270 TO SH. 12

0121 — +30V 200, 0V -22V 201, 202 — A B C — L M N — HI REF — IVC — D G E H F J — 1298 SH. 12 — 203 (HREF) — 508 OVERSPEED REFERENCE SH. 5

SEE SH. 10 — 5MR — 322 — 0217 — D 201 — F,H,K,M — E,G,J,L 300 — 6V, RMS 3KHZ SH. 4,5,8,9 LVDT EXCITATION — 60SC AMPLIFIER — A B C 200 +30V, 201 0V, 202 -22V

VOLTAGE COMPARATORS (IVC) ARE PICKED UP AND WILL DROP OUT AT OPERATING POINT OR SPEED.

0219 — L M N — 105% SPEED — IVC — D G E H F J — 320 SH. 11(105X) — 203 +24V — A B C 200 +30V, 201 0V, 202 -22V

0221 — L M N — 200 RPM SPEED — IVC — D G E H F J — 316 — 1250 TO LSS SH. 5 — 1249 TO SH. 12 — 514 SH. 5 — 510 SH. 5 — A B C 200 +30V, 201 0V, 202 -22V

▲ INDICATES ITEM MOUNTED ON REMOTE CONTROL PANEL

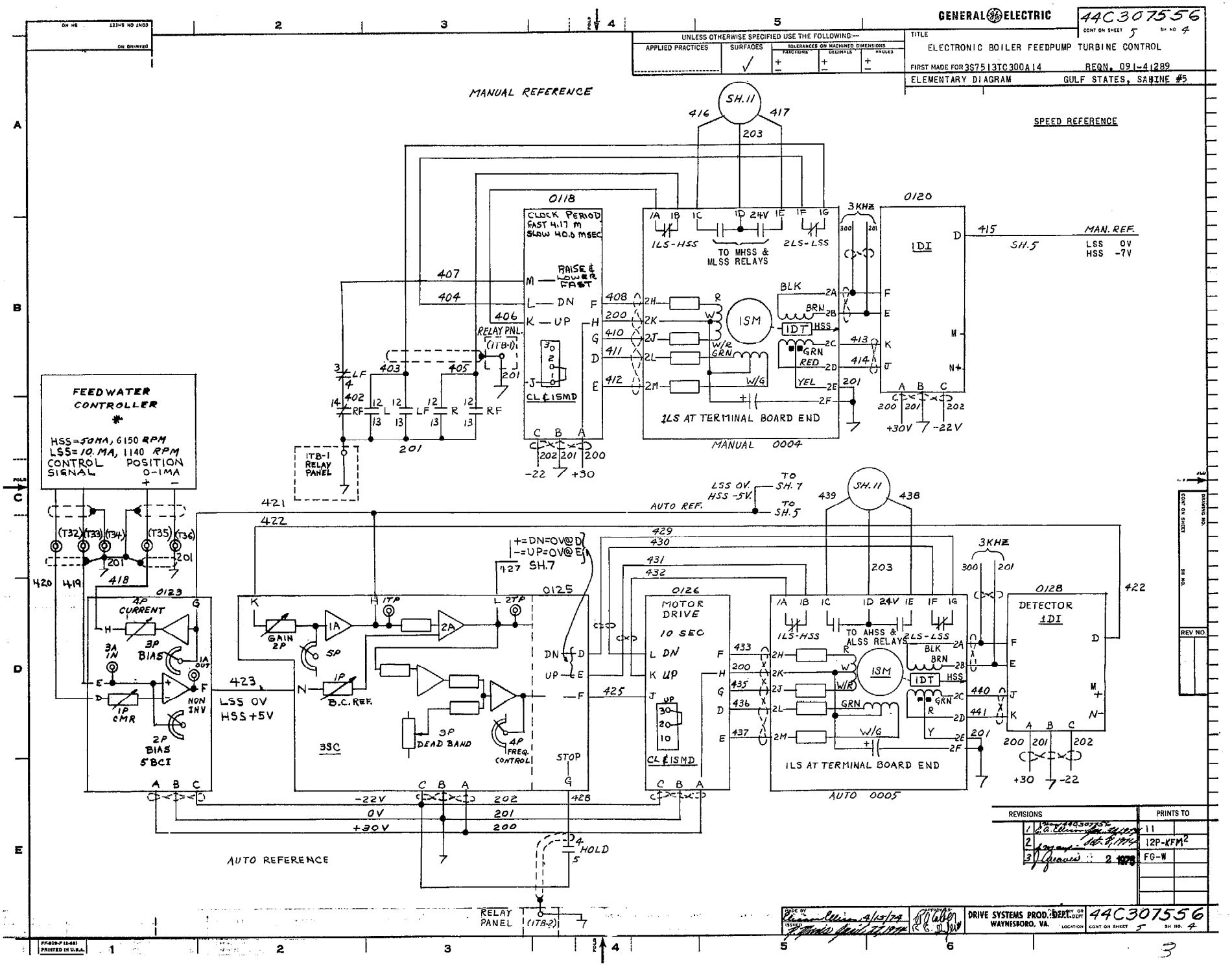
REVISIONS

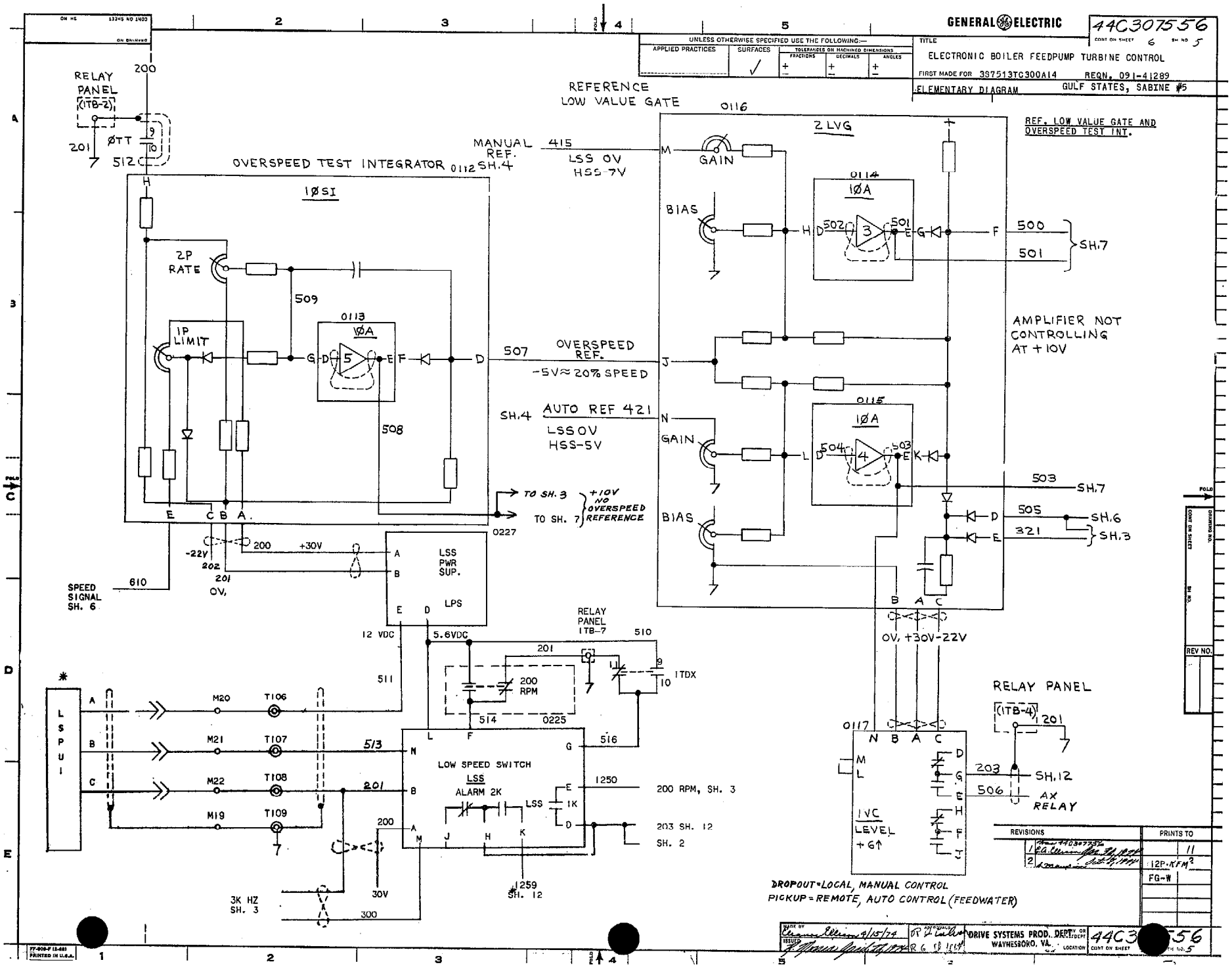
1	Mod 44C307556	
2		12P-KFM2
3	1975	FG-W

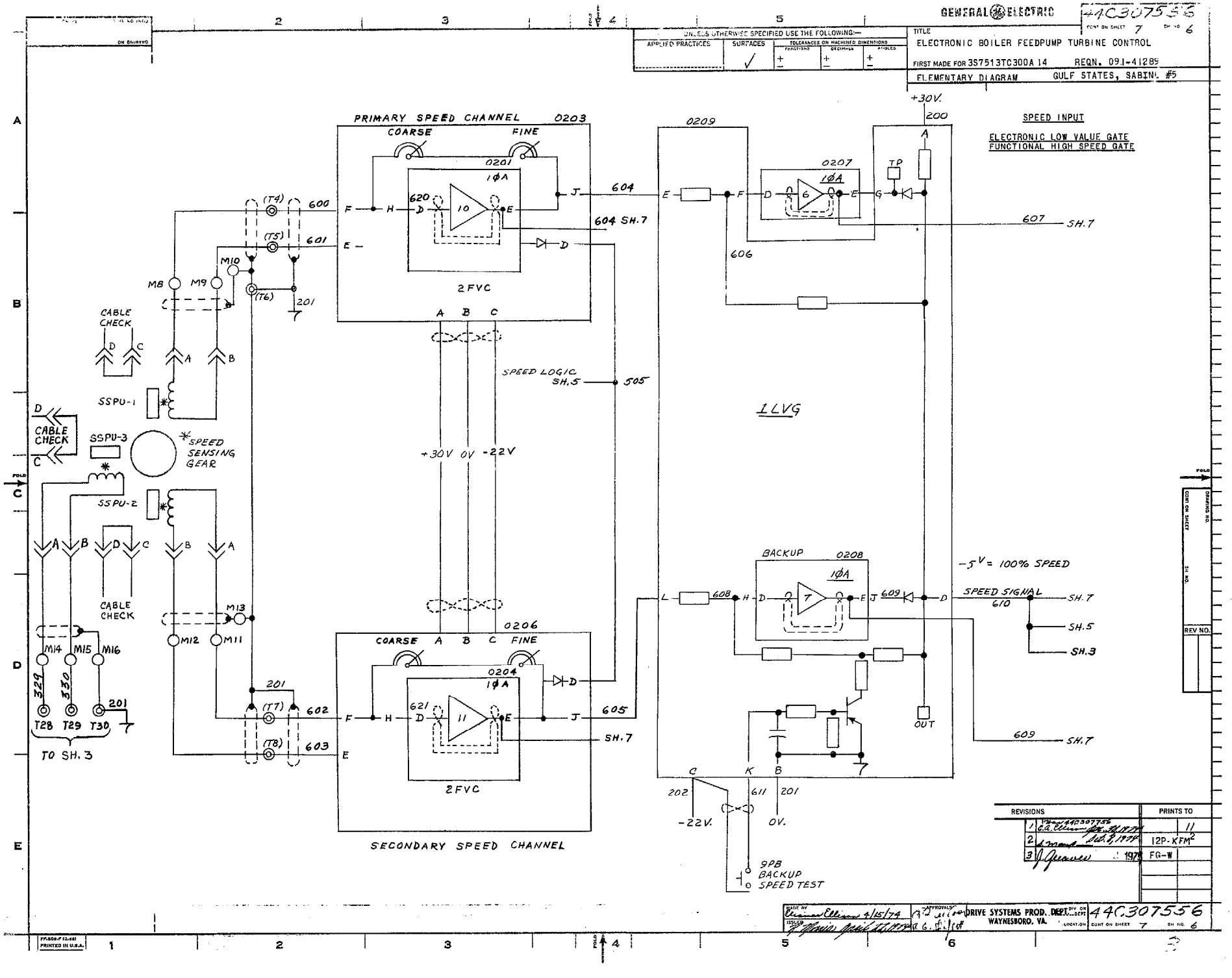
PRINTS TO 11

MADE BY ISSUED 4/15/74 — APPROVALS R.A. Gilbert — DRIVE SYSTEMS PROD. DEPT. OR DEPT. WAYNESBORO, VA. LOCATION — 44C307556 CONT ON SHEET 4 SH NO. 3

77-609-F 68-791 PRINTED IN U.S.A.


GENERAL ELECTRIC
44C307556
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
ELEMENTARY DIAGRAM
GULF STATES, SABINE #5
MANUAL REFERENCE
SPEED REFERENCE
FEEDWATER CONTROLLER
HSS=50MA, 6150 RPM
LSS=10. MA, 1140 RPM
CONTROL SIGNAL
POSITION 0-1MA
CLOCK PERIOD FAST 4.17 M SLOW 40.0 MSEC
RAISE & LOWER FAST
DN
UP
RELAY PNL. (1TB-1)
CL & ISMD
ILS-HSS
TO MHSS & MLSS RELAYS
2LS-LSS
ILS AT TERMINAL BOARD END
MANUAL 0004
ISM
IDT
HSS
3KHZ
IDI
MAN. REF.
LSS 0V
HSS -7V
+30V
-22V
ITB-1 RELAY PANEL
AUTO REF.
LSS 0V. HSS -5V.
TO SH. 7
TO SH. 5
+=DN=0V@D
-=UP=0V@E
GAIN 2P
B.C. REF.
3P DEAD BAND
4P FREQ. CONTROL
3SC
MOTOR DRIVE 10 SEC
L DN
K UP
STOP
TO AHSS & ALSS RELAYS
DETECTOR 1DI
AUTO 0005
LSS 0V HSS +5V
-22V
0V
+30V
HOLD
AUTO REFERENCE
RELAY PANEL (1TB-2)
REVISIONS
PRINTS TO
11
12P-KFM2
FG-W
DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.
44C307556


GENERAL ELECTRIC
44C307556
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
ELEMENTARY DIAGRAM
GULF STATES, SABINE #5
REFERENCE LOW VALUE GATE
0116
2 LVG
REF. LOW VALUE GATE AND OVERSPEED TEST INT.
MANUAL REF.
415
LSS 0V
HSS-7V
OVERSPEED TEST INTEGRATOR 0112 SH.4
1ØSI
2P RATE
1P LIMIT
509
0113
1ØA
508
507
OVERSPEED REF.
-5V≈20% SPEED
SH.4 AUTO REF 421
LSS 0V
HSS-5V
0114
0115
GAIN
BIAS
500
501
503
505
321
SH.7
SH.6
SH.3
AMPLIFIER NOT CONTROLLING AT +10V
TO SH. 3 +10V NO OVERSPEED REFERENCE
TO SH. 7 REFERENCE
0227
RELAY PANEL (ITB-2)
200
201
512
ØTT
SPEED SIGNAL SH. 6
610
-22V
202
200
+30V
201
0V,
LSS PWR SUP.
LPS
12 VDC
5.6VDC
RELAY PANEL ITB-7
510
511
201
200 RPM
514
0225
1TDX
516
LSPUI
M20
T106
M21
T107
M22
T108
M19
T109
5/3
201
LOW SPEED SWITCH
LSS
ALARM 2K
LSS
1K
1250
200 RPM, SH. 3
203 SH. 12
SH. 2
1259
SH. 12
3K HZ
SH. 3
30V
300
0V, +30V -22V
0117
IVC
LEVEL
+6↑
RELAY PANEL
(ITB-4)
201
203
506
SH.12
AX RELAY
DROPOUT=LOCAL, MANUAL CONTROL
PICKUP=REMOTE, AUTO CONTROL (FEEDWATER)
REVISIONS
PRINTS TO
12P-KFM2
FG-W
DRIVE SYSTEMS PROD. DEPT
WAYNESBORO, VA.
44C307556


GENERAL ELECTRIC
44C307556
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A 14
REQN. 09.1-41285
ELEMENTARY DIAGRAM
GULF STATES, SABINE #5
PRIMARY SPEED CHANNEL 0203
COARSE
FINE
0201
1ΦA
2FVC
SECONDARY SPEED CHANNEL
0206
0204
0209
0207
0208
BACKUP
+30V.
200
SPEED INPUT
ELECTRONIC LOW VALUE GATE
FUNCTIONAL HIGH SPEED GATE
1LVG
CABLE CHECK
SSPU-1
SSPU-2
SSPU-3
SPEED SENSING GEAR
SPEED LOGIC SH.5
+30V 0V -22V
-5V = 100% SPEED
SPEED SIGNAL
TO SH. 3
-22V.
0V.
9PB
BACKUP
SPEED TEST
OUT
REVISIONS
PRINTS TO
12P-KFM2
FG-W
DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.
44C307556

GENERAL ELECTRIC

44C307556 CONT ON SHEET 8 SH NO 7

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

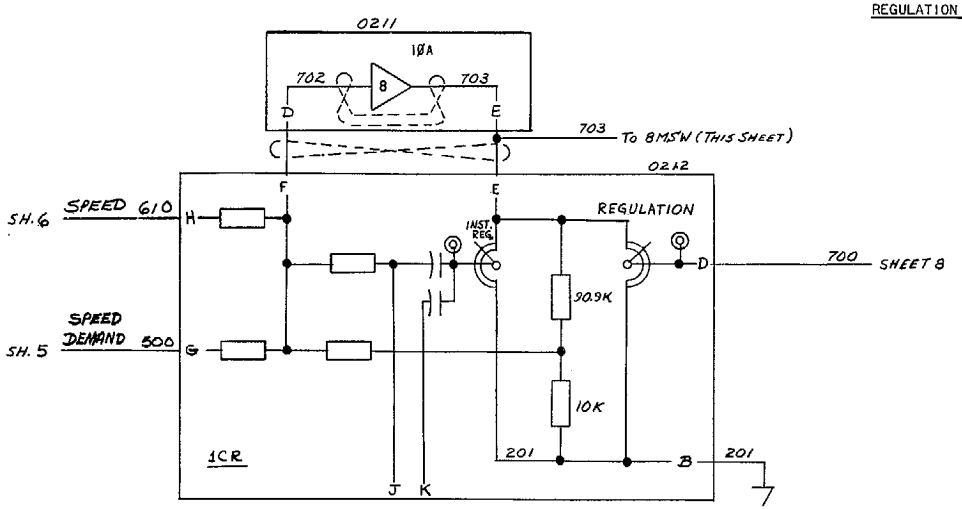
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

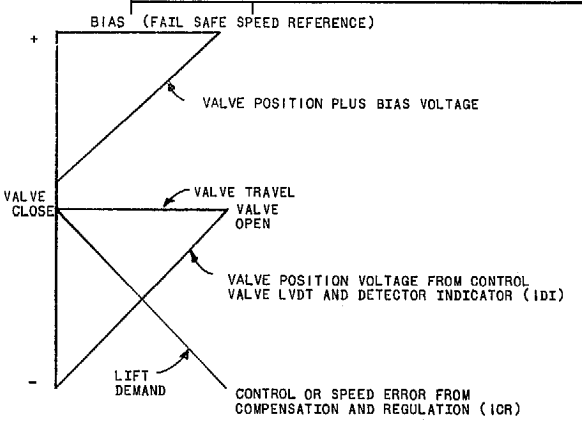
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

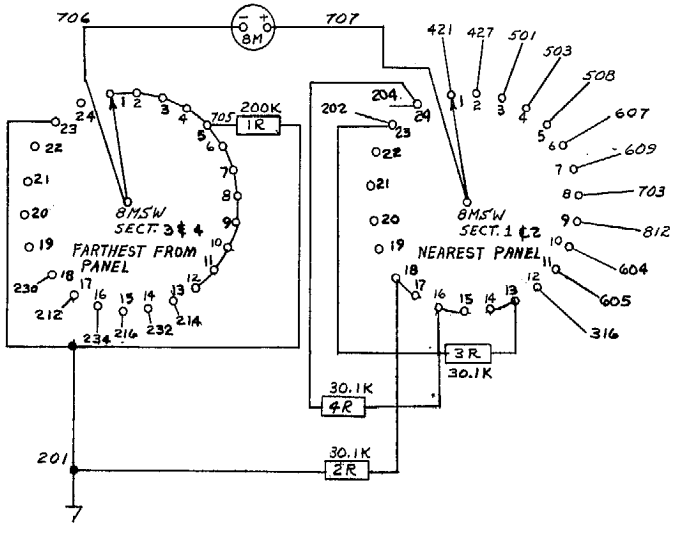
FIRST MADE FOR 3S7513TC300A14 REQN. 091-41289

ELEMENTARY DIAGRAM GULF STATES, SABINE #5

REGULATION AND STABILIZATION







AMPLIFIER

SWITCH POSITION	NUMBER	SYMBOL	FUNCTION	RACK LOCATION	APPROXIMATE OUTPUT VOLT RANGE	SHEET NO.
1	1	3SC	SIG. CONV.	0125	0 TO -5VDC	4
2	2	3SC	SIG. CONV.	0125	10VDC TO -10VDC	4
3	3	2LVG	MAN. REF.	0114	10VDC TO -5VDC	5
4	4	2LVG	AUTO REF.	0115	10VDC TO -5VDC	5
5	5	0'SPD	OVERSPEED	0113	10VDC TO -5VDC	5
6	6	1LVG	PRI. SPD.	0207	10VDC TO -7VDC	6
7	7	1LVG	SEC. SPD.	0208	10VDC TO -7VDC	6
8	8	1CR	COMP. & REG.	0211	10VDC TO -10VDC	7
9	9	2SA	SERVO AMP.	0214	7VDC TO -7VDC	8
10	10	2FVC	PRI. SPD.	0201	10VDC TO -7VDC	6
11	11	2FVC	SEC. SPD.	0204	0 TO -7VDC	6
12	12	2FVC	SPD. IND.	0222	0 TO -7VDC	3

POWER SUPPLY

					APPROXIMATE OUTPUT CURRENT RANGE	
13	LINE A	1SR	-22VDC	0316	5 AMPS	2
14	LINE B	1SR	-22VDC	0318		2
15	LINE A	1SR	+24VDC	0322		2
16	LINE B	1SR	+24VDC	0328		2
17	LINE A	1SR	+30VDC	0308		2
18	LINE B	1SR	+30VDC	0310	5 AMPS	2

REVISIONS

PRINTS TO 12P-KFM2 FG-W

MADE BY ISSUED 4/15/74 APPROVALS DRIVE SYSTEMS PROD. DEPT WAYNESBORO, VA. LOCATION

44C307556 CONT ON SHEET 8 SH NO. 7

FP-909-P (2-69) PRINTED IN U.S.A.

GENERAL ELECTRIC

44C307556 CONT ON SHEET 9 SH NO 8

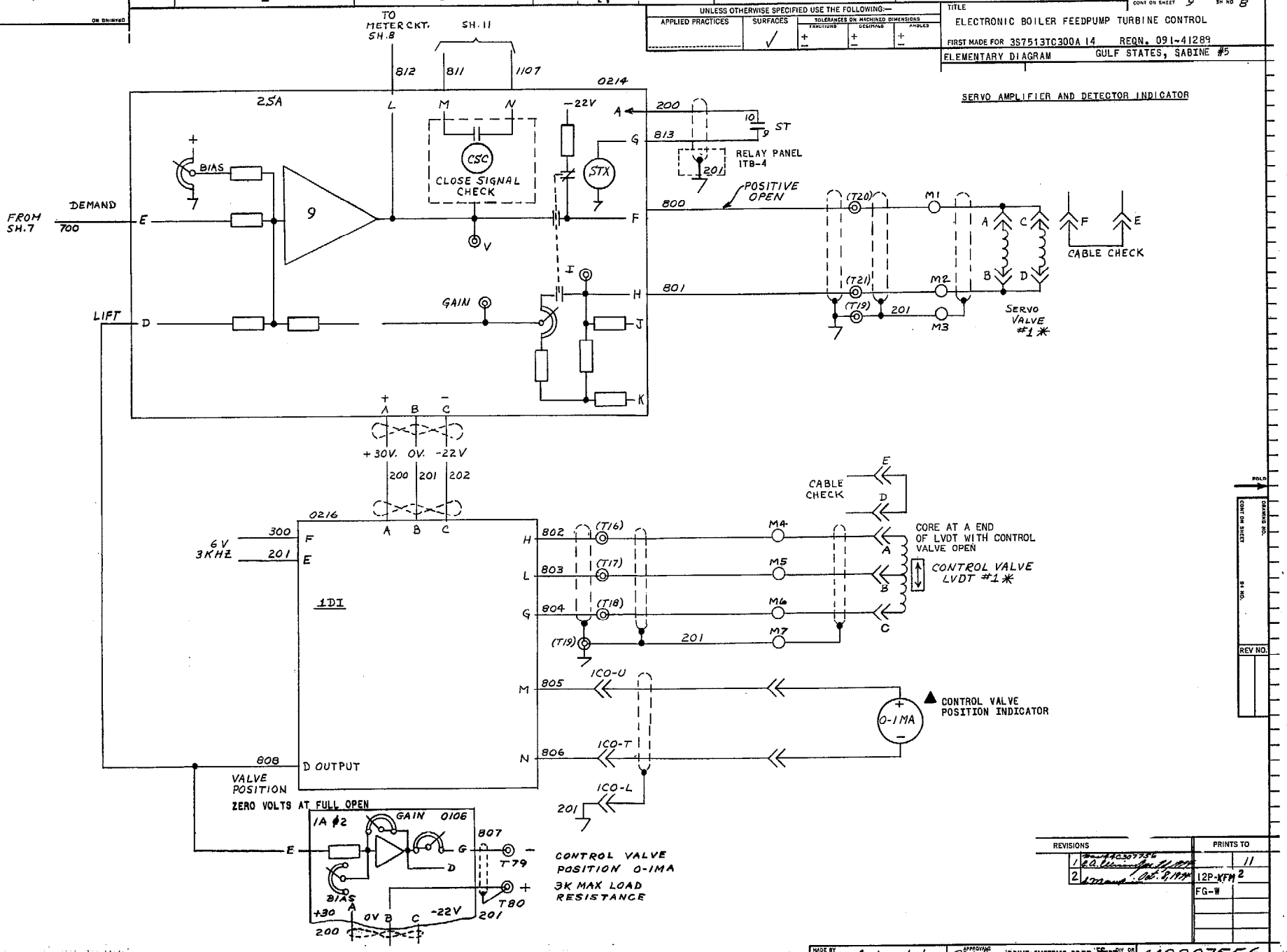
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A 14 REQN. 091-41289

ELEMENTARY DIAGRAM GULF STATES, SABINE #5

SERVO AMPLIFIER AND DETECTOR INDICATOR



▲ INDICATES ITEM MOUNTED ON REMOTE CONTROL PANEL.

REVISIONS

1	Per 44C307556 C.D. Clemens 10-11-74	
2	A Manuel Oct. 9, 1974	

PRINTS TO: 11 | 12P-XFM 2 | FG-W

MADE BY C. D. Clemens 9/15/74 ISSUED

APPROVING R. G. Lee 9-22-74

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C307556 CONT ON SHEET 9 SH NO. 8

3

GENERAL ELECTRIC

44C307556 CONT ON SHEET 10 SH NO 9

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

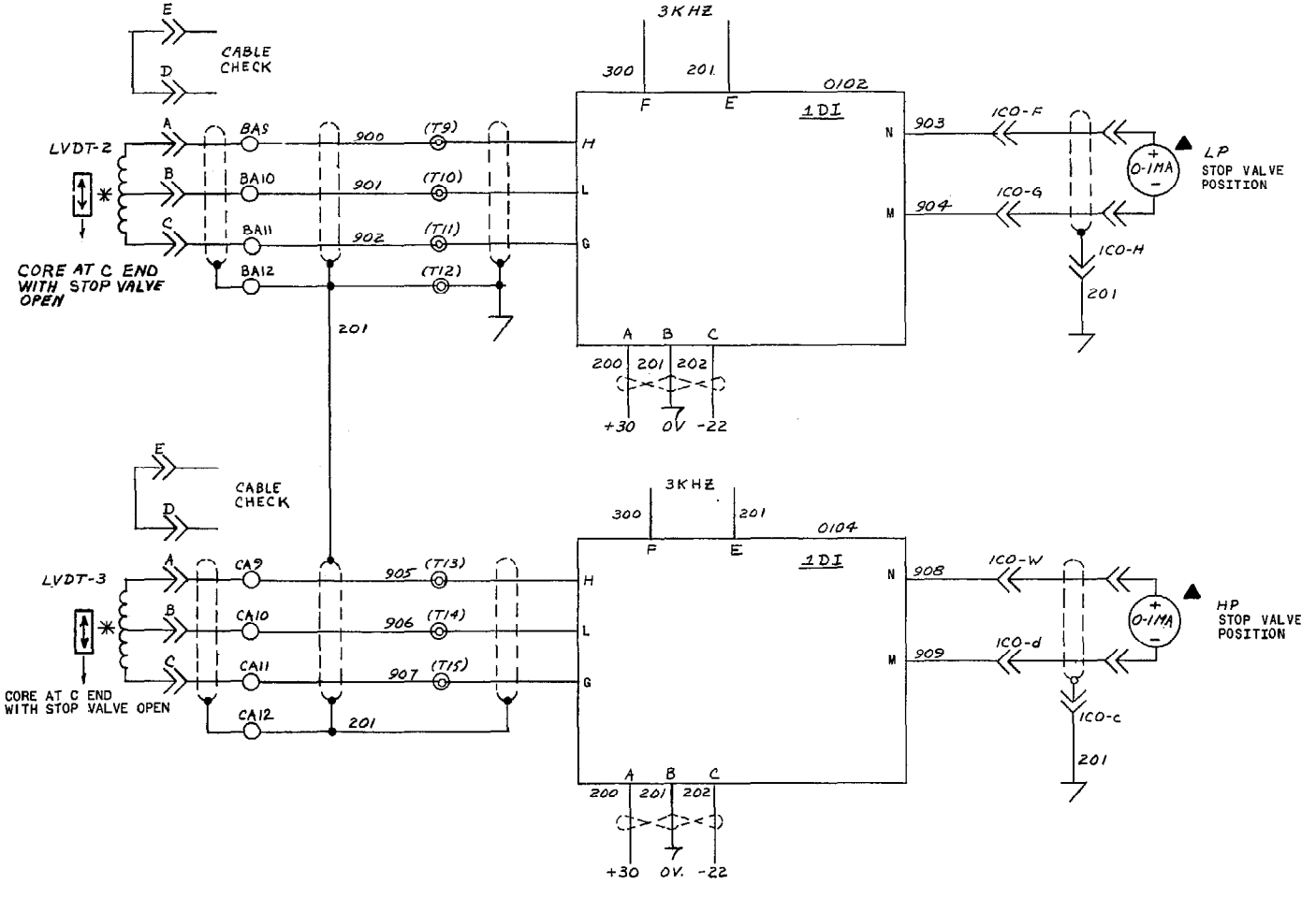
TITLE

ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A14 REQN. 091-41289

ELEMENTARY DIAGRAM GULF STATES, SABINE #5

STOP VALVE POSITION



▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL

REVISIONS		PRINTS TO	
1	44C307756 E.A. Clemens 9/19/74		11
2	Same as 9/19/74	L2P-KFM2	
		FG-W	

MADE BY E.A. Clemens 9/15/74 APPROVED R.C. Fuller

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C307556 CONT ON SHEET 10 SH NO 9

FY-809-P (8-68) PRINTED IN U.S.A.

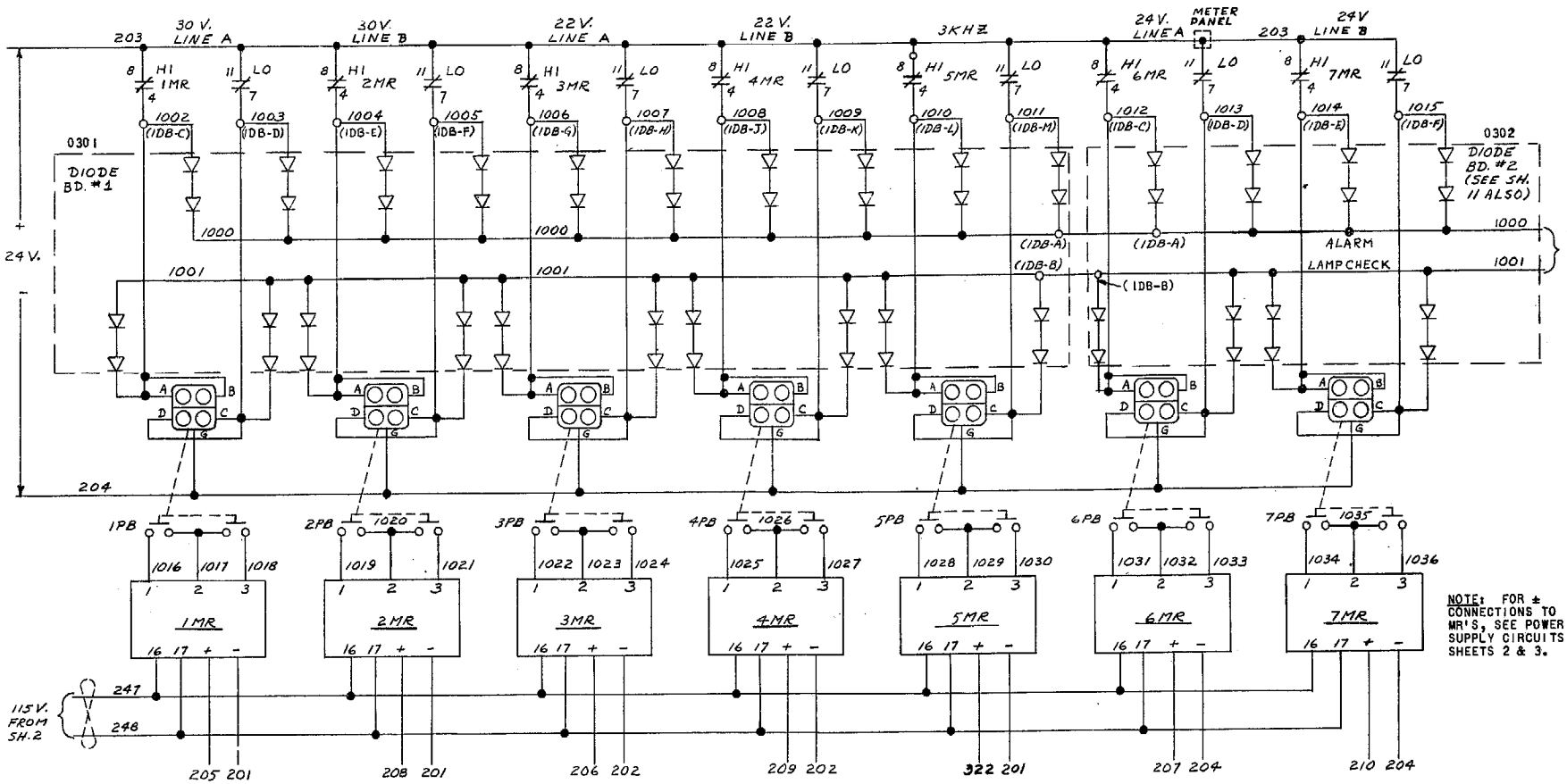
GENERAL ELECTRIC

44C307556 CONT ON SHEET 11 SH NO 10

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A14 REQN. 091-41289
ELEMENTARY DIAGRAM GULF STATES, SABINE #5

MONITOR



NOTE: FOR ± CONNECTIONS TO MR'S, SEE POWER SUPPLY CIRCUITS SHEETS 2 & 3.

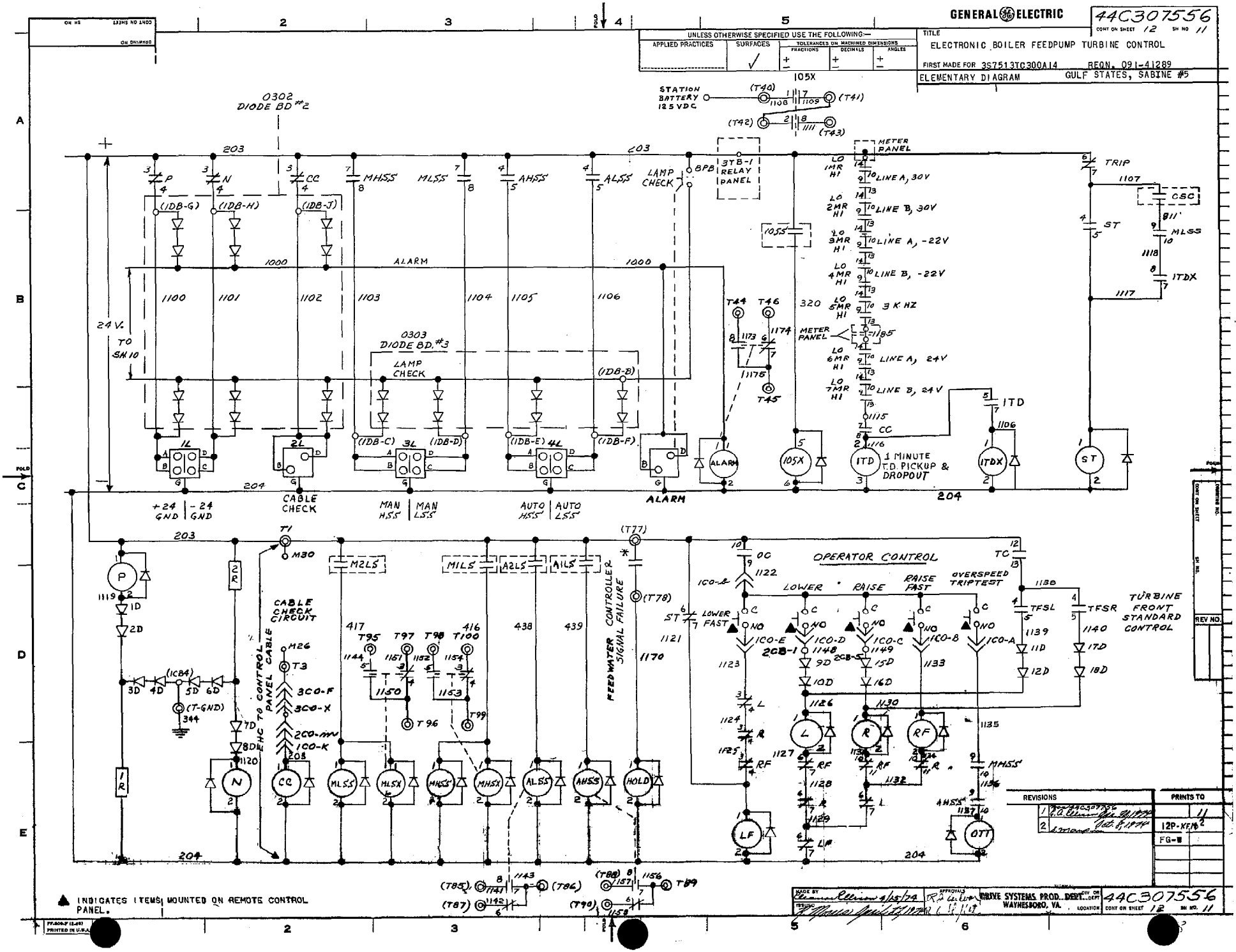
REVISIONS
1 P.O. Clemons 44C307736 7/18/74

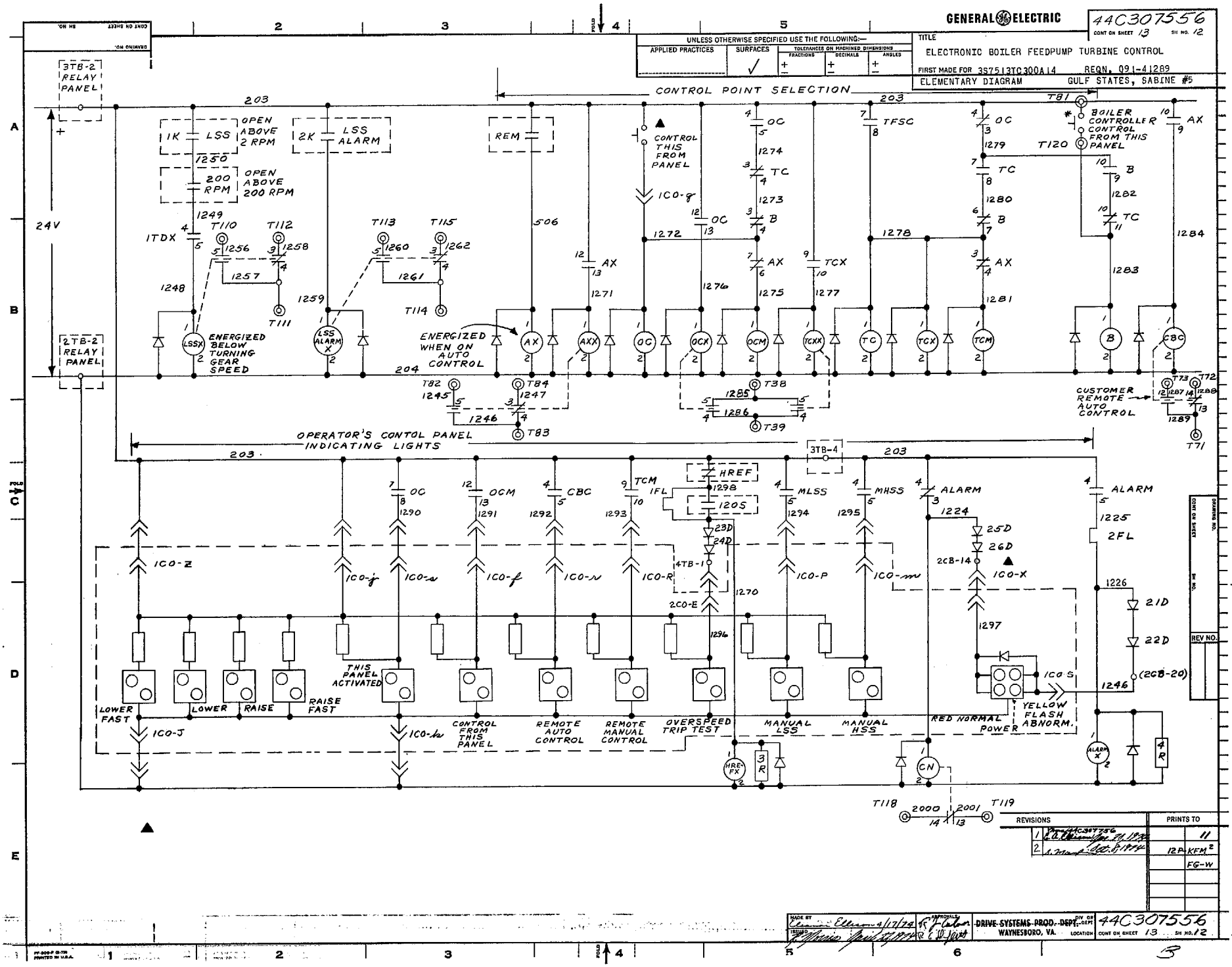
PRINTS TO 11
12P-KFM²
FG-W

MADE BY P.O. Clemons 4/15/74 APPROVED R.J. Weber
ISSUED P.O. Clemons 7/18/74 C.H. Hunt
DRIVE SYSTEMS PROD. DEPT. DEPT. LOCATION WAYNESBORO, VA.
44C307556 CONT ON SHEET 11 SH NO 10

77-804-F (2-68) PRINTED IN U.S.A.

3


GENERAL ELECTRIC
44C307556
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
ELEMENTARY DIAGRAM
GULF STATES, SABINE #5
105X
STATION BATTERY 125 VDC
0302 DIODE BD #2
0303 DIODE BD. #3
24 V. TO SM10
LAMP CHECK
ALARM
3TB-1 RELAY PANEL
METER PANEL
LINE A, 30V
LINE B, 30V
LINE A, -22V
LINE B, -22V
3 K HZ
LINE A, 24V
LINE B, 24V
1 MINUTE T.D. PICKUP & DROPOUT
CABLE CHECK
MAN HSS
MAN LSS
AUTO HSS
AUTO LSS
CABLE CHECK CIRCUIT
TO CONTROL PANEL CABLE
FEEDWATER CONTROLLER SIGNAL FAILURE
OPERATOR CONTROL
LOWER
RAISE
RAISE FAST
OVERSPEED TRIPTEST
TURBINE FRONT STANDARD CONTROL
▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL.
DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.


GENERAL ELECTRIC
44C307556
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
ELEMENTARY DIAGRAM
GULF STATES, SABINE #5
CONTROL POINT SELECTION
OPERATOR'S CONTOL PANEL INDICATING LIGHTS
3TB-2 RELAY PANEL
2TB-2 RELAY PANEL
3TB-4
24V
1K LSS OPEN ABOVE 2 RPM
2K LSS ALARM
200 RPM OPEN ABOVE 200 RPM
REM
CONTROL THIS FROM PANEL
BOILER CONTROLLER CONTROL FROM THIS PANEL
ENERGIZED BELOW TURNING GEAR SPEED
ENERGIZED WHEN ON AUTO CONTROL
CUSTOMER REMOTE AUTO CONTROL
THIS PANEL ACTIVATED
LOWER FAST
LOWER
RAISE
RAISE FAST
CONTROL FROM THIS PANEL
REMOTE AUTO CONTROL
REMOTE MANUAL CONTROL
OVERSPEED TRIP TEST
MANUAL LSS
MANUAL HSS
RED NORMAL POWER
YELLOW FLASH ABNORM.
REVISIONS
PRINTS TO

GENERAL ELECTRIC

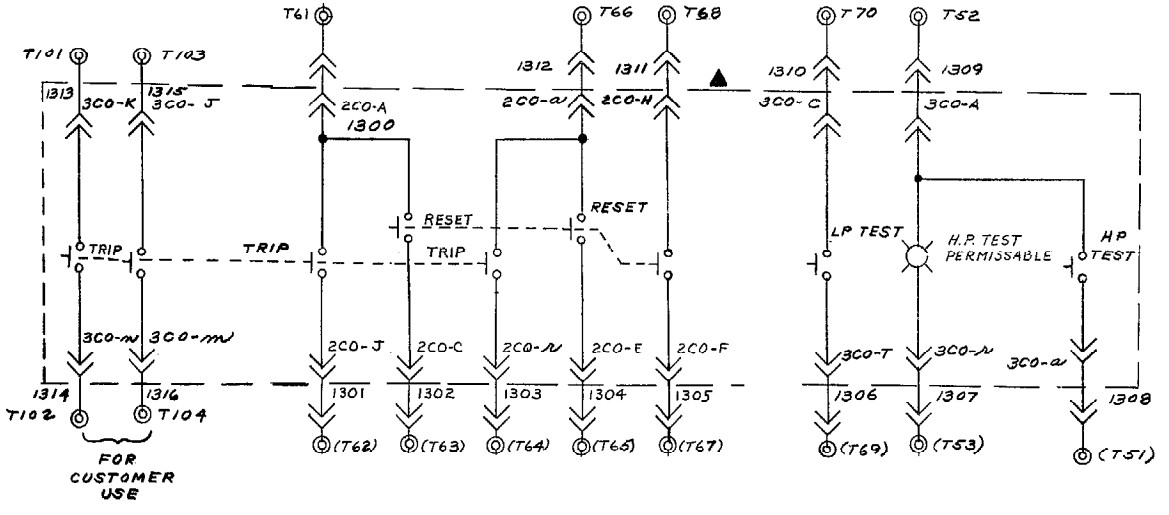
44C3075556 CONT ON SHEET FNL, SH NO 13

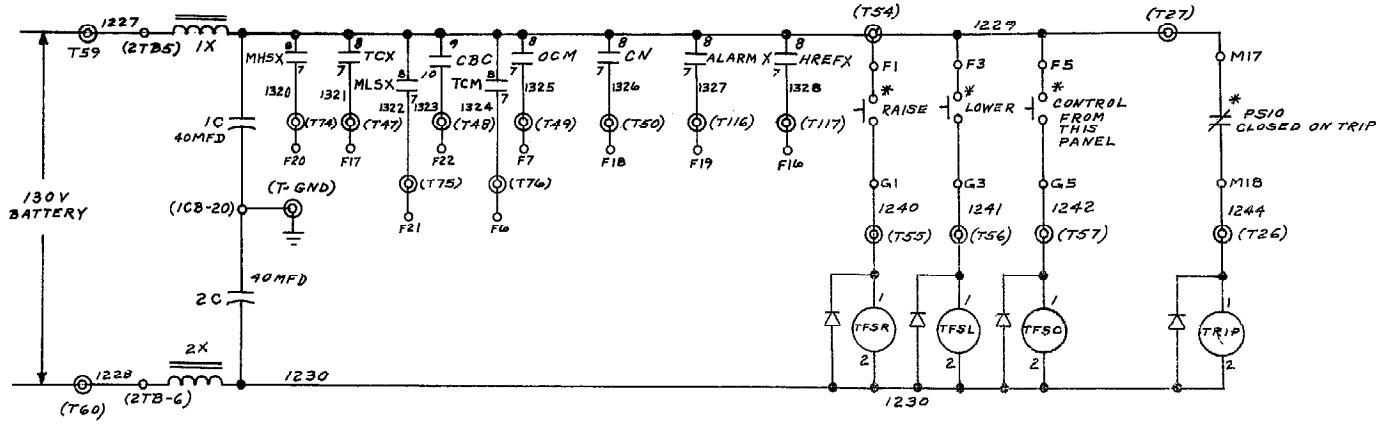
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A14 REQN. 091-41289

ELEMENTARY DIAGRAM GULF STATES, SABINE #5





▲ INDICATES ITEMS MOUNTED ON CONTROL PANEL

REVISIONS
1 [illegible] 7/11/74
2 [illegible]

PRINTS TO
11
12P-KFM2
FG-W

MADE BY [illegible] 4/15/74
ISSUED [illegible]
APPROVED [illegible]
DRIVE SYSTEMS PROD. DEPT
WAYNESBORO, VA.
DW OR DEPT LOCATION
44C3075556 CONT ON SHEET FNL SH NO 13

FY-609-P (5-68) PRINTED IN U.S.A.

3

GENERAL ELECTRIC

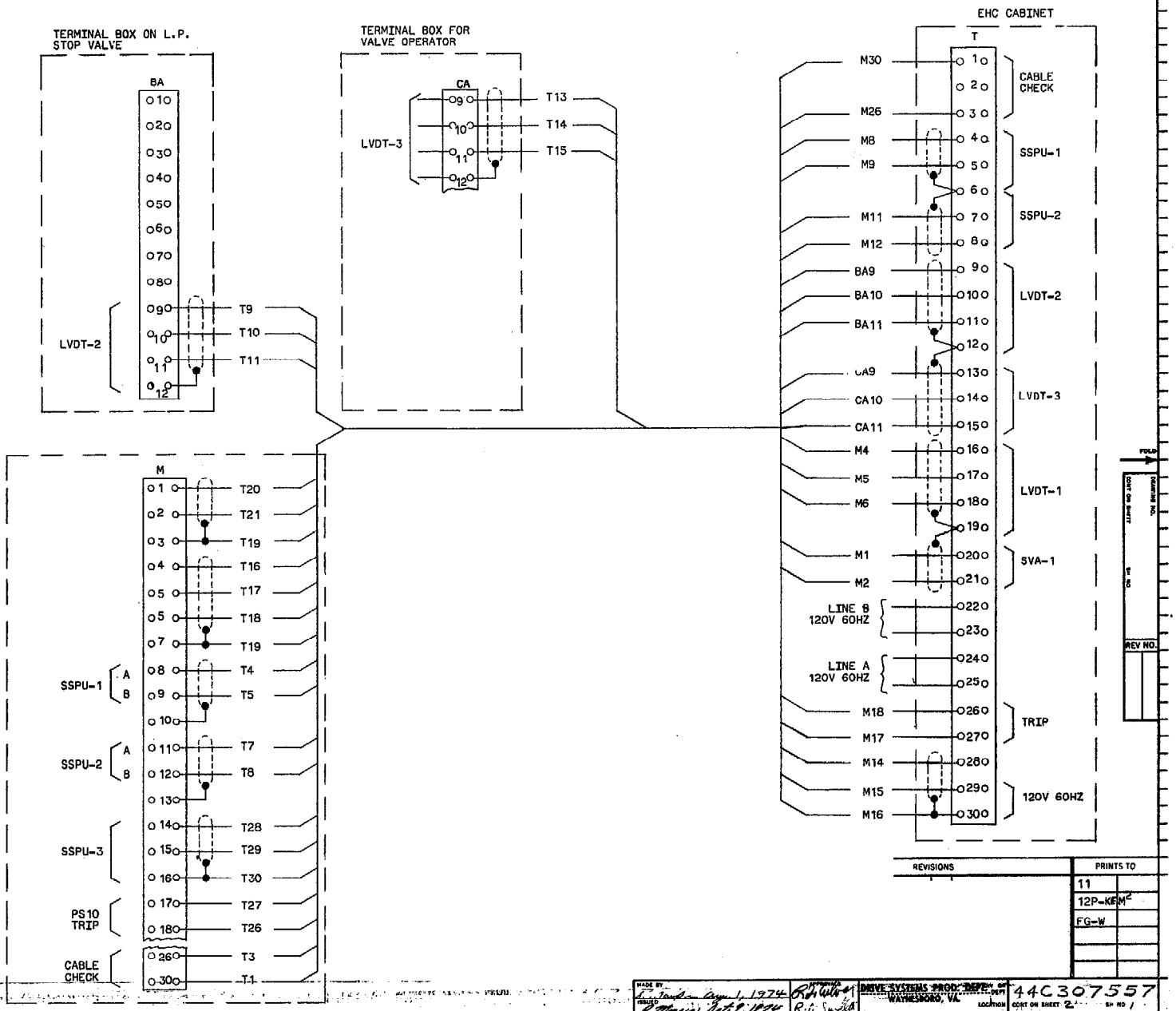
44C307557 CONT ON SHEET 2 SH NO. 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
INTERCONNECTION DIAGRAM
ELECTRONIC FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A14 TURBINE: 12?816
GULF STATE UTILITIES SABINE #5

NOTE:
1. REFER TO EHC ELEMENTARY 44C307556 AND TURBINE WIRING DIAGRAM 1348550EE FOR REFERENCE.



REVISIONS

PRINTS TO
11
12P-KEM
FG-W

MADE BY: L. Franks April 1, 1974
ISSUED: R. Masters April 1, 1974
APPROVED: R.D. Wilson R.C. Smith
DRIVE SYSTEMS PROD. DEPT. GE WAYNESBORO, VA.
44C307557 CONT ON SHEET 2 SH NO. 1

3

GENERAL ELECTRIC

44C307557 CONT ON SHEET 3 SH NO. 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
INTERCONNECTION DIAGRAM
ELECTRONIC FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A14 TURBINE; 122816
GULF STATE UTILITIES SABINE #5

EHC CABINET

CB
T51
D
T53
E
T43
T52
F
T54
T49
T47
T50
T116
T48
G
T55
T56
T57
T

ANALOG FEEDWATER CONTROL SIGNAL
10MA = LSS
50MA = HSS
FOR CUSTOMER USE
AUTO SPEED REFERENCE
CX — TCXX
105X CONTACTS OPEN ABOVE
105% SPEED
E7
FOR CUST. USE
EHC ABNORMAL ALARM
F17
F22
F7
F18
CB7
E30
D28
F1
G1
G3
G5
STATION BATTERY 125V +

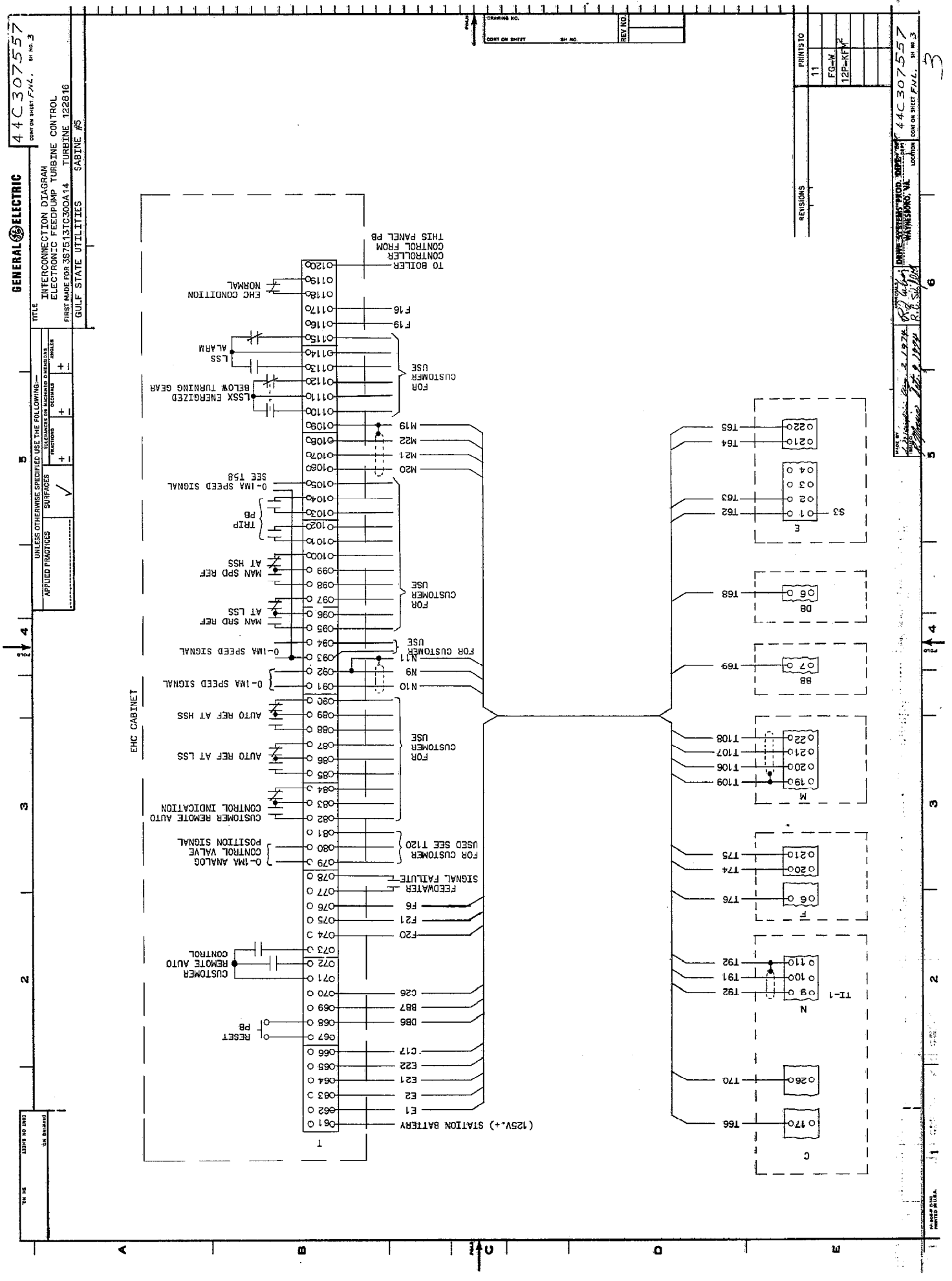
CONTROL PANEL CONN. DIAG. 44D306687

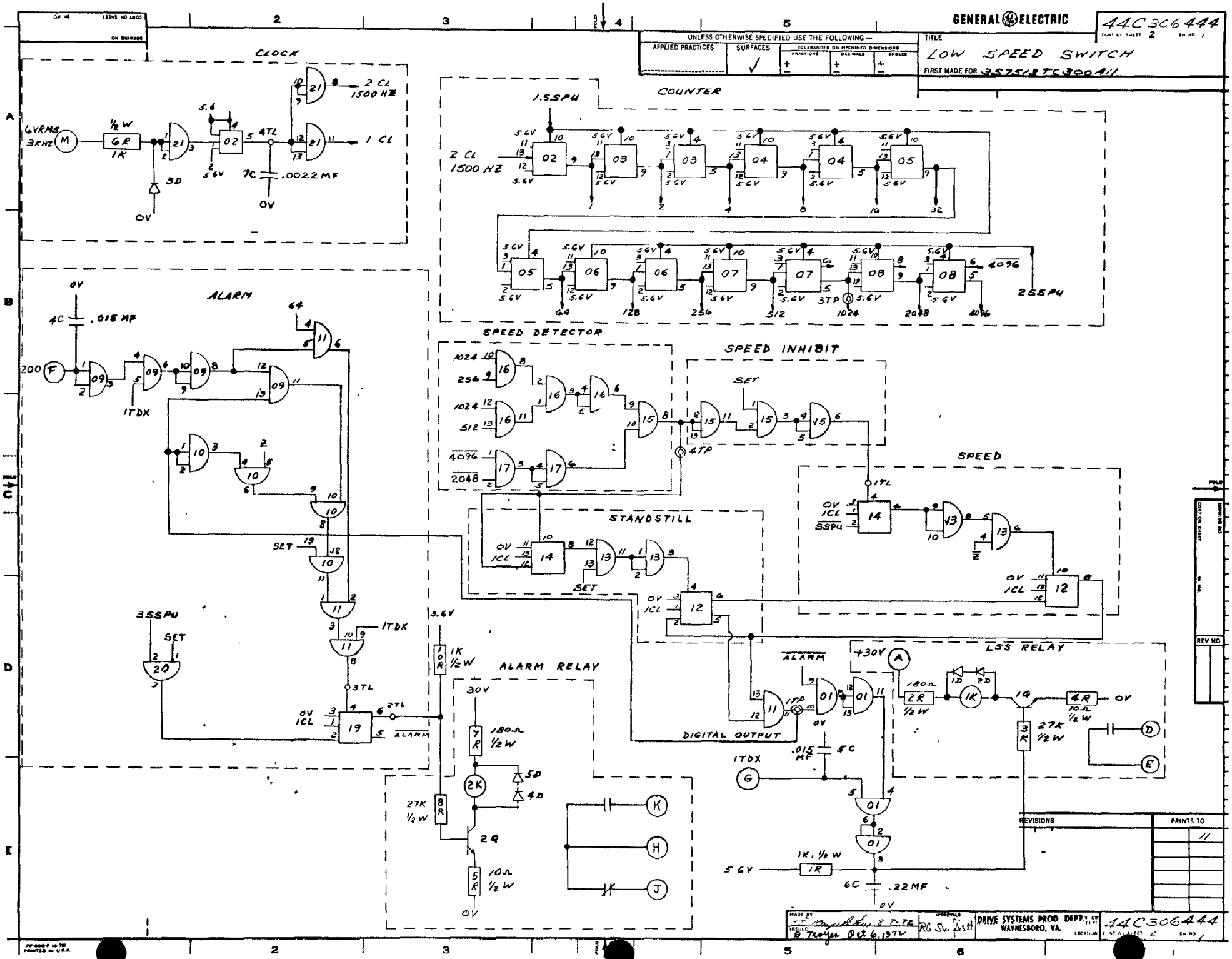
1CO 9/16 1 31/32"
2CO 13/16 1 31/32"
3CO 13/16 1 31/32"
THREE MULTICONDUCTOR 50 FT. CABLES WITH PLUGS FURNISHED WITH EHC.

REVISIONS

PRINTS TO
11
FG-W
12PKFM2

MADE BY R. J. Mullin Aug. 1, 1974
ISSUED R. J. Mullin Sept. 9, 1974
APPROVALS R. J. Culver R. L. Hall
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C307557 CONT ON SHEET 3 SH NO 2


GENERAL ELECTRIC
44C307557
INTERCONNECTION DIAGRAM
ELECTRONIC FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 357513TC300A14 TURBINE 122816
GULF STATE UTILITIES SABINE #5
EHC CABINET
T
TO BOILER CONTROLLER CONTROL FROM THIS PANEL PB
EHC CONDITION NORMAL
F16
F19
LSS ALARM
FOR CUSTOMER USE
LSSX ENERGIZED BELOW TURNING GEAR
M19
M22
M21
M20
0-1MA SPEED SIGNAL SEE T5B
TRIP PB
MAN SPD REF AT HSS
MAN SRD REF AT LSS
0-1MA SPEED SIGNAL
FOR CUSTOMER USE
N11
N9
N10
0-1MA SPEED SIGNAL
AUTO REF AT HSS
AUTO REF AT LSS
FOR CUSTOMER USE
CUSTOMER REMOTE AUTO CONTROL INDICATION
0-1MA ANALOG CONTROL VALVE POSITION SIGNAL
FOR CUSTOMER USED SEE T120
FEEDWATER SIGNAL FAILURE
F6
F21
F20
CUSTOMER REMOTE AUTO CONTROL
RESET PB
D86
BB7
C26
C17
E22
E21
E2
E1
(125V.+) STATION BATTERY
T65
T64
T63
T62
S3
E
T68
DB
T69
BB
T108
T107
T106
T109
M
T75
T74
T76
F
T92
T91
T90
TB-1
N
T70
C
PRINTS TO
11
FG-W
12P-KFW
REVISIONS
DRAWING NO.
CONT ON SHEET
REV NO.
APPROVED
DRAWN
LOCATION
SCALE
SHEET 3


GENERAL ELECTRIC
44C306444
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
LOW SPEED SWITCH
FIRST MADE FOR 3S7518TC300A11
CLOCK
6VRMS 3KHZ
½ W
6R 1K
3D
7C .0022MF
4TL
2 CL 1500 HZ
1 CL
1.5SPU
COUNTER
2 CL 1500 HZ
2.5SPU
ALARM
4C .015 MF
ITDX
SPEED DETECTOR
SPEED INHIBIT
4TP
SPEED
STANDSTILL
1TL
3.5SPU
ALARM RELAY
1K ½W
LSS RELAY
DIGITAL OUTPUT
1TD
2TL
3TL
2Q
5D
4D
1K, ½ W
6C .22MF
REVISIONS
PRINTS TO
DRIVE SYSTEMS PROD DEPT.
WAYNESBORO, VA.
44C306444

GENERAL ELECTRIC

44[illegible]3CE444

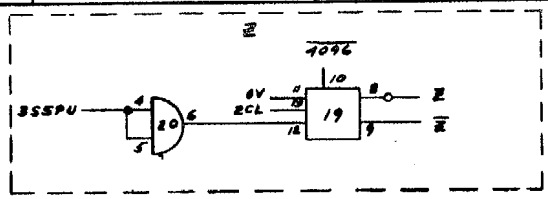
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

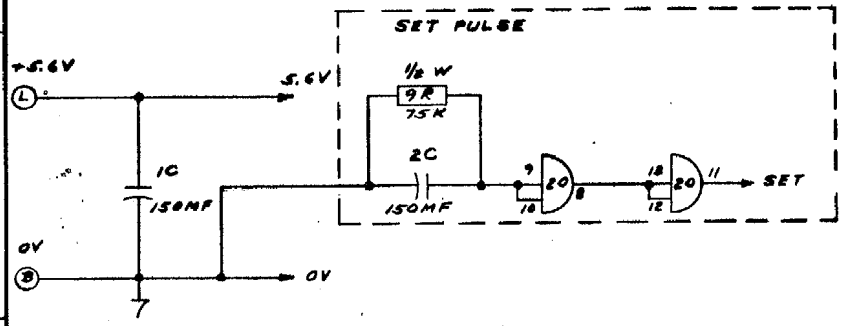
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ±	DECIMALS ±	ANGLES ±
	✓			

TITLE

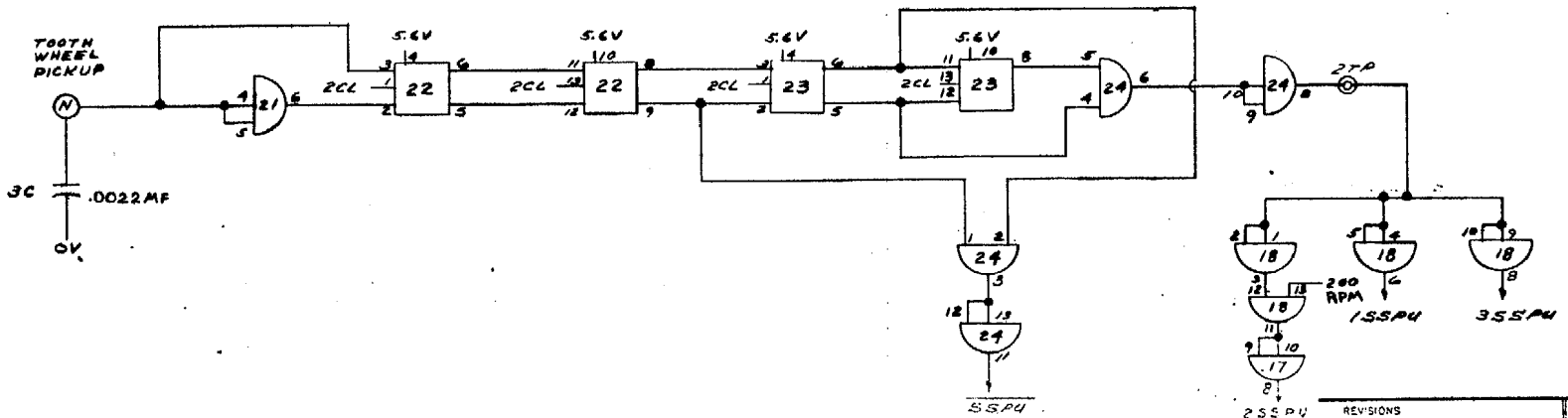
LOW SPEED SWITCH

FIRST MADE FOR 3S75/3TC300A11





SHAFT SPEED PICK-UP PULSE



REVISIONS

D. Thorpe Oct. 6, 1972

GENERAL ELECTRIC

44C305210

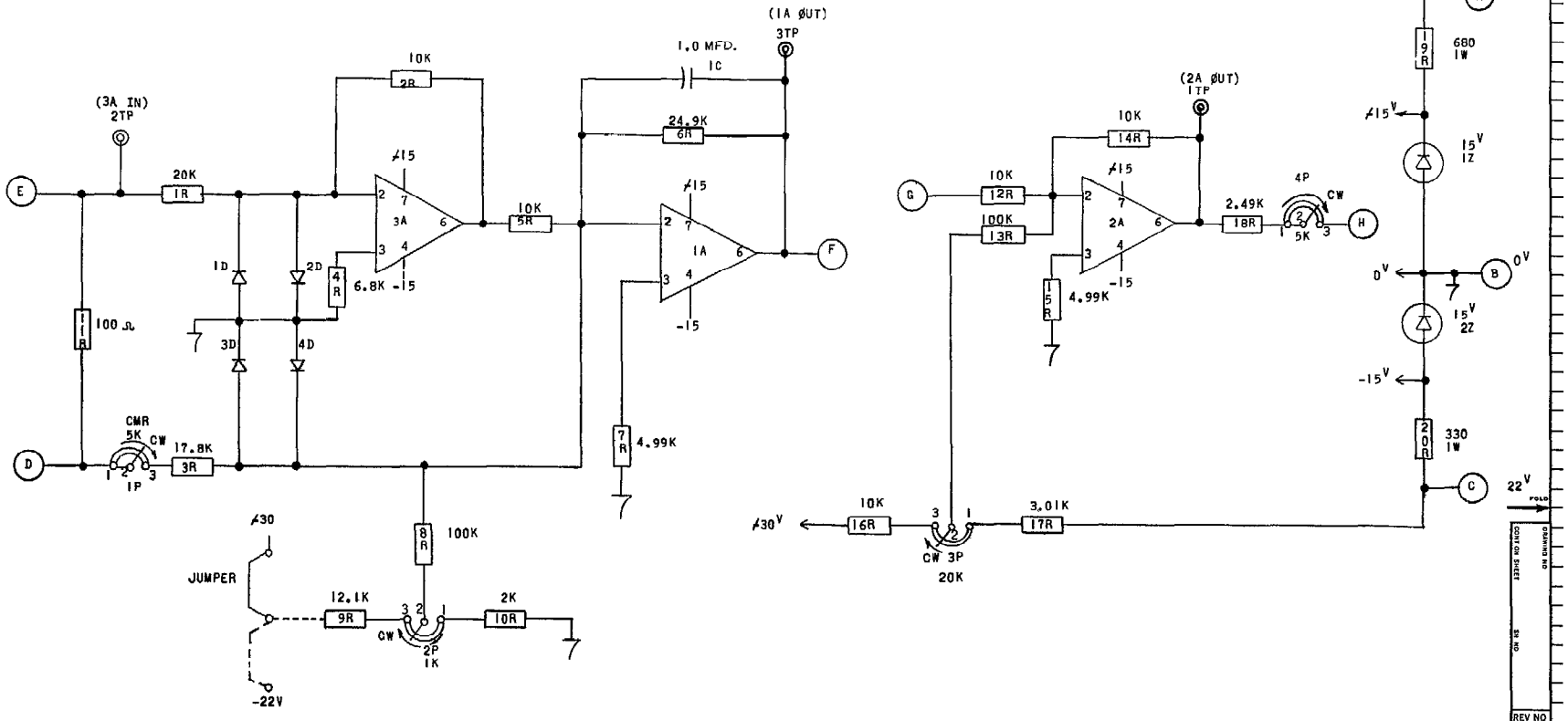
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	√	±	±	±

TITLE

5BCI
10-50-MA INPUT

FIRST MADE FOR 44B331746 G01



NOTES:

1. 2TP INPUT

LSS	HSS
10MA	50MA

3TP OUTPUT

LSS	HSS
0V	5V

2. CONNECTIONS

A ±30 VDC
B 0V
C -22 VDC
D BOILER CONTROL COMMON
E INPUT FROM BOILER CONTROL
F OUTPUT
G EHC AUTO SPEED REFERENCE
H AUTO REF POSITION TO BOILER CONTROL

REVISIONS		PRINTS TO
1	Nov 17, 1971	12P-DFB
		FG-W

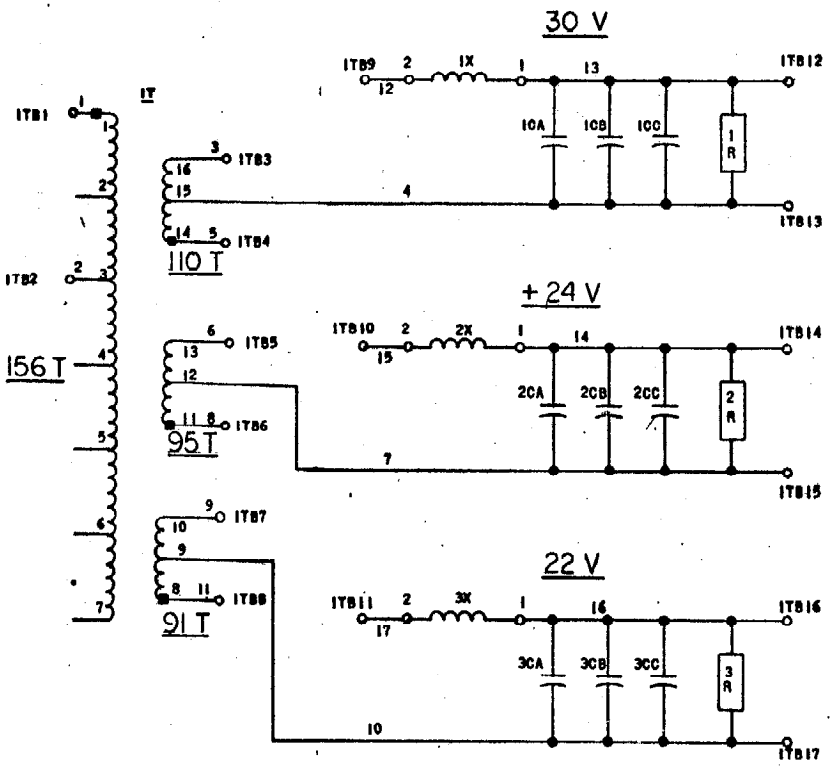
MADE BY	APPROVED	COMM. & CONT DEVICES	44C305210
ISSUED 9-21-71		WAYNESBORO, VA	CONT ON SHEET / SH NO

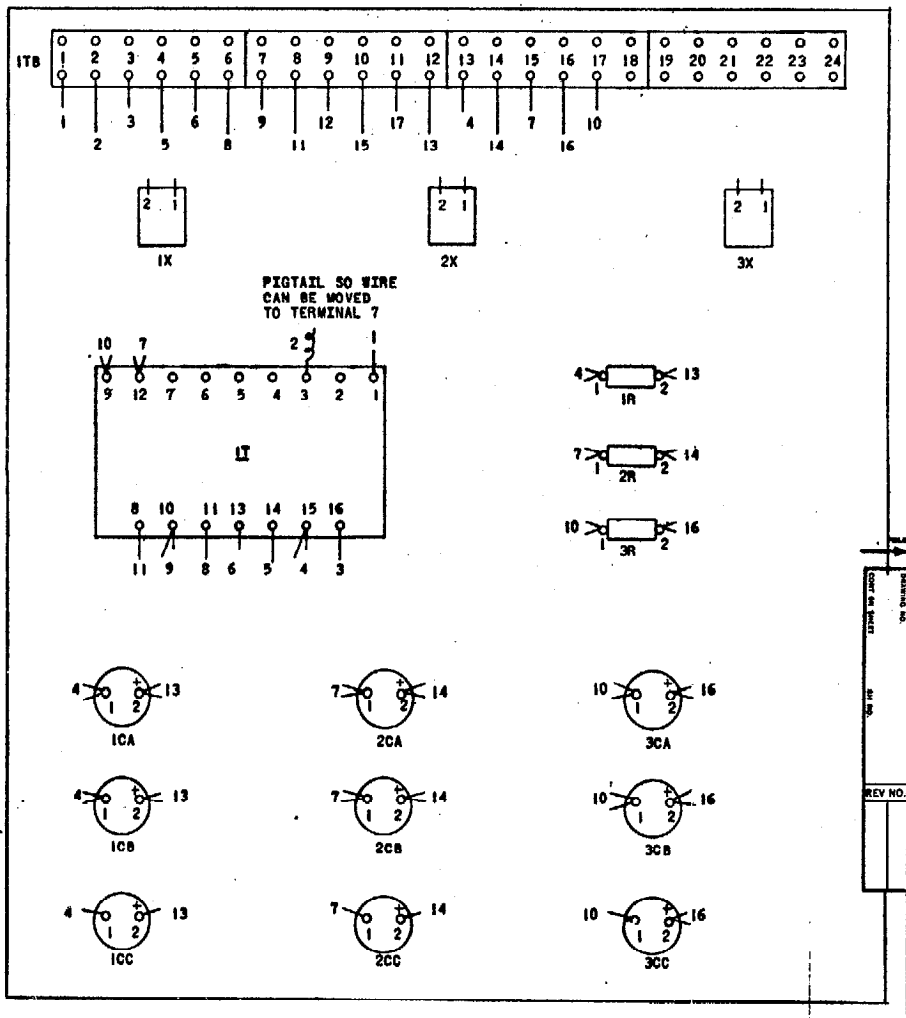
GENERAL ELECTRIC

44C306526

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES OF MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
LINE POWER SUPPLY
FIRST MADE FOR 3S75I3KF300A5





CKT. NO.	WIRE SEQUENCE, WIRE M9729494P1918 #18 WH.
1	ITB(1),IT(1)
2	ITB(2),IT(3)
3	ITB(3),IT(16)
4	1CC(1),1CB(1),1CA(1),IT(15),1R(1),ITB(13)
5	IT(14),ITB(4)
6	IT(13),ITB(5)
7	2CC(1),2CB(1),2CA(1),2R(1),IT(12),ITB(15)
8	IT(11),ITB(6)
9	IT(10),ITB(7)
10	3CC(1),3CB(1),3CA(1),3R(1),IT(9),ITB(17)
11	IT(8),ITB(8)
13	1CC(2),1CB(2),1CA(2),1R(2),ITB(12)
14	2CC(2),2CB(2),2CA(2),2R(2),ITB(14)
16	3CC(2),3CB(2),3CA(2),3R(2),ITB(16)

1X

CKT.	LEAD NO.	WIRE TO
13	1	ITB(12)
12	2	ITB(9)

2X

CKT.	LEAD NO.	WIRE TO
14	1	ITB(14)
15	2	ITB(10)

3X

CKT.	LEAD NO.	WIRE TO
16	1	ITB(16)
17	2	ITB(11)

REVISIONS

1	C.C.P.	Nov. 3, 1971
2	E.C. Calkins	May 10, 1972
3	W.H. Chapman	May 30, 1972

PRINTS TO 11

REV NO.

MADE BY A. Tripolitano 10-12-72
ISSUED Oct. 20, 1972 B. Mayer
CHECKED A. Carlson
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C306526

FF-609 P 14.72 PRINTED IN U.S.A.

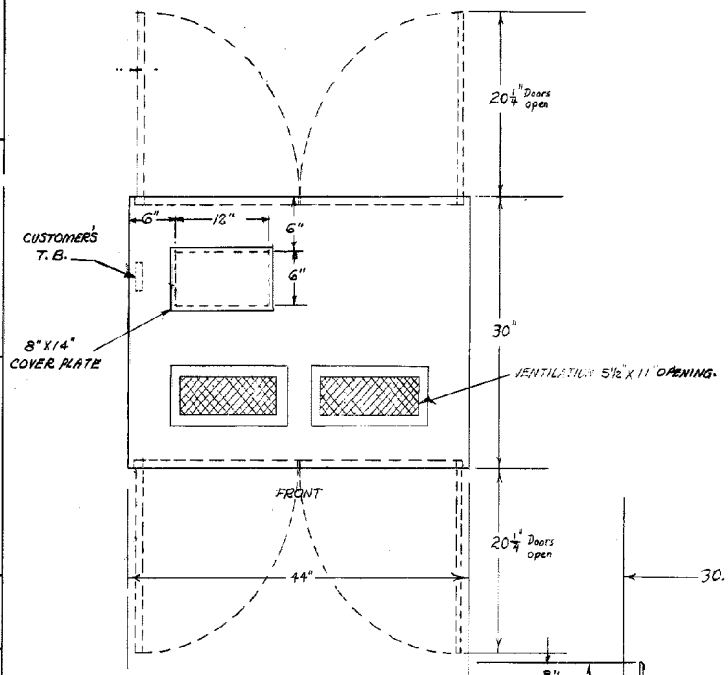
GENERAL ELECTRIC — 44D300043

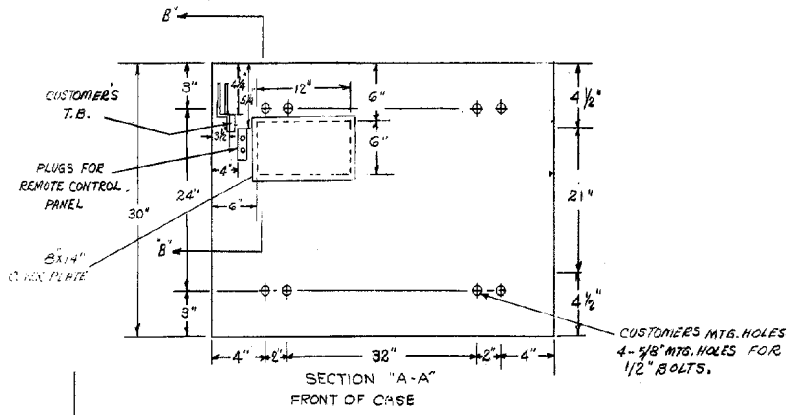
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

44D300043

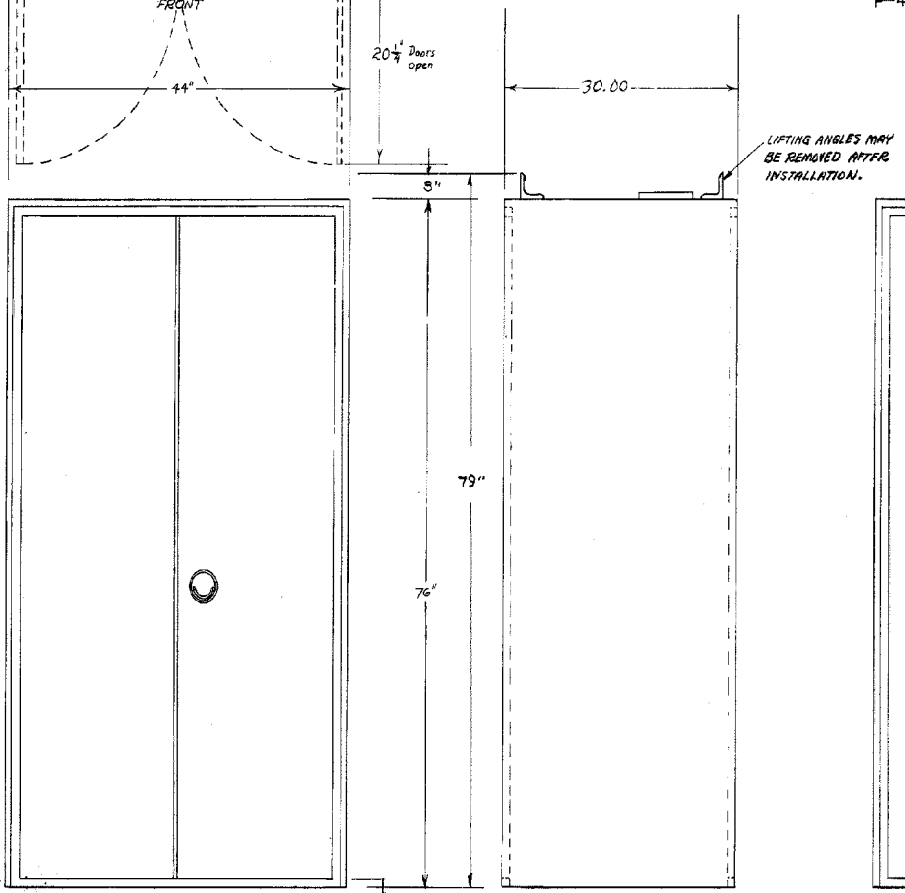
TITLE: ELECTRONIC FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S75L17C300A2
CABINET OUTLINE

NOTES:
1. ONE CABINET REQUIRED FOR EACH BOILER FEED PUMP TURBINE.
2. ESTIMATED WT. 1500 LBS.
3. MAXIMUM AMBIENT TEMP. 40°C
4. FINISH OUTSIDE: ASA49 GRAY
   INSIDE : GLOSS WHITE LACQUER
   DOORS : BLUE AIR DRY LACQUER
5. CABINET HAS LOOSE PIN, REMOVABLE HINGED DOORS.
6. DISCONNECTION OF CABINET/REMOTE CONTROL CABLE ENTRANCE EITHER FROM TOP ☐ OR BOTTOM ☐ AS CUSTOMER DESIRES. (CUSTOMER TO CHECK ONE BLOCK.)
7. CUSTOMER'S PROVISIONS FOR STEERING CABLE RUN NEXT TO CUSTOMER'S TERMINAL BOARDS.
8. IF NON-STANDARD FINISH REQUIRED SPECIFY ON ORDERING DRAWING.
9. POWER CONSUMPTION/DISSIPATION 750W.

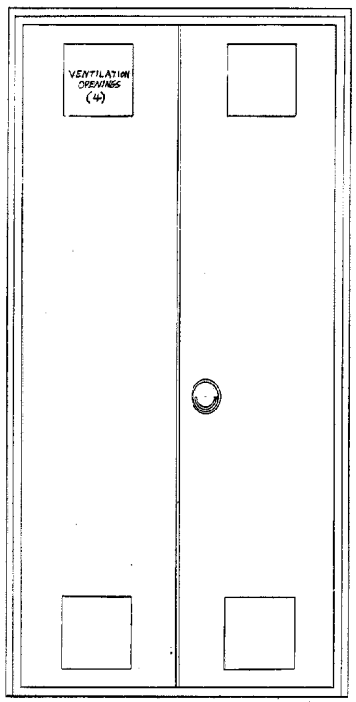




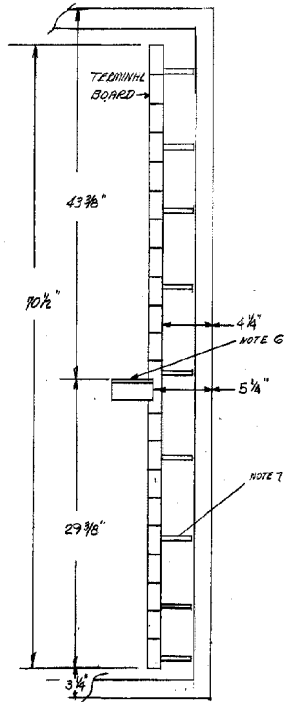
SECTION "A-A"
FRONT OF CASE



FRONT VIEW — "A" — SIDE VIEW



BACK VIEW



SECTION "B"-"B"

REVISIONS			PRINTS TO
1	S.C.	DEC/11/1967	11
2			
3			
4			
5			
6			
7	V.A.	JUL 13 1973	

MADE BY: [illegible] — ISSUED: NOV 21 1967 — APPROVED: [illegible]

SPECIALTY CONTROL DEPT. — WAYNESBORO, VA. — 44D300043

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	√	±	±	±

44D306685 — DRAWING NO. — 44D306685 CONT ON SHEET

TITLE: EHC BOILER FEEDPUMP ALIGNMENT DIAGRAM

FIRST MADE FOR 357313TC906A1A, 091-41289

GULF STATES — SABINE #5

PRIMARY CONTROL-5.00V
BACK-UP CONTROL-5.05V
AT RATED SPEED 5SPU 4000 HZ

ALIGNMENT INSTRUCTIONS

TURBINE RATED SPEED: 5700RPM@ 4000 HZ 5SPU

VOLTAGE COMPARATOR SETTINGS

LOC.	FUNCTION	OPERATING POINT
0218	10% SPEED	DO 400HZ 5SPU-3
0219	105% SPEED	DO 4200HZ 5SPU-3
0220	120% SPEED	DO4800HZ 5SPU-1
0117	REMOTE/LOCAL	DO+6.5V N-TP
0121	HI REF	DO+5V L-TP
0221	200 RPM	DO 122HZ 5SPU-3

VALVE LIFT	CONTROL	STOP L P	STOP H P	"OUT" VOLTS (DI, 0216
C.O.T.	0"	0"	0"	(1)
EFF.O	0.5			
FULL OPEN	10.6"	4.9"	4.25	(2)
FULL LOAD LIFT	3.823 FROM EFF. ZERO			

GAIN = (2) − (1) VOLT / 13 INCH =

REF. STOPS 2LVG, 0113 OUTPUT TP

	MAN	AUTO
LSS	0V	1.021V
HSS	5.271V	5.271V

COMPENSATION & REG. - 0212

TERMINALS J-K - NO CONNECTION
INST. REG. EFF. ROT. %
REG. 6% - SEE INSTRUCTIONS

OVERSPEED INTEGRATOR, 0109

RATE 12½ 2% / MINUTE — SEE INSTRUCTIONS
LIMIT 120% SPEED

SERVO AMP - 0214

BIAS — SEE INSTRUCTIONS
GAIN

VOLTS LIFT=GAIN (SEE ABOVE)

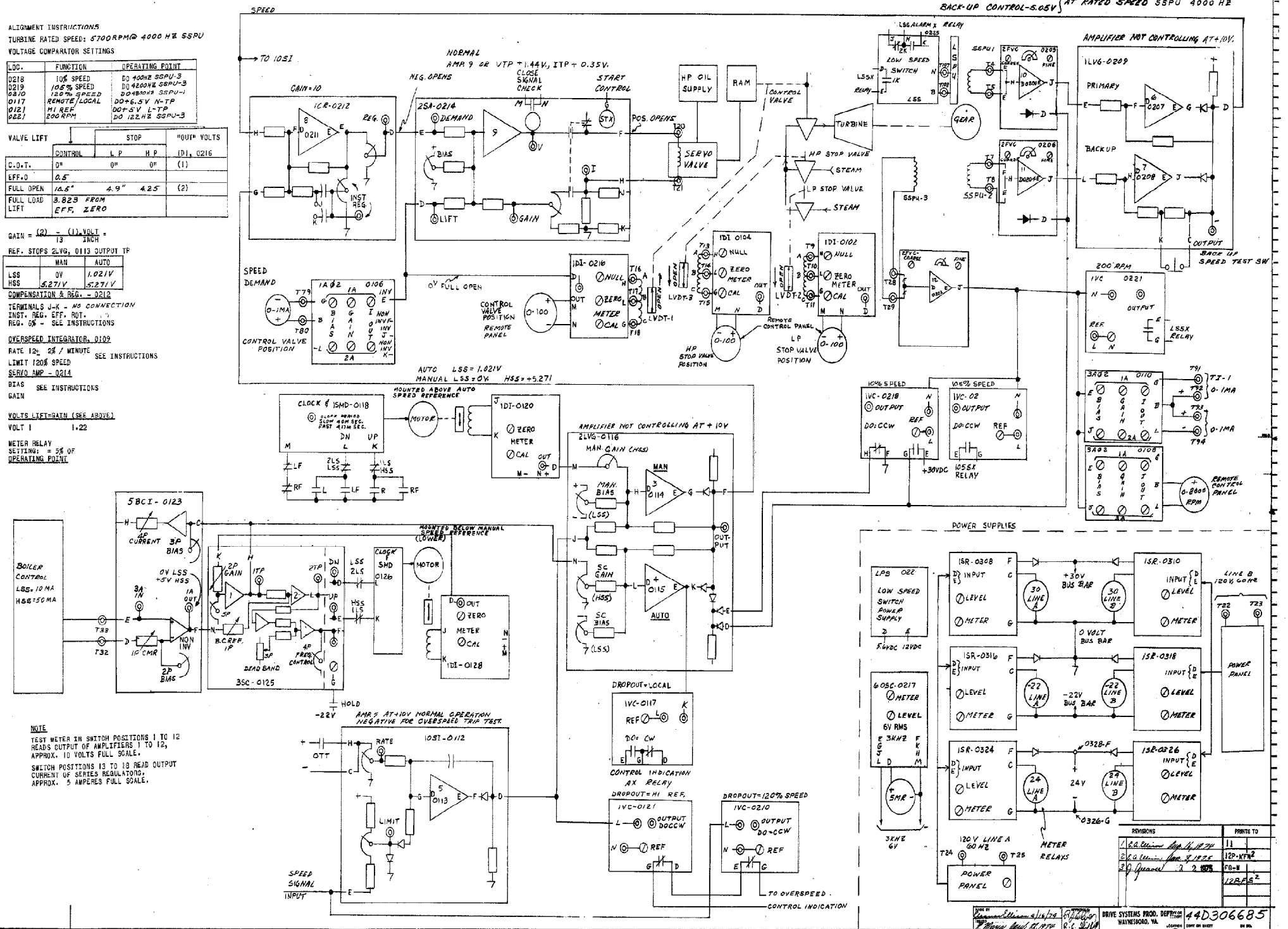
VOLT 1 1.22

METER RELAY SETTING: = 5% OF OPERATING POINT

NOTE

TEST METER IN SWITCH POSITIONS 1 TO 12 READS OUTPUT OF AMPLIFIERS 1 TO 12, APPROX. 10 VOLTS FULL SCALE.

SWITCH POSITIONS 13 TO 18 READ OUTPUT CURRENT OF SERIES REGULATORS, APPROX. 5 AMPERES FULL SCALE.



REVISIONS	PRINTS TO
1 P.A. Clinton Sept 11, 1974	11
2 P.A. Clinton June 7, 1976	12P-KTH2
3 J. Greene 2 2 1978	FO-M
	12RFE2

DRAWN BY P. Clinton 4/16/74 — APPROVED — DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA — 44D306685

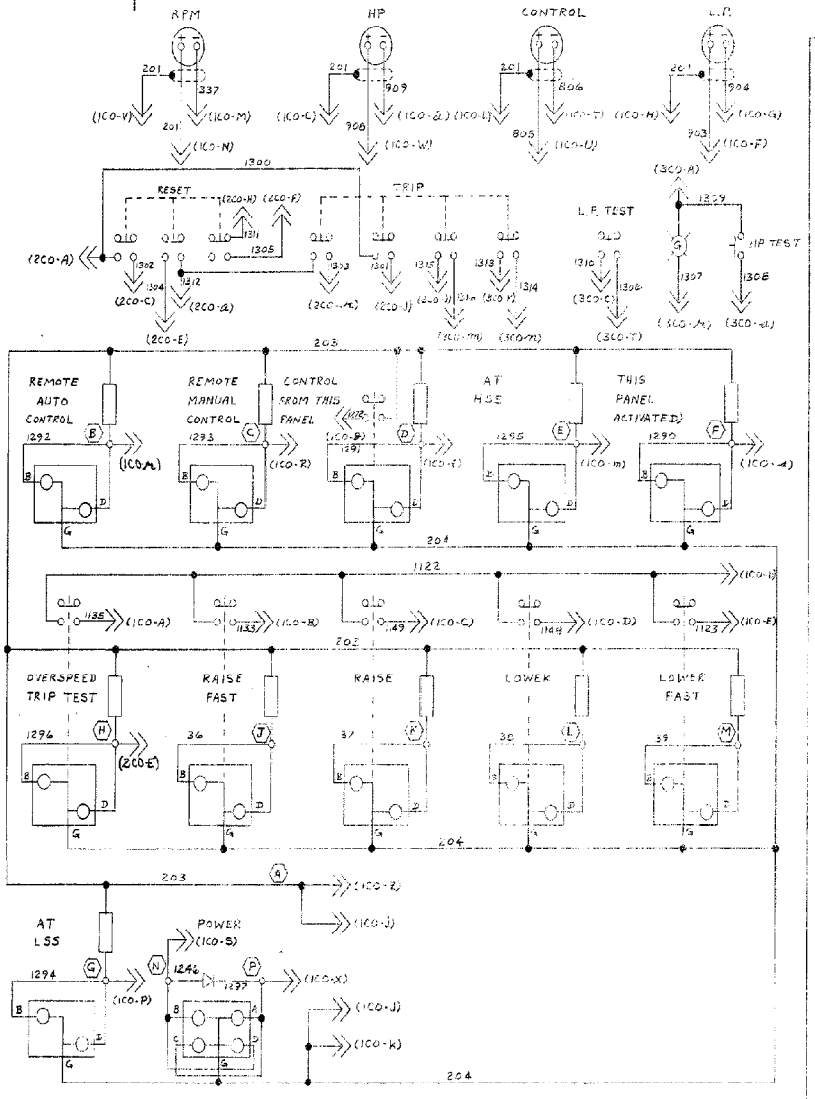
GENERAL ELECTRIC

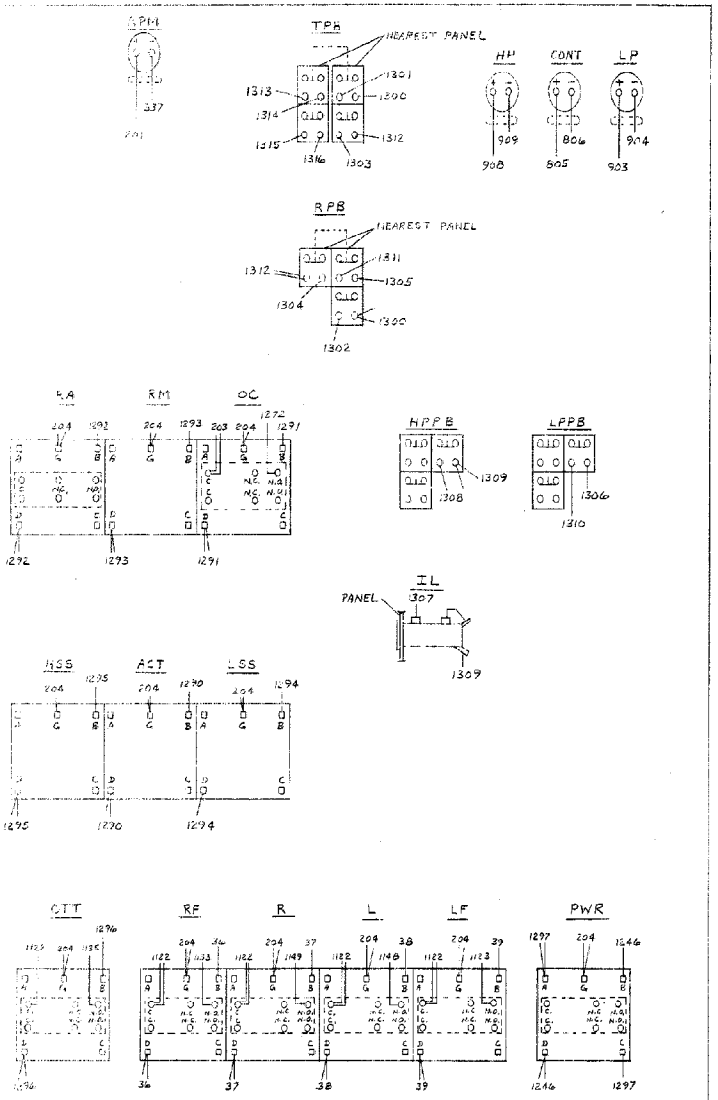
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-									
APPLIED PRACTICES:	SURFACES	TOLERANCES ON MACHINED DIMENSIONS							
		FRACTIONS		DECIMALS		ANGLES			
	√	+	−	+	−	+	−		

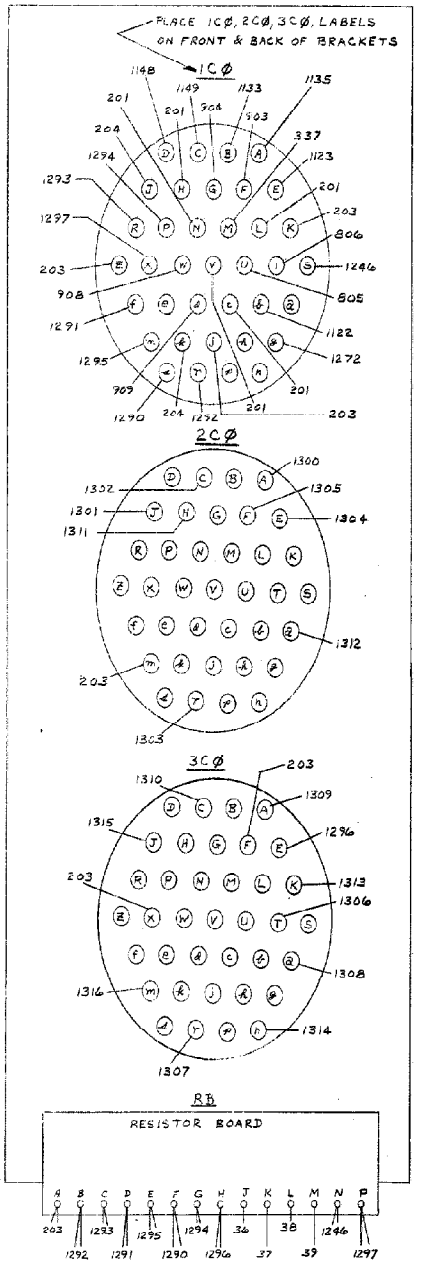
44D3066687 CONT ON SHEET DR NO.

TITLE: CONTROL PANEL
FIRST MADE FOR 3675/13KL300A12

44D3066687 CONT ON SHEET DR NO.







USE 44A310526-001 SHIELDED CABLE & #20 WIRE, M9729494P1819

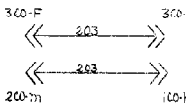
CIRCUIT	WIRE SIZE	CONNECTIONS
SHIELDED CABLE 201	WHITE	1CØ,RPM
	BLACK	1CØ,RPM
	SHIELD	1CØ,-V
SHIELDED CABLE 805	WHITE	1CØ,CONT
	BLACK	1CØ,CONT
	SHIELD	1CØ,L
SHIELDED CABLE 903	WHITE	1CØ,LP
	BLACK	1CØ,LP
	SHIELD	1CØ,-C
SHIELDED CABLE 908	WHITE	1CØ,HP
	BLACK	1CØ,HP
	SHIELD	1CØ,-C
36	#20	RF,RB
37		R,RB
38		LF,RB
39		LF,RB
203		100-J,-RB,OC,100-Z
1122		100-A,LF,-R,RF,QTT
1133		300-F,300-X
204		100-J,PWR,LF,-R,RF,QTT,HSS,ACT, LSS,OC,RM,PA,100-A
1123		LF,10Ø
1153		RF,10Ø
1135		10Ø,QTT
1148		L,10Ø
1149		R,10Ø
1246		100-S,RM,PWR,MR
1272		100-S,TC
203	#20	2CO-M,1CO-K

201	#20	100-,RF,ACT,ACT
1201		100-,RB,OC,OC
1206		100-,RB,RA,RA
1234		100-R,RB,RM,RM
1296		100-P,RB,LSS,LSS
1295		100-,RB,HSS,HSS
1206		200-E,RB,QTT,QTT
1297		100-X,RB,PWR,PWR

#16 WIRE, M9729494P2118

CIRCUIT	WIRE SIZE	CONNECTIONS
1300	#16	TPB,RPB,20Ø
1301		TPB,2CØ
1302		RPB,2CØ
1303		TPB,2CØ
1304		RPB,2CØ
1305		RPB,2CØ
1306		LPPB,3CØ
1307		IL,3CØ
1308		HPPB,3CØ
1309		IL,HPPB,3CØ
1310		LPPB,3CØ
1311		RPB,2CØ
1312		TPB,RPB,2CO
1313		TPB,3CO
1314		TPB,3CO
1315		TPB,3CO
1316		TPB,3CO

INDICATES RESISTOR BOARD TERMINALS



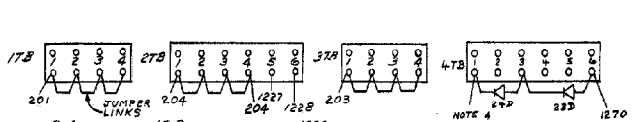
REVISIONS	PRINTS TO
	12RKPM
	FG-W

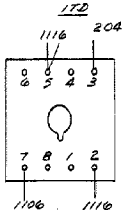
DESIGNED BY	APPROVED	DRIVE SYSTEMS PROD. DEPT.	44D3066687
LOCATION		WAYNESBORO, VA	CONT ON SHEET DR NO.

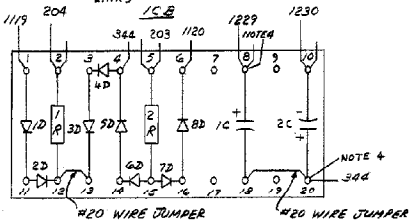
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES: SURFACES: √ TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± DECIMALS ± ANGLES ± REV NO. 44D308103 CONT ON SHEET SH NO.

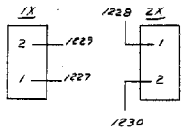
TITLE: RELAY PANEL

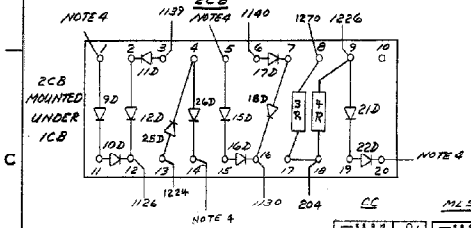
FIRST MADE FOR 9S7513 KF907A15

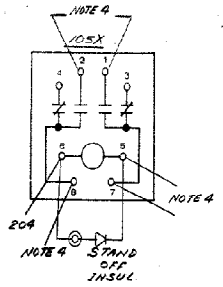


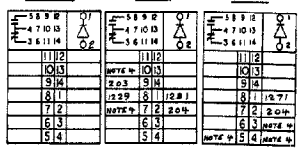


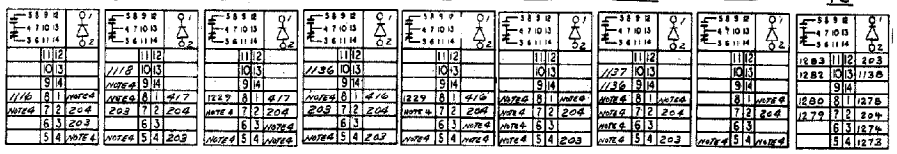


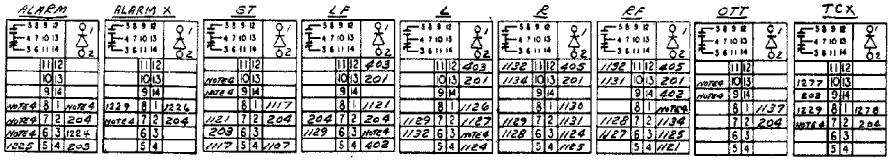


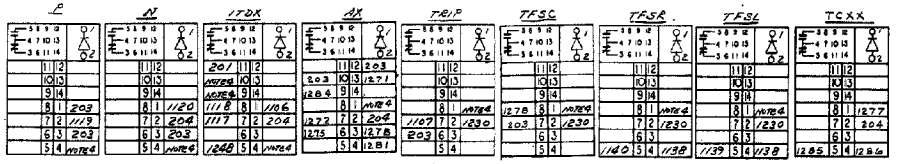


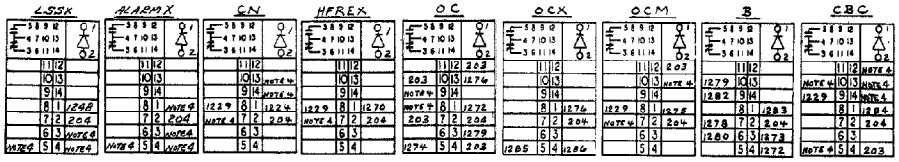














WIRE TABLE

CIRCUIT	WIRE SEQUENCE
201	RF,R,L,LF,1TDX,1TB
203	CC,MLSS,MLSX,MHSS,MHSX,ALSS,AHSS,TC,TCX,ST ALARM,P,N,AX,AX TRIP,TFSC,CBC,OCM,OC,OC OC,OC,TCM,1CB,3TB
204	105X,CC,MLSS,MLSX,MHSS,MHSX,ALSS,AHSS,HOLD, TC,TCX,OTT,LF,LF,ST,ALARM X,ALARM,N,1TDX,AX, LSSX,LSS ALARM X,CN,HREFX,OC,OCX,OCM,B,CBC, TCXX,AXX,TCM,1TD3,2TB
204	2CB-18,1CB-2,2TB
344	1CB-4,1CB-20
402	LF,RF
403	L,LF
405	R,RF
416	MHSS,MHSX
417	MLSS,MLSX
1106	1TD,1TDX
1107	TRIP,ST
1116	1TD,1TD,CC
1117	ST,ST,1TDX
1118	1TDX,MLSS
1119	1CB,P
1120	N,1CB
1121	RF,LF,ST
1124	L,R
1125	R,RF
1126	L,2CB
1127	L,RF
1128	RF,R
1129	R,L,LF
1130	R,2CB
1131	R,RF
1132	RF,R,L
1134	RF,R
1136	MHSS,AHSS
1137	AHSS,OTT
1138	TC,TFSR,TFSL
1139	TFSL,2CB
1140	TFSR,2CB
1225	ALARM,2FL
1226	2FL,2CB,ALARM X
1230	TRIP,TFSC,TFSR,TFSL,1CB
1248	LSSX,1TCX
1224	2CB,ALARM,CN
1270	1FL,HREFX,2CB,4TB
1271	AX,AXX
1272	OC,B,AX
1273	B,TC
1274	TC,OC
1275	CBC,AX
1276	OC,OCX
1277	TCXX,TCX
1278	TC,TCX,TFSC,AX,B
1279	OC,B,TC
1280	TC,B
1281	AX,TCM
1282	B,TC
1283	B,TC
1284	AX,CBC
1229	1CB,TCM,MHSX,MLSX,ALARM X,TCX,CBC,OCM, HREFX,CN
1285	TCXX,OCX
1286	TCXX,OCX

FACTORY NOTES:

1. USE #20 WIRE, W9729494P1618
2. DO NOT CORD-TIE WIRES. INTERCONNECTING WIRES TO BE ADDED AFTER PANEL IS MOUNTED IN CASE ASM.
3. CONNECT DIODES TO EACH RELAY SOCKET AS SHOWN IN RELAY SYMBOL.
4. WIRE CONNECTION SHOWN ON MASTER WIRE TABLE.

REACTOR WIRE TABLE

CIRCUIT	CONNECTIONS
1227	1X, 2TB
1228	2X, 2TB
1229	1X, 1CB
1230	2X, 1CB

REVISIONS: 1 APR 22 1975

PRINTS TO: 11, 12DT, FG4

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44D308103 CONT ON SHEET SH NO.

SCS-204 (4-55)

GENERAL ELECTRIC CO. WAYNESBORO, VA.	REGULATOR SUMMARY July 24 1974	44X306880
COMPILED BY: RJ Culver	DATE: 7/23/74	SHEETS; SHEET 1 of 1
CUSTOMER: GULF STATES, SABINE #5 (MDTPD)		
BFTB 122816	Requisition 071-41289	

ITEM	MATERIAL	QUAN.
1	3S7513TC300A14 Electronic Boiler Feedpump Turbine Control	1
	Outline Drawing 44D300043	
	Wire Table 44A307257	
	Interconnection Drawing 44C307557	
	Elementary Drawing 44C307556	
	Alignment Drawing 44D306685	
	Alignment Instructions 44A307258	
	Renewal Parts 44A335558	
2	3S7513KL300A12	1
	Outline 44B307044	
	Connection Diagram 44D306687	
3	Cabinet-Control Panel Power Cable 44B306148	2
4	Cabinet-Control Panel Control Cable 44B300423	1

REVISIONS	DISTRIBUTION	
1/Chgd. Pts.Nos. on Intercomm. Drwg. & Ele. Drwg./Culver 8/19/74	11	
	FG-W	
	IPP FS	2

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS PRODUCT DEPARTMENT
CONTROL DEVICES OPERATION • WAYNESBORO, VIRGINIA 22980

GENERAL ELECTRIC

7/75 (50)

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